

Fording: A Sound Shallow Translation of Indexed Inductive Families into First-Order Logic

AI draft of July 16, 2026

Abstract

Hammer-style proof automation for the Calculus of Inductive Constructions (CIC) translates goals and premises into untyped first-order logic (FOL) and runs classical automated theorem provers on the result. A companion paper defines such a translation, factored through program extraction, for a monomorphic fragment with non-indexed inductive datatypes, and proves it sound against a proof-irrelevant term model containing ill-typed “junk” elements. The present paper extends the architecture, and its soundness proof, to *indexed inductive families* — vectors, finite sets, bounded numbers, `reflect`-style views, and user-declared equality-like singletons. The mechanism is McBride’s *fording*: every constructor $c : \forall \vec{a}. I \vec{t}(\vec{a})$ is treated as a constructor of a non-indexed type carrying the equational premises $\vec{u} \approx \vec{t}(\vec{a})$, so that index constraints become propositional payload, stated — never solved — at translation time. The translation remains *shallow*: index constraints arrive at the prover as first-order equations inside guard formulas and inversion axioms, matches on families compile to per-constructor rewrite equations, and no inhabitation predicate mediates the logical structure. We define the source calculus (a stratified fragment of CIC with indexed datatypes, indexed inductive predicates, forced dependent case analysis, and elimination of index-determined propositional singletons), the erasure to an untyped functional calculus, the emitted first-order theory, and a term model with junk in which every emitted axiom is proved valid; soundness and consistency of the translation follow. The model construction doubles as an arbiter for the design questions documented in implementations of related translations: the guard of a family’s inversion axiom is not merely necessary — for families that specialise indices to constructor forms its omission is *inconsistent*; forgetting the index arguments of guard predicates reproduces, and is refuted by, the type-indexed-equality unsoundness observed in F* (issue #1542); per-constructor match equations need no guards even at statically impossible branches; index equations for

forced arguments are redundant; per-occurrence inlining of the forced inversion disjunction is sound in both polarities and interderivable with the axiomatic form; and the unconditional collapse of singleton eliminations is exactly the point where the model’s validation of uniqueness of identity proofs is spent. Every obstruction catalogued in the dependent-pattern-matching literature — heterogeneous equality, Streicher’s K, green slime — is either validated by the model or dissolved by the untyped target.

Contents

1	Introduction	3
1.1	The idea: fording	4
1.2	Contributions	5
1.3	What soundness means here	7
1.4	Relation to the implemented translation	8
2	Fording, and what an untyped target changes	8
2.1	Indexed families and their eliminations	8
2.2	Fording	9
2.3	What the untyped target dissolves	10
2.4	What the untyped target does not dissolve	11
3	The source calculus CIC_{ix}^-	12
3.1	Syntax	12
3.2	Environments	14
3.3	Typing	16
3.4	Reduction and conversion	20
3.5	The shallow fragment	21
3.6	Programming with forced case analysis	21
3.7	Relation to CIC	23
4	The intermediate language and program extraction	24
4.1	The calculus $\lambda_{\square}$	24
4.2	Erasure	26
5	The translation	27
5.1	Signature	28
5.2	Compilation of erased programs	28
5.3	Specification extraction	31
5.4	Structural axioms: the ford of a family	32
5.5	Forced-argument simplification	35

6	Semantics	36
6.1	The domain	36
6.2	Interpretation of types, propositions and predicates	37
6.3	The first-order structure $\mathfrak{M}$	40
6.4	Adequacy of the term translation and of compilation	41
6.5	Coincidence: the translation reflects the semantics	42
7	Soundness	43
7.1	Guard disciplines	43
7.2	The fundamental lemma	44
7.3	Validity of the emitted theory	48
7.4	Main theorem and corollaries	49
8	Fording at the level of formulas: design questions settled	51
8.1	Per-occurrence expansion of forded guards	51
8.2	Which guards are load-bearing	53
8.3	Match equations for indexed scrutinees	55
8.4	Indices, elements, and injectivity of type formers	57
8.5	Singleton collapse and uniqueness of identity proofs	58
9	Shallowness	61
10	Related work	63
11	Limitations and future work	65
11.1	Dependent index telescopes	65
11.2	Polymorphism, large indices, large elimination	65
11.3	Well-founded recursion and Acc-style predicates	66
11.4	Acyclicity, induction, completeness	66
11.5	Functional extensionality	66
11.6	Reconstruction	67
11.7	Other restrictions	67
A	Confluence of $\lambda_{\square}$	72

1 Introduction

A *hammer* translates a goal of a proof assistant, together with a selection of library facts, into the input logic of automated theorem provers (ATPs), and reconstructs any proof the ATPs find [BKPU16]. For proof assistants based on the Calculus of Inductive Constructions (CIC) — Rocq (formerly Coq) and its relatives — the reference tool is CoqHammer [CK18, CK26], whose target is untyped classical first-order logic (FOL). The efficiency of such a translation hinges on it being *shallow* [Cza18]: source propositions must

become first-order *formulas* whose connectives, quantifiers and equalities are the native first-order ones, so that clausification, superposition and term indexing do real logical work on the translated problem, instead of simulating the object logic through an inhabitation predicate one bridge axiom at a time.

A companion paper [Ext26] carries this programme out for a monomorphic fragment of CIC with *non-indexed* inductive datatypes: the translation of a definition is factored into *program extraction* (the Letouzey-style erasure [Let03, Let04] of the body, compiled to per-constructor rewrite equations) and *specification extraction* (a realizability reading of the type as a first-order formula about the translated constant), and both components are proved sound against a single first-order model: a proof-irrelevant, classical term model over the erasure calculus whose domain deliberately contains ill-typed “junk” elements. The model decides, uniformly, which guards are semantically mandatory and which may be dropped.

That development stops at a well-marked boundary: its datatypes have no indices. *Indexed inductive families* — length-indexed vectors $\text{vec } A : \text{nat} \rightarrow \text{Set}$, finite sets $\text{fin} : \text{nat} \rightarrow \text{Set}$, bounded numbers, the `ssreflect` view type $\text{reflect } P : \text{bool} \rightarrow \text{Set}$, and the user-declared equality-like and `le`-like predicates — are the workhorses of dependently typed libraries, and they are exactly the point where CIC’s typing discipline diverges most sharply from first-order logic: the type of a constructor *constrains* the indices at which it may appear, case analysis must *unify* the indices of the scrutinee’s type with those of each constructor, and eliminations of propositional families into computational content are governed by the singleton criterion, Streicher’s K, and their refinements [GMM06, CDP14, GCST19]. The present paper extends the companion’s calculus, translation and soundness proof to indexed families, and settles the design questions specific to them.

1.1 The idea: fording

The mechanism is due to McBride [McB99, §3.5], who named it after Henry Ford’s “any colour ... so long as it is black”: a constructor of an indexed family,

$$c : \forall \vec{a} : \vec{A}. I \vec{t}(\vec{a}),$$

is equivalent to a constructor of a *non-indexed* family that accepts arbitrary indices $\vec{u}$ but carries equational premises restoring the lost specificity:

$$c^{\text{Ford}} : \forall \vec{u}. \forall \vec{a} : \vec{A}. (\vec{u} = \vec{t}(\vec{a})) \rightarrow I^{\text{Ford}} \vec{u}.$$

Index constraints thereby become *propositional payload*, on the same footing as the refinement payloads $P x$ of subset types $\{x:A \mid P x\}$, which the companion translation already expands shallowly at every occurrence. Every

system that elaborates dependent pattern matching performs this reduction internally — Coq’s *inversion* [CT96, Mon10], the Goguen–McBride–McKinna translation of pattern matching to eliminators (“the equations on the indices are exactly those we must unify” [GMM06]), Agda’s unifier [CDP14, CD18], and the Equations package [SM19]. The present paper performs it in a translation to untyped first-order logic, *per occurrence and per constructor*, and proves the result sound.

Two classical obstacles make fording delicate inside type theory, and both dissolve in an untyped first-order target. First, telescopic index equations relate terms whose *types* are only equal after earlier equations have been substituted; stating them requires heterogeneous (“John Major”) equality, whose eliminator is exactly Streicher’s K [McB99, CDP14]. Untyped first-order equality states them directly. Second, defined functions in index positions — McBride’s *green slime* [McB14] — make elaboration-time unification stuck: $0 \approx \text{plus}(m, n)$ is neither a positive nor a negative unification success because *plus* is not a rigid symbol. A saturation prover simply *keeps* the equation and reasons with the defining equations of *plus*; specialization by unification becomes ordinary superposition, and branches that a type-theoretic elaborator must leave stuck can be closed by proof. The translation of this paper accordingly never solves index constraints; it only states them.

What does *not* dissolve by itself is soundness. The literature on translations of inductive types into untyped or weakly typed logics is a catalogue of documented accidents precisely at the three points this paper must get right: unguarded constructor-exhaustiveness bounds the cardinality of the whole untyped universe (Meng and Paulson’s two-element universe [MP08, §2.3]; the monotonicity analysis of [BBPS16] identifies exhaustiveness axioms as exactly the sentences that can never lose their guards); encoding a *type-indexed* equality by the native untyped equality of the target lets an equation established at one index leak to another (the F^* unsoundness #1542 [Ase18], where pointwise function equality proved at domain *nat* escaped to domain *int*, yielding *False*); and axioms that are individually plausible — selector totality, rank/height orderings, box/unbox roundtrips — can interact inconsistently on degenerate or empty instances (the Verus unsoundness #1366 [The24b]). Our response to all three is the companion’s discipline, extended: *every* emitted axiom is proved valid in one fixed model, constructed once, so no combination of emitted axioms can be inconsistent (Corollary 7.10), and the model arbitrates which guards are load-bearing. Where a design variant is unsound, we do not merely avoid it: we prove its unsoundness inside the framework, as a specification for implementations (Section 8).

1.2 Contributions

This paper is self-contained: all definitions and proofs are given in full, including the parts shared with the companion paper [Ext26], which are re-developed here in their generalized form. Concretely:

1. We define CIC_{ix}^- (Section 3), a stratified, monomorphic, first-order fragment of CIC whose datatypes are *indexed inductive families* with dependent constructor telescopes and propositional constructor premises, subsuming the non-indexed datatypes, refinement types and indexed inductive predicates of the companion fragment. Case analysis on a family is *forded*: each branch binds, besides the constructor’s arguments and premises, proofs of the index equations $\vec{u} = \vec{t}_i(\vec{y}_i)$, so that dependently typed programming idioms (impossible branches, index-directed results) are expressible with first-order motives, without large elimination. The calculus also has elimination of *index-determined* propositional singletons — the image of CIC’s singleton elimination for equality-like families, with McBride-style forcing in place of CIC’s parameter side condition — and a constructor-discrimination proof former.
2. We define the erasure of CIC_{ix}^- into the untyped calculus $\lambda_{\square}$ and prove the two syntactic pillars of extraction — substitution commutation and simulation of reduction — together with confluence of $\lambda_{\square}$ (Section 4, Appendix A). Erasure is index-blind; in particular proofs occurring *inside index terms* vanish, which is the translation-level form of canonicalizing erased data to a single token [Atk18, Wer08].
3. We define the translation (Section 5). Guard predicates become $(1+m)$ -ary — the guard of an occurrence $I(\vec{u})$ is the atom $\hat{I}(w, \mathcal{C}(\mathcal{E}(\vec{u})))$, carrying the translated indices — and the structural axioms of a family are the first-order image of its ford: constructor typing with index conclusions, and a guarded *inversion* axiom whose disjuncts carry constructor payloads *and index equations*. Pattern-matching definitions compile, unchanged, to unguarded per-constructor equations. We prove the forced-argument simplification (index equations solvable for a bare variable are redundant) as a first-order equivalence (Lemma 5.14).
4. We construct the model (Section 6): the domain is again the set of closed $\lambda_{\square}$ terms modulo conversion, junk included; an indexed family denotes a set of (index-tuple, element) pairs built in stages. The key new lemma is *index reflection* (Lemma 6.5): every member of the interpretation decomposes uniquely as a constructor form whose index tuple equals the denotation of the constructor’s index terms — the semantic content of fording, and the point where the model validates what uniqueness of identity proofs validates in type theory. Ranks become element ranks, which index reflection makes well defined.

5. We prove soundness (Section 7): every axiom of the emitted theory holds in the model (Theorem 7.8), first-order provability of the translated goal implies truth of the goal in the proof-irrelevant classical semantics (Theorem 7.9), the theory is consistent (Corollary 7.10), and no refutable goal becomes provable (Corollary 7.11). The fundamental lemma (Lemma 7.6) is proved in full; its new cases — forced case analysis, singleton elimination, fixpoints over families — are exactly where index reflection is spent. On premise-free families with first-order arguments the model is standard: $\llbracket \text{vec} \rrbracket$ at index n is in bijection with length- n lists (Proposition 7.12).
6. We settle the indexed-specific design questions (Section 8), each as a theorem about the model or the theory:
 - the inversion axiom’s guard is mandatory in a strong sense — for every family that specialises an index to a constructor form, the unguarded axiom is *inconsistent* with the discrimination axioms, not merely unsound (Proposition 8.7);
 - the index arguments of guard predicates are mandatory: the index-forgetting variant proves a goal that is false in the source semantics — the precise analogue of $F^* \#1542$ (Proposition 8.9);
 - per-constructor match equations remain valid without any guard or index premise, *including* equations for statically impossible branches; the index-premised variant is strictly weaker and both are sound, so the choice is operational (Propositions 8.11 and 8.12); pruning rigidly clashing branches is sound and prunes only semantically dead equations (Proposition 8.13);
 - per-occurrence inlining of the forced inversion disjunction — the shallow “expansion” of classified occurrences — is sound in both polarities and interderivable with the axiomatic form (Theorem 8.3 and Proposition 8.4);
 - index uniqueness ($\widehat{I}(w, \vec{z}) \wedge \widehat{I}(w, \vec{z}') \rightarrow \vec{z} \approx \vec{z}'$) is a first-order consequence of the emitted axioms, while injectivity of the type former itself is neither expressible in the signature nor emitted — as it must not be, being anti-classical (Proposition 8.15, Remark 8.16);
 - the unconditional singleton collapse is the exact point where the model’s proof irrelevance is spent as uniqueness of identity proofs; Werner’s guarded singleton reduction is the premise-carrying alternative, and both variants are sound (Fact 8.17, Proposition 8.18).
7. We re-verify the shallowness checklist of the companion paper on the extended translation and extend it with an indexed-specific item: index constraints arrive as first-order equations computed at translation time

— never as unification problems, never as type-term encodings, never through an inhabitation predicate (Section 9).

1.3 What soundness means here

As in the companion paper, soundness is model-theoretic: if the emitted theory first-order proves the translated goal, the goal is true in a fixed proof-irrelevant, classical semantics of the source development (Theorem 7.9). This is the property with operational consequences for a hammer — no refutable goal acquires a phantom proof, premise-selection learning is not poisoned, and the emitted theory is consistent — while proof reconstruction inside the proof assistant remains the final arbiter of each individual proof. The semantics validates excluded middle and proof irrelevance by construction (the ATPs are classical; erasure forgets proofs); consequently it also validates uniqueness of identity proofs, and Section 8.5 makes precise where the translation spends that strength and what the corresponding reconstruction obligations are. The converse property, completeness, is deliberately traded away; Section 11 delimits what is lost.

1.4 Relation to the implemented translation

CIC_{ix}^- and its translation are an idealisation of a design for CoqHammer’s translation pipeline, not a description of its current code. The current implementation refuses to classify indexed families for erasure-based treatment; declaration-level inversion axioms for inductive predicates already carry index equations, but they are mediated by an inhabitation atom rather than expanded per occurrence; a user-written match on an equality proof receives no definitional axiom at all (only a fixed list of transport constants is special-cased); and per-constructor case equations for indexed matches are emitted without index premises, including for statically impossible branches. Every construction of this paper has a direct implementation site in that pipeline: the forced constructor telescopes of Section 5.4 correspond to classifying the forced form of a family; the expansion theorem of Section 8.1 to per-occurrence guard expansion; the singleton elimination of Section 3 to the erasure of matches on equality-like proofs; and the propositions of Section 8.3 to the choice points (index premises, branch pruning) that the implementation exposes as options. We indicate the correspondences where relevant but keep the development self-contained.

2 Fording, and what an untyped target changes

This section reviews the type-theoretic situation the paper starts from, fixes the reading of *fording* used throughout, and identifies which parts of the classical machinery an untyped first-order target renders unnecessary — and

which soundness obligations it creates instead. It contains no formal development; the definitions begin in Section 3.

2.1 Indexed families and their eliminations

An inductive family in the sense of Dybjer [Dyb94] is an inductive definition $I : \vec{\sigma} \rightarrow s$ whose constructors target *instances* $I \vec{t}(\vec{a})$ determined by the constructor’s own arguments. The paradigmatic informative examples are

$$\text{vnil} : \text{vec } A \text{ } \mathbf{O}, \quad \text{vcons} : \forall n : \text{nat}. A \rightarrow \text{vec } A \text{ } n \rightarrow \text{vec } A \text{ } (\mathbf{S } n),$$

and the paradigmatic propositional ones are the equality family and `le`. Three features distinguish families from plain datatypes in a way that any translation must confront.

1. *Constructor typing constrains indices.* A constructor inhabits only the instances of the form $I \vec{t}(\vec{a})$; an arbitrary instance $I \vec{u}$ is inhabited by c-forms only when the constraint $\vec{u} = \vec{t}(\vec{a})$ is solvable.
2. *Case analysis specializes indices.* A dependent match on a scrutinee of type $I \vec{u}$ must reconcile $\vec{u}$ with each constructor’s $\vec{t}_i(\vec{y}_i)$. Type-theoretic elaborations do so by first-order unification with constructors as the rigid symbols [GMM06]; branches whose constraints are refutable are discharged, and solvable constraints instantiate the branch.
3. *Propositional families may or may not eliminate into computations.* CIC permits a match on a `Prop`-sorted scrutinee in an informative position only for empty and *singleton* inductives (one constructor, all arguments propositional, parameters aside) [The26, Let03]; the equality family qualifies because its constructor has no arguments beyond parameters. The criterion is known to be simultaneously too strong and too weak [GCST19]: it rejects families whose non-parameter arguments are *forced* by the indices, and accepting it together with proof irrelevance makes uniqueness of identity proofs (UIP) hold.

2.2 Fording

McBride’s thesis [McB99, §3.5] introduces the constraint technique this paper is built on: to use an elimination rule whose motive ranges over the whole “aperture” $I \vec{t}$ at a more specific instance, one abstracts the full aperture and *constrains* it with pointwise equations $\vec{t} \simeq \vec{u}$ — “the patterns to which it applies are subject to equational constraints which recover their specificity.” Goguen, McBride and McKinna’s elimination of dependent pattern matching [GMM06] runs on the same device: in their specialization by unification, the method for constructor `c` receives hypotheses $\vec{u} \simeq \vec{t}_c(\vec{y})$ and $c \vec{y} \simeq x$ — “the equations on the indices are exactly those we must unify

to allow the instantiation.” At the level of declarations the transformation is folklore, stated compactly by Zhang [Zha23]: a constructor $c : X \rightarrow F a$ fords to $c^{\text{Ford}} : \forall a'. (a = a') \rightarrow X \rightarrow F a'$, after which “ F is no longer indexed” — the family has become a *parameterized* type whose constructors carry equational premises, and index specialization has become ordinary propositional reasoning that a consumer may perform whenever convenient.

This paper uses fording in three places, always as a lens and never as a literal re-declaration:

- in the *source calculus*: the branches of a case analysis on $I(\vec{u})$ bind proofs of the index equations $\vec{u} = \vec{t}_i(\vec{y}_i)$ (Figure 2), which is what lets dependently typed idioms — impossible branches, results depending on indices — be written with first-order motives;
- in the *emitted theory*: the inversion axiom of a family quantifies over an arbitrary index tuple and disjoins, per constructor, the payload *and the index equations* (Definition 5.10) — the first-order image of the forded declaration;
- in the *semantics*: the interpretation of a family is a set of (index tuple, element) pairs, and the index-reflection lemma (Lemma 6.5) states that membership holds exactly when a constructor decomposition with matching index denotations exists — fording read as a property of one model.

One point deserves emphasis, because it explains why no “forded re-declaration” appears anywhere in the formal development. Read as a CIC declaration, the forded family has its erstwhile indices as *non-uniform parameters*: recursive constructor arguments still occur at other instantiations (vcons at n , conclusion at $\text{S } n$). Parameter-versus-index is a property of constructor *conclusions* only, and elimination confronts only conclusions: since every forded constructor returns the generic instance, case analysis needs no index unification, and each branch simply receives its equations as hypotheses. Constraint discharge is pay-as-you-go, one constructor layer per match. In the untyped first-order output the parameter/index distinction vanishes entirely — there are only function symbols, predicates and equations — which is why fording can remain a per-occurrence, per-constructor lens.

2.3 What the untyped target dissolves

The type-theoretic literature attaches three warning labels to fording and to dependent case analysis generally. All three are analysed precisely in the body of the paper; we preview the outcome.

Heterogeneous equality. Telescopic index equations relate terms whose types agree only after earlier equations are substituted: $v_1 = v_2$ at types $\text{vec } A n_1$ and $\text{vec } A n_2$ cannot even be *stated* homogeneously before $n_1 = n_2$ is substituted. McBride’s heterogeneous $\simeq$ exists to state such equations, and its eliminator is exactly K [McB99, CDP14]. In an untyped target the formation problem does not exist: equality is a single binary predicate on one universe, and telescopic equations are stated pointwise (Definition 5.10). The paper’s source calculus sidesteps the issue differently — its index telescopes are non-dependent (Definition 3.3), so source-level equations are homogeneous — and Section 11 discusses the residual gap.

K, UIP, and the singleton criterion. The without-K analysis of pattern matching [CDP14, CD18] isolates exactly two unification steps that require K: deletion of reflexive equations, and injectivity at indices that are non-linear or not self-unifiable; Gilbert et al. [GCST19] show that eliminating an indexed propositional singleton under proof irrelevance *is* UIP. Every one of these obstructions is a proof-relevance obstruction. The model of Section 6 is proof-irrelevant by construction, hence validates UIP outright, so the *unrestricted* rule set is sound for the translation: deletion, non-linear injectivity and singleton collapse need no side conditions (Section 8.5). What survives of the without-K literature is its *negative* space: discrimination only for fully applied constructor forms (under-applied conflict is refutable under functional extensionality — Counterexample 35 of [CD18]), and no injectivity for type formers (Remark 8.16).

Green slime. Defined functions in index positions make unification-based elaboration stuck [McB14]: $0 \simeq \text{plus } m n$ is neither a clash nor a solution. The standard type-theoretic workarounds (fold the offending index, relational re-expression, derived eliminators) all amount to *keeping the equation as data*. The translation does exactly that, uniformly, for every index: the emitted equation $z \approx \widehat{\text{plus}}(m, n)$ sits in an inversion disjunct or a guard formula next to the defining equations of $\widehat{\text{plus}}$, and a saturation prover performs specialization by unification as ordinary equational reasoning — including closing branches that elaboration-time unification cannot (a refutation of $0 \approx \text{plus}(m, n)$ under $m \approx S(k)$ is one paramodulation step away from the defining equations). Nothing in the soundness proof distinguishes rigid from non-rigid index terms; decidability is neither claimed nor needed (Remark 5.16).

2.4 What the untyped target does not dissolve

Three families of documented accidents mark the soundness obligations that an untyped target *creates*; the formal development treats each as a theorem rather than a guideline.

First, *exhaustiveness*: in a one-universe target, “every x is a constructor form” bounds the cardinality of the whole universe [MP08, §2.3], and the monotonicity analysis of [BBPS16] shows such sentences are exactly the ones whose guards can never be erased. For indexed families the situation is strictly worse: for every family that specialises an index to a constructor form — vectors, finite sets, and their relatives — an *unguarded* inversion axiom is not merely false in the intended model but inconsistent with the discrimination axioms the translation must also emit, its own index equations identifying distinct constructor forms (Proposition 8.7); enumeration-shaped families detonate by cardinality instead (Remark 8.8).

Second, *index erasure in atoms*: a type-indexed notion encoded by an index-blind target notion lets facts established at one index leak to another. The F^* unsoundness #1542 [Ase18] is the documented instance — extensional function equality at domain `nat`, encoded into the SMT solver’s untyped `=`, leaked to domain `int` and proved **False**. The analogue inside this paper’s design space is a guard predicate that forgets its index arguments; Proposition 8.9 shows the resulting translation proves a semantically false goal. The translation of Section 5 therefore keeps the indices as arguments of every guard atom, and its native untyped `≈` is sound for a different reason than F^* ’s was unsound: the *semantics of source equality* in the model is itself untyped conversion-class equality, so no typed reading exists to be violated — at the documented price that the model is intensional and functional extensionality is not validated (Remark 8.19).

Third, *degenerate instances*: axioms asserting existence or structure at empty instances — selector totality, box/unbox roundtrips, rank orderings — have produced inconsistencies in deployed systems (Verus #1366 [The24b]). The translation emits no rank axioms, no selectors and no unboxing roundtrips; at an empty instance the guard atom is simply false in the model and every structural axiom is vacuously satisfied there, while refutations of inhabitation (“`fin(0)` is empty”) become one-step consequences of the guarded inversion axiom and discrimination (Example 5.12).

3 The source calculus CIC_{ix}^-

This section defines the source calculus: a monomorphic, first-order fragment of the Calculus of Inductive Constructions whose inductive datatypes are *indexed families*. As in the companion paper [Ext26], the presentation is *stratified*: set types, propositions, predicate kinds, terms and proofs are separate syntactic categories, so the outcome of CIC’s sort analysis — which subterms are computational, which are logical — is built into the grammar, and the erasure of Section 4 and the translation of Section 5 are definable by structural recursion on raw syntax, with no typing oracle. The fragment of [Ext26] is literally the sub-calculus in which every datatype has zero in-

dices and premise-free constructors with closed argument types; Section 3.7 discusses how the extended fragment is carved out of full CIC.

3.1 Syntax

We fix countably infinite, pairwise disjoint sets of *term variables* (written $x, y, z, w, f, \dots$), of *proof variables* ($p, q, h, \dots$) and of *predicate variables* ($X, \dots$), and disjoint sets of names for *datatypes* (I), for *datatype constructors* (c, d), for *inductive predicates* (J), for their *clause constructors* (k), and for *constants* (e).

Definition 3.1 (Syntax of CIC_{ix}^-). The syntactic categories of *set types* σ, τ , *propositions* φ, ψ , *predicate kinds* K , *predicate expressions* P , *terms* t, a, b, u, v and *proofs* π are defined by the following grammar.

$$\begin{aligned}
\sigma, \tau &::= I(\vec{a}) \mid \Pi x:\sigma. \tau \mid \varphi \rightarrow \tau \mid \{x:\sigma \mid \varphi\} \\
\varphi, \psi &::= \top \mid \perp \mid \varphi \wedge \psi \mid \varphi \vee \psi \mid \varphi \rightarrow \psi \mid \forall x:\sigma. \varphi \mid \exists x:\sigma. \varphi \\
&\quad \mid \forall X:K. \varphi \mid a =_{\sigma} b \mid J(\vec{a}) \mid X(\vec{a}) \\
K &::= (\sigma_1, \dots, \sigma_n) \rightarrow \text{Prop} \quad (n \geq 1) \\
P &::= X \mid \lambda(x_1:\sigma_1, \dots, x_n:\sigma_n). \varphi \\
t, a, b &::= x \mid c \mid t a \mid t \pi \mid \lambda x:\sigma. t \mid \lambda p:\varphi. t \mid c(\vec{a}; \vec{\pi}) \\
&\quad \mid \text{case}(a; \vec{z}.z.\tau; \vec{br}) \mid \text{fix } f:T. t \mid \text{refine}(t, \pi) \mid \text{val}(t) \\
&\quad \mid \text{split}(\pi; p_1 p_2. t) \mid \text{cast}(\pi; z.\tau; t) \mid \text{scase}_J(\pi; \vec{z}.\tau; \vec{p}.t) \mid \text{efq}(\pi; \sigma) \\
\pi &::= p \mid c \mid \lambda p:\varphi. \pi \mid \pi \pi' \mid \lambda x:\sigma. \pi \mid \pi a \mid \Lambda X:K. \pi \mid \pi P \mid \langle \pi_1, \pi_2 \rangle \mid \pi.1 \mid \pi.2 \\
&\quad \mid \iota_1(\pi) \mid \iota_2(\pi) \mid \text{vcase}(\pi; p_1.\pi_1; p_2.\pi_2) \mid \langle a, \pi \rangle_{\exists x:\sigma. \varphi} \mid \text{ecase}(\pi; x p. \pi') \\
&\quad \mid \text{tt} \mid \text{efq}(\pi; \varphi) \mid \text{refl}_{\sigma}(a) \mid \text{rew}(\pi; z.\psi; \pi') \mid \text{discr}(\pi) \mid k(\vec{a}; \vec{\pi}) \\
&\quad \mid \text{ind}_J(\vec{z}.\psi; \vec{\pi}; \pi) \mid \text{ind}_I(\vec{z}.z.\psi; \vec{\Pi}; a) \mid \text{valid}(t)
\end{aligned}$$

where:

- a branch list $\vec{br}$ has the form $\{c_1(\vec{y}_1; \vec{p}_1; \vec{q}_1) \Rightarrow t_1 \mid \dots \mid c_M(\vec{y}_M; \vec{p}_M; \vec{q}_M) \Rightarrow t_M\}$ with one branch for every constructor of the datatype concerned; in the branch for c_i , the term variables $\vec{y}_i$ bind the constructor's informative arguments, the proof variables $\vec{p}_i$ its propositional premises, and the proof variables $\vec{q}_i$ (one per index of the datatype) the *index equations* of the ford;
- in $\text{case}(a; \vec{z}.z.\tau; \vec{br})$ the term variables $\vec{z}$ (one per index) and z bind into the motive τ ; in $\text{scase}_J(\pi; \vec{z}.\tau; \vec{p}.t)$ the variables $\vec{z}$ (one per index of J) bind into τ and the proof variables $\vec{p}$ bind into t ; in ind_I each clause proof Π_i has the form $\vec{y}_i \vec{p}_i \vec{h}_i. \pi_i$, binding the constructor telescope, its premises, and one induction hypothesis per recursive argument;

- T in $\text{fix } f:T. t$ ranges over set types; $\{x:\sigma \mid \varphi\}$ binds x in φ ; the binding structure of the remaining forms is the evident one.

Terms of the form $\text{refine}(t, \pi)$ carry a refinement-type annotation, and $\langle a, \pi \rangle$ carries its $\exists$ -type; we omit the annotations when irrelevant. We write $\neg\varphi$ for $\varphi \rightarrow \perp$ and $\varphi \leftrightarrow \psi$ for $(\varphi \rightarrow \psi) \wedge (\psi \rightarrow \varphi)$. All categories are identified up to α -equivalence; substitution is capture-avoiding.

Notation 3.2. For an m -indexed datatype or predicate we abbreviate index tuples as $\vec{u} = u_1, \dots, u_m$, and write $\vec{u} =_{\vec{\sigma}} \vec{v}$ for the list of propositions $u_1 =_{\sigma_1} v_1, \dots, u_m =_{\sigma_m} v_m$; a proof-variable telescope $\vec{q} : (\vec{u} =_{\vec{\sigma}} \vec{v})$ declares $q_h : u_h =_{\sigma_h} v_h$ for $1 \leq h \leq m$. When a constructor has no premises and the datatype no indices we write $c(\vec{a})$ for $c(\vec{a};)$ and $c(\vec{y}) \Rightarrow t$ for a branch, recovering the notation of [Ext26].

Compared with the companion calculus, the new material is: datatype occurrences $I(\vec{a})$ applied to *index terms*; constructor terms $c(\vec{a}; \vec{\pi})$ carrying proofs of the constructor's propositional premises; the *forded* case former, whose branches bind index-equation proofs $\vec{q}_i$; the singleton elimination $\text{scase}_J(; ;)$ for index-determined propositional singletons (Definition 3.6); the discrimination proof former discr ; and the generalized datatype induction ind_I . The three function spaces, the refinement formers, the propositional singleton eliminations $\text{split} (; ;)$, $\text{cast} (; ;)$, $\text{efq} (;)$ and the proof-level logic are unchanged. Dependency of types on terms now enters through *three* channels: refinement payloads, propositions embedded in types, and index positions of datatype occurrences. There are still no type-level functions, no universes and no large elimination, so set types remain *rigid* (Lemma 3.11).

3.2 Environments

Definition 3.3 (Declarations and environments). A *global environment* $\mathbb{E}$ is a finite list of declarations of the following three kinds.

1. A *datatype declaration*

$$\text{data } I : (\sigma_1, \dots, \sigma_m) \rightarrow \text{Set} := \{ c_1 : K_1; \dots; c_M : K_M \}$$

with $m \geq 0$ *indices* and $M \geq 1$ constructors, where the index types σ_h are closed set types over the preceding environment, and each *constructor scheme* K_i has the form

$$K_i = \forall(y_1:A_{i1}) \cdots (y_{r_i}:A_{ir_i}). \varphi_{i1} \Rightarrow \cdots \Rightarrow \varphi_{il_i} \Rightarrow I(t_{i1}, \dots, t_{im})$$

subject to: each *argument type* A_{ij} is either a *recursive occurrence* $I(\vec{v})$ with $\vec{v}$ terms over the preceding environment and $y_1, \dots, y_{j-1}$ (of the index types, in the sense of Figure 1), or a set type over the preceding environment and $y_1, \dots, y_{j-1}$ in which I does not occur; each *premise*

φ_{il} is a proposition over the preceding environment and $\vec{y}_i$ in which I does not occur; and each *index term* t_{ih} is a term over the preceding environment and $\vec{y}_i$ with $t_{ih} : \sigma_h$. We call $\vec{t}_i = t_{i1}, \dots, t_{im}$ the *index pattern* of c_i and say the argument position j is *recursive* if A_{ij} is a recursive occurrence.

2. An *inductive-predicate declaration* $\text{pred } J : (\vec{\sigma}) \rightarrow \text{Prop}$ with a finite list of clauses

$$k_j : \forall \vec{x}_j : \vec{\tau}_j. \varphi_{j1} \rightarrow \dots \rightarrow \varphi_{j\ell_j} \rightarrow J(\vec{u}_j)$$

where the index types $\vec{\sigma}$ are closed set types over the preceding environment, the $\vec{\tau}_j$ are set types over the preceding environment (possibly mentioning $\vec{x}_j$ bound earlier in the same telescope) in which J does not occur, the indices $\vec{u}_j$ are terms over $\vec{x}_j$ with $u_{jh} : \sigma_h$, and each premise φ_{ji} is either of the form $J(\vec{v})$ (a *recursive premise*, $\vec{v}$ terms over the telescope of the index types) or a proposition over the preceding environment and $\vec{x}_j$ in which J does not occur.

3. A *definition*, either a *term definition* $c := t : \sigma$ with σ a closed set type and t a closed term, or a *lemma* $c := \pi : \varphi$ with φ a closed proposition and π a closed proof.

All names declared in $\mathbb{E}$ are distinct. An environment in which every constant has a body is *definitional*; this paper considers definitional environments (see Remark 7.13 for postulates). A datatype with $m = 0$ and premise-free constructors with closed argument types is exactly a datatype of [Ext26]; we say I is a *family* when $m > 0$.

Example 3.4 (Running declarations). Fix nat with $\text{O} : \text{nat}$ -constructors as in [Ext26] (formally: $\text{data nat} : () \rightarrow \text{Set} := \{\text{O} : \text{nat}; \text{S} : \forall y : \text{nat}. \text{nat}\}$, where we write nat for the 0-index occurrence $\text{nat}()$), bool likewise, and a datatype A (the element type, e.g. nat). The running families are:

```
data vec : (nat) → Set :=
  { vnul : vec(O); vcons : ∀(n:nat)(a:A)(v:vec(n)). vec(S n) }
data fin : (nat) → Set := { fz : ∀n:nat. fin(S n); fs : ∀(n:nat)(k:fin(n)). fin(S n) }
data bnd : (nat) → Set := { B : ∀(n:nat)(k:nat). It(k, n) ⇒ bnd(n) }
data rflφ0 : (bool) → Set :=
  { RT : φ0 ⇒ rflφ0(true); RF : ¬φ0 ⇒ rflφ0(false) }
pred le : (nat, nat) → Prop :=
  { len : ∀n:nat. le(n, n); leS : ∀(nm:nat). le(n, m) → le(n, S m) }
pred eqσ : (σ, σ) → Prop := { refleq : ∀x:σ. eqσ(x, x) }
```

where It is some previously declared inductive predicate and φ_0 a closed proposition. vec and fin are informative recursive families; bnd is a subset-shaped family (one constructor, informative carrier, propositional payload,

index); rfl_{φ_0} is an enumeration-shaped family (constant constructors, propositional payloads, distinct rigid index patterns) — the monomorphic image of ssreflect 's $\text{reflect } P \ b$ at $P = \varphi_0$; le is a predicate family; eq_σ is a user-declared equality — the primitive $=_\sigma$ remains available, and Example 5.13 compares the two.

Definition 3.5 (Contexts). A *context* Γ is a finite list of declarations $x : \sigma$, $p : \varphi$ and $X : K$ with pairwise distinct variables, each type, proposition or kind well formed over the preceding declarations (Figure 1).

3.3 Typing

The judgements are: $\vdash \mathbb{E} \text{ wf}$; and, relative to a fixed well-formed $\mathbb{E}$ left implicit, $\Gamma \vdash \sigma \text{ type}$, $\Gamma \vdash \varphi \text{ prop}$, $\Gamma \vdash P : K$, $\Gamma \vdash t : \sigma$, and $\Gamma \vdash \pi : \varphi$. The rules are given in Figures 1 to 5. Environment well-formedness requires each declaration to be well formed over its prefix: for a datatype declaration, each index type must satisfy $\vdash \sigma_h \text{ type}$, and for each constructor scheme, working in the telescope $y_1 : A_{i1}, \dots$ extended left to right, each argument type must satisfy $\Gamma_j \vdash A_{ij} \text{ type}$ (where for a recursive occurrence $I(\vec{v})$ this means $\Gamma_j \vdash v_h : \sigma_h$ for all h , the formation rule for I -occurrences being available at this point for the declaration being processed), each premise $\Gamma_{r_i} \vdash \varphi_{il} \text{ prop}$, and each index term $\Gamma_{r_i} \vdash t_{ih} : \sigma_h$; predicate declarations are analogous; a definition $c := t : \sigma$ (resp. $c := \pi : \varphi$) requires $\vdash \sigma \text{ type}$ and $\vdash t : \sigma$ (resp. $\vdash \varphi \text{ prop}$ and $\vdash \pi : \varphi$) over the prefix.

Definition 3.6 (Forced singletons). An inductive predicate $J : (\vec{\sigma}) \rightarrow \text{Prop}$ is a *forced singleton* if it has exactly one clause

$$\mathbf{k} : \forall \vec{x} : \vec{\tau}. \varphi_1 \rightarrow \dots \rightarrow \varphi_\ell \rightarrow J(\vec{u}^0)$$

and there is a function $j : \{1, \dots, |\vec{x}|\} \rightarrow \{1, \dots, m\}$, called a *solution*, with $u_{j(i)}^0 = x_i$ (syntactic identity) for every i . Note that j is automatically injective ($u_{j(i)}^0$ determines x_i), that repeated variables among other index positions are permitted (non-linear index patterns), and that the remaining index positions may hold arbitrary terms. For an occurrence $J(\vec{u})$ the induced *solution substitution* is $\theta = [u_{j(1)}/x_1, \dots, u_{j(|\vec{x}|)}/x_{|\vec{x}|}]$. In the $\text{scase}_J(; ;)$ rule of Figure 3, $\theta(\vec{\varphi})$ and $\theta(\vec{u}^0)$ denote the simultaneous application of θ . When the elimination is used we consider j fixed once per declaration (any choice works; Remark 3.8).

Example 3.7. eq_σ is a forced singleton: the clause $\text{refl}_{\text{eq}} : \forall x : \sigma. \text{eq}_\sigma(x, x)$ has $\vec{u}^0 = (x, x)$ and $j(1) = 1$ (or 2). For an occurrence $\text{eq}_\sigma(a, b)$, the solution substitution is $\theta = [a/x]$, so $\text{scase}_{\text{eq}_\sigma}(\pi; z_1 z_2. \tau; .t)$ requires $t : \tau[a/z_1, a/z_2]$ and produces $\tau[a/z_1, b/z_2]$ — precisely transport in the second index. The primitive $\text{cast} (; ;)$ is the same elimination for the primitive equality. The

$$\begin{array}{c}
\frac{I : (\sigma_1, \dots, \sigma_m) \rightarrow \mathbf{Set} \in \mathbb{E} \quad \Gamma \vdash a_h : \sigma_h \ (1 \leq h \leq m)}{\Gamma \vdash I(\vec{a}) \text{ type}} \quad \frac{\Gamma \vdash \sigma \text{ type} \quad \Gamma, x:\sigma \vdash \tau \text{ type}}{\Gamma \vdash \Pi x:\sigma. \tau \text{ type}} \\
\frac{\Gamma \vdash \varphi \text{ prop} \quad \Gamma \vdash \tau \text{ type}}{\Gamma \vdash \varphi \rightarrow \tau \text{ type}} \quad \frac{\Gamma \vdash \sigma \text{ type} \quad \Gamma, x:\sigma \vdash \varphi \text{ prop}}{\Gamma \vdash \{x:\sigma \mid \varphi\} \text{ type}} \\
\frac{\Gamma \vdash \top \text{ prop} \quad \Gamma \vdash \perp \text{ prop}}{\Gamma \vdash \sigma \text{ type} \quad \Gamma, x:\sigma \vdash \varphi \text{ prop}} \quad \frac{\Gamma \vdash \varphi \text{ prop} \quad \Gamma \vdash \psi \text{ prop}}{\Gamma \vdash \varphi \otimes \psi \text{ prop}} \quad \otimes \in \{\wedge, \vee, \rightarrow\} \\
\frac{\Gamma \vdash Qx:\sigma. \varphi \text{ prop}}{\Gamma \vdash a : \sigma \quad \Gamma \vdash b : \sigma} \quad Q \in \{\forall, \exists\} \quad \frac{\Gamma, X:K \vdash \varphi \text{ prop} \quad \Gamma \vdash K \text{ kind}}{\Gamma \vdash \forall X:K. \varphi \text{ prop}} \\
\frac{\Gamma \vdash a =_\sigma b \text{ prop}}{\Gamma \vdash a : \sigma \quad \Gamma \vdash b : \sigma} \quad \frac{J : (\vec{\sigma}) \rightarrow \mathbf{Prop} \in \mathbb{E} \quad \Gamma \vdash \vec{a} : \vec{\sigma}}{\Gamma \vdash J(\vec{a}) \text{ prop}} \\
\frac{X : (\vec{\sigma}) \rightarrow \mathbf{Prop} \in \Gamma \quad \Gamma \vdash \vec{a} : \vec{\sigma}}{\Gamma \vdash X(\vec{a}) \text{ prop}} \quad \frac{\Gamma \vdash \sigma_i \text{ type} \ (1 \leq i \leq n)}{\Gamma \vdash (\sigma_1, \dots, \sigma_n) \rightarrow \mathbf{Prop} \text{ kind}} \quad \frac{X : K \in \Gamma}{\Gamma \vdash X : K} \\
\frac{\Gamma, \vec{x}:\vec{\sigma} \vdash \varphi \text{ prop}}{\Gamma \vdash \lambda(\vec{x}:\vec{\sigma}). \varphi : (\vec{\sigma}) \rightarrow \mathbf{Prop}}
\end{array}$$

Figure 1: Well-formedness of set types, propositions, kinds and predicate expressions. $\Gamma \vdash \vec{a} : \vec{\sigma}$ abbreviates $\Gamma \vdash a_h : \sigma_h$ for all h ; since index types are closed, no substitution occurs in index telescopes.

$$\begin{array}{c}
\frac{x : \sigma \in \Gamma}{\Gamma \vdash x : \sigma} \quad \frac{(c := t : \sigma) \in \mathbb{E}}{\Gamma \vdash c : \sigma} \quad \frac{\Gamma \vdash t : \Pi x:\sigma. \tau \quad \Gamma \vdash a : \sigma}{\Gamma \vdash ta : \tau[a/x]} \\
\frac{\Gamma \vdash t : \varphi \rightarrow \tau \quad \Gamma \vdash \pi : \varphi}{\Gamma \vdash t\pi : \tau} \quad \frac{\Gamma, x:\sigma \vdash t : \tau}{\Gamma \vdash \lambda x:\sigma. t : \Pi x:\sigma. \tau} \quad \frac{\Gamma, p:\varphi \vdash t : \tau}{\Gamma \vdash \lambda p:\varphi. t : \varphi \rightarrow \tau} \\
\frac{\begin{array}{c} \mathbf{c}_i : \forall \vec{y}:\vec{A}. \varphi_1 \Rightarrow \dots \Rightarrow \varphi_\ell \Rightarrow I(\vec{t}) \in I \\ \Gamma \vdash a_j : A_j[\vec{a}_{<j}/\vec{y}_{<j}] \ (1 \leq j \leq r) \quad \Gamma \vdash \pi_l : \varphi_l[\vec{a}/\vec{y}] \ (1 \leq l \leq \ell) \end{array}}{\Gamma \vdash \mathbf{c}_i(\vec{a}; \vec{\pi}) : I(\vec{t}[\vec{a}/\vec{y}])} \quad \frac{\Gamma \vdash t : \{x:\sigma \mid \varphi\}}{\Gamma \vdash \text{val}(t) : \sigma} \\
\frac{\begin{array}{c} \Gamma \vdash a : I(\vec{u}) \quad \Gamma, \vec{z}:\vec{\sigma}, z:I(\vec{z}) \vdash \tau \text{ type} \\ \Gamma, \vec{y}_i:\vec{A}_i, \vec{p}_i:\vec{\varphi}_i, \vec{q}_i:(\vec{u} =_{\vec{\sigma}} \vec{t}_i) \vdash t_i : \tau[\vec{t}_i/\vec{z}, \mathbf{c}_i(\vec{y}_i;\vec{p}_i)/z] \quad (\text{all } i) \end{array}}{\Gamma \vdash \text{case}(a; \vec{z}.z.\tau; \vec{br}) : \tau[\vec{u}/\vec{z}, a/z]} \\
\frac{\Gamma \vdash \{x:\sigma \mid \varphi\} \text{ type} \quad \Gamma \vdash t : \sigma \quad \Gamma \vdash \pi : \varphi[t/x]}{\Gamma \vdash \text{refine}(t, \pi) : \{x:\sigma \mid \varphi\}} \quad \frac{\Gamma \vdash \pi : \perp \quad \Gamma \vdash \sigma \text{ type}}{\Gamma \vdash \text{efq}(\pi; \sigma) : \sigma}
\end{array}$$

Figure 2: Typing of terms, part I. In the constructor rule, argument types are instantiated left to right and premises at the full argument list. In the case rule, $\vec{br}$ abbreviates the branch list $\{\mathbf{c}_i(\vec{y}_i;\vec{p}_i;\vec{q}_i) \Rightarrow t_i\}_i$, and $\vec{u} =_{\vec{\sigma}} \vec{t}_i$ are the index equations of the ford (Notation 3.2); with $m = 0$ and premise-free constructors the rule degenerates to the case rule of [Ext26].

$$\begin{array}{c}
\frac{\Gamma \vdash \pi : \varphi_1 \wedge \varphi_2 \quad \Gamma, p_1 : \varphi_1, p_2 : \varphi_2 \vdash t : \tau}{\Gamma \vdash \text{split}(\pi; p_1 p_2.t) : \tau} \quad \frac{\Gamma \vdash t : \sigma \quad \sigma \simeq \sigma' \quad \Gamma \vdash \sigma' \text{ type}}{\Gamma \vdash t : \sigma'} \\
\frac{\Gamma \vdash \pi : a =_{\sigma} b \quad \Gamma, z : \sigma \vdash \tau \text{ type} \quad \Gamma \vdash t : \tau[a/z]}{\Gamma \vdash \text{cast}(\pi; z.\tau; t) : \tau[b/z]} \\
\begin{array}{l}
J \text{ a forced singleton with clause } k : \forall \vec{x} : \vec{\tau}. \varphi_1 \rightarrow \dots \rightarrow \varphi_{\ell} \rightarrow J(\vec{u}^0) \\
\text{and solution } j \text{ (Definition 3.6); } \quad \theta = [u_{j(i)}/x_i]_i \\
\Gamma \vdash \pi : J(\vec{u}) \quad \Gamma, \vec{z} : \vec{\sigma} \vdash \tau \text{ type} \quad \Gamma, \vec{p} : \theta(\vec{\varphi}) \vdash t : \tau[\theta(\vec{u}^0)/\vec{z}]
\end{array} \\
\hline
\Gamma \vdash \text{scase}_J(\pi; \vec{z}.\tau; \vec{p}.t) : \tau[\vec{u}/\vec{z}] \\
\begin{array}{l}
T = \Pi x_1 : \sigma_1 \dots \Pi x_n : \sigma_n. \tau \quad \sigma_k = I(\vec{v}) \quad \Gamma \vdash T \text{ type} \\
t = \lambda \vec{x} : \vec{\sigma}. \text{case}(x_k; \vec{z}.z.\tau_0; \vec{b}r) \quad \tau = \tau_0[\vec{v}/\vec{z}, x_k/z] \\
\Gamma, f : T, \vec{x} : \vec{\sigma}, \vec{y}_i : \vec{A}_i, \vec{p}_i : \vec{\varphi}_i, \vec{q}_i : (\vec{v} =_{\vec{\sigma}'} \vec{t}_i) \\
\vdash b_i : \tau_0[\vec{t}_i/\vec{z}, c_i(\vec{y}_i; \vec{p}_i)/z] \quad (\text{all } i) \\
\text{grd}_k(f; \vec{y}_i^{\text{rec}}; b_i) \quad (\text{all } i)
\end{array} \\
\hline
\Gamma \vdash \text{fix } f : T. t : T
\end{array}$$

Figure 3: Typing of terms, part II. In the $\text{scase}_J(\cdot; \cdot; \cdot)$ rule, J , its clause data and j are as in Definition 3.6. In the fix rule, $\vec{b}r$ abbreviates the branch list with bodies b_i ; x_k is the designated *guard argument*; $\vec{\sigma}'$ are the index types of I ; $\vec{y}_i^{\text{rec}} \subseteq \vec{y}_i$ denotes the recursive argument positions of c_i ; and the *guard condition* $\text{grd}_k(f; V; b)$ requires every occurrence of f in b to occur in a subterm of the form $f a_1 \dots a_n$ with $a_k \in V$. We further require $f, \vec{x}, \vec{y}_i$ not to be shadowed in b_i .

$$\begin{array}{c}
\frac{p : \varphi \in \Gamma}{\Gamma \vdash p : \varphi} \quad \frac{(c := \pi : \varphi) \in \mathbb{E}}{\Gamma \vdash c : \varphi} \quad \frac{\Gamma, p : \varphi \vdash \pi : \psi}{\Gamma \vdash \lambda p : \varphi. \pi : \varphi \rightarrow \psi} \\
\frac{\Gamma \vdash \pi : \varphi \rightarrow \psi \quad \Gamma \vdash \pi' : \varphi}{\Gamma \vdash \pi \pi' : \psi} \quad \frac{\Gamma, x : \sigma \vdash \pi : \varphi}{\Gamma \vdash \lambda x : \sigma. \pi : \forall x : \sigma. \varphi} \quad \frac{\Gamma \vdash \pi : \forall x : \sigma. \varphi \quad \Gamma \vdash a : \sigma}{\Gamma \vdash \pi a : \varphi[a/x]} \\
\frac{\Gamma, X : K \vdash \pi : \varphi}{\Gamma \vdash \Lambda X : K. \pi : \forall X : K. \varphi} \quad \frac{\Gamma \vdash \pi : \forall X : K. \varphi \quad \Gamma \vdash P : K}{\Gamma \vdash \pi P : \varphi[P/X]} \\
\frac{\Gamma \vdash \pi_1 : \varphi_1 \quad \Gamma \vdash \pi_2 : \varphi_2}{\Gamma \vdash \langle \pi_1, \pi_2 \rangle : \varphi_1 \wedge \varphi_2} \quad \frac{\Gamma \vdash \pi : \varphi_1 \wedge \varphi_2}{\Gamma \vdash \pi.i : \varphi_i} \quad \frac{\Gamma \vdash \pi : \varphi_i}{\Gamma \vdash \iota_i(\pi) : \varphi_1 \vee \varphi_2} \\
\frac{\Gamma \vdash \pi : \varphi_1 \vee \varphi_2 \quad \Gamma, p_i : \varphi_i \vdash \pi_i : \psi \quad (i = 1, 2)}{\Gamma \vdash \text{vcase}(\pi; p_1.\pi_1; p_2.\pi_2) : \psi} \quad \frac{\Gamma \vdash a : \sigma \quad \Gamma \vdash \pi : \varphi[a/x]}{\Gamma \vdash \langle a, \pi \rangle : \exists x : \sigma. \varphi} \\
\frac{\Gamma \vdash \pi : \exists x : \sigma. \varphi \quad \Gamma, x : \sigma, p : \varphi \vdash \pi' : \psi \quad x, p \notin \text{FV}(\psi)}{\Gamma \vdash \text{ecase}(\pi; x p. \pi') : \psi} \quad \Gamma \vdash \text{tt} : \top \\
\frac{\Gamma \vdash \pi : \perp \quad \Gamma \vdash \varphi \text{ prop}}{\Gamma \vdash \text{efq}(\pi; \varphi) : \varphi} \quad \frac{\Gamma \vdash a : \sigma}{\Gamma \vdash \text{refl}_{\sigma}(a) : a =_{\sigma} a} \\
\frac{\Gamma \vdash \pi : a =_{\sigma} b \quad \Gamma, z : \sigma \vdash \psi \text{ prop} \quad \Gamma \vdash \pi' : \psi[a/z]}{\Gamma \vdash \text{rew}(\pi; z. \psi; \pi') : \psi[b/z]}
\end{array}$$

Figure 4: Typing of proofs, part I.

$$\begin{array}{c}
\frac{\Gamma \vdash \pi : c(\vec{a}; \vec{\pi}_1) =_{\sigma} d(\vec{b}; \vec{\pi}_2) \quad c \neq d \text{ constructors of a datatype } I}{\Gamma \vdash \text{discr}(\pi) : \perp} \\
\frac{\Gamma \vdash t : \{x:\sigma \mid \varphi\}}{\Gamma \vdash \text{valid}(t) : \varphi[\text{val}(t)/x]} \quad \frac{\Gamma \vdash \pi : \varphi \quad \varphi \simeq \varphi' \quad \Gamma \vdash \varphi' \text{ prop}}{\Gamma \vdash \pi : \varphi'} \\
\frac{\begin{array}{c} k : \forall \vec{x}:\vec{\tau}. \varphi_1 \rightarrow \dots \rightarrow \varphi_{\ell} \rightarrow J(\vec{u}) \in \mathbb{E} \\ \Gamma \vdash a_i : \tau_i[\vec{a}_{<i}/\vec{x}_{<i}] \text{ (all } i) \quad \Gamma \vdash \pi_l : \varphi_l[\vec{a}/\vec{x}] \text{ (all } l) \end{array}}{\Gamma \vdash k(\vec{a}; \vec{\pi}) : J(\vec{u}[\vec{a}/\vec{x}])} \\
\frac{\Gamma \vdash \pi : J(\vec{a}) \quad \Gamma, \vec{z}:\vec{\sigma} \vdash \psi \text{ prop} \quad \Gamma \vdash \pi_j : \Xi_j(\vec{z}.\psi) \text{ (all clauses } j)}{\Gamma \vdash \text{ind}_J(\vec{z}.\psi; \vec{\pi}; \pi) : \psi[\vec{a}/\vec{z}]} \\
\frac{\begin{array}{c} \Gamma \vdash a : I(\vec{u}) \quad \Gamma, \vec{z}:\vec{\sigma}, z:I(\vec{z}) \vdash \psi \text{ prop} \\ \Gamma, \vec{y}_i:\vec{A}_i, \vec{p}_i:\vec{\varphi}_i, \vec{h}_i:(\psi[\vec{v}_{ij}/\vec{z}, y_{ij}/z])_{j \in \text{rec}(i)} \\ \vdash \pi_i : \psi[\vec{t}_i/\vec{z}, c_i(\vec{y}_i; \vec{p}_i)/z] \text{ (all } i) \end{array}}{\Gamma \vdash \text{ind}_I(\vec{z}.z.\psi; (\vec{y}_i \vec{p}_i \vec{h}_i.\pi_i)_i; a) : \psi[\vec{u}/\vec{z}, a/z]}
\end{array}$$

Figure 5: Typing of proofs, part II. For a predicate clause $k_j : \forall \vec{x}:\vec{\tau}. \varphi_1 \rightarrow \dots \rightarrow \varphi_{\ell} \rightarrow J(\vec{u})$, the *clause obligation* is $\Xi_j(\vec{z}.\psi) = \forall \vec{x}:\vec{\tau}. \varphi_1 \rightarrow \dots \rightarrow \varphi_{\ell} \rightarrow \psi[\vec{v}_1/\vec{z}] \rightarrow \dots \rightarrow \psi[\vec{v}_s/\vec{z}] \rightarrow \psi[\vec{u}/\vec{z}]$, where $J(\vec{v}_1), \dots, J(\vec{v}_s)$ are the recursive premises among the φ_i . In the datatype induction ind_I , $\text{rec}(i)$ is the set of recursive argument positions of c_i , and for $j \in \text{rec}(i)$ with $A_{ij} = I(\vec{v}_{ij})$ the hypothesis h_{ij} has the displayed type; the clause proofs bind $\vec{y}_i, \vec{p}_i, \vec{h}_i$ because the conclusion instance mentions the constructor term $c_i(\vec{y}_i; \vec{p}_i)$, which requires names for the premise proofs.

predicate `le` is not a forced singleton (two clauses); `Acc`-style predicates are excluded from CIC_{ix}^- altogether because their recursive premises sit under quantifiers (Section 3.7).

Remark 3.8 (Choice of solution). If j and j' are both solutions, the two instantiations of the `sca` rule are interderivable: from $\pi : J(\vec{u})$ one obtains, by ind_J with motive $\vec{z}. (z_{j(i)} =_{\sigma_{j(i)}} z_{j'(i)})$ — whose clause obligation is $u_{j(i)}^0 = u_{j'(i)}^0$, i.e. $x_i = x_i$, discharged by `refl` — a proof of $u_{j(i)} = u_{j'(i)}$ for every i , and `cast` along these equations mediates between the two premise and motive instantiations. (The equation is well formed because both index types are convertible to the type of x_i .) Semantically both instantiations are validated by the same argument (Lemma 7.6), so fixing j loses no generality.

Remark 3.9 (Relation to CIC’s singleton criterion). CIC permits a match on a `Prop`-sorted scrutinee in an informative position when the inductive is empty or a *singleton*: one constructor whose arguments are all propositional, *parameters put aside* [The26, Let03]. Under the monomorphic stratification of Section 3.7, the parameters of an equality-like family become clause variables that occur verbatim among the result indices — forced variables in the sense of Brady et al. [BMM04] and Gilbert et al. [GCST19] — and the remaining CIC constructor arguments become the premises $\vec{\varphi}$. Every CIC singleton elimination is therefore an instance of `sca`. The converse fails: Definition 3.6 does not require the forced variables to occur *uniformly* in recursive premises, so it accepts clauses that CIC’s parameter mechanism cannot express (e.g. $k : \forall x:\text{nat}. J(\text{S } x, \text{O}) \rightarrow J(x, \text{S O})$). The permissiveness is deliberate and harmless: soundness is proved against the model (Lemma 7.6), which validates the general rule; only proof *reconstruction* of uses of the extra generality requires inversion-style reasoning rather than a primitive CIC match (Section 11.6). Empty elimination needs no new former: for a predicate J declared with zero clauses, ind_J with motive $\perp$ and no clause obligations yields $\perp$ from $J(\vec{a})$, and `efq` concludes.

3.4 Reduction and conversion

Definition 3.10 (Reduction). The relation $\rightarrow$ on terms is the contextual closure — in all term, proof, type, proposition and predicate positions — of the following root rules:

$$\begin{array}{ll}
(\lambda x:\sigma.t) a \rightarrow t[a/x] & (\lambda p:\varphi.t) \pi \rightarrow t[\pi/p] \\
\text{fix } f:T. t \rightarrow t[\text{fix } f:T. t/f] & \text{val}(\text{refine}(t, \pi)) \rightarrow t \\
c \rightarrow t \quad \text{if } (c := t : \sigma) \in \mathbb{E} & \text{cast}(\text{refl}_\sigma(a); z.\tau; t) \rightarrow t \\
\text{split}(\langle \pi_1, \pi_2 \rangle; p_1 p_2. t) \rightarrow t[\pi_1/p_1][\pi_2/p_2] & \text{sca}_J(k(\vec{a}; \vec{\pi}); \vec{z}.\tau; \vec{p}.t) \rightarrow t[\vec{\pi}/\vec{p}]
\end{array}$$

together with the rule for case analysis meeting a constructor,

$$\text{case}(c_i(\vec{a}; \vec{\pi}); \vec{z}.z.\tau; \{\dots\}) \rightarrow t_i[\vec{a}/\vec{y}_i, \vec{\pi}/\vec{p}_i, \overrightarrow{\text{refl}}(\vec{t}_i[\vec{a}/\vec{y}_i])/\vec{q}_i],$$

where $\overrightarrow{\text{refl}}(\vec{t}_i[\vec{a}/\vec{y}_i])$ abbreviates the proof list

$$\text{refl}_{\sigma_1}(t_{i1}[\vec{a}/\vec{y}_i]), \dots, \text{refl}_{\sigma_m}(t_{im}[\vec{a}/\vec{y}_i])$$

substituted for the index-equation variables. Source *conversion* $\simeq$ is the least congruence (on each syntactic category, closing under all formers) containing $\rightarrow$.

Reduction is untyped and unrestricted. As in the companion paper, no metatheoretic property of $\rightarrow$ itself is needed anywhere: $\simeq$ enters only through the conversion typing rules, and the semantics of Section 6 is invariant under it (Lemma 6.8). Note that the index-equation variables $\vec{q}_i$ receive *reflexivity* proofs when a case meets a constructor: at the redex, the occurrence indices are convertible to the constructor's index pattern, and the conversion typing rule absorbs the mismatch. Since proofs are erased (Definition 4.5), this substitution is invisible in the target.

Lemma 3.11 (Rigidity of type formers). *If $\sigma \simeq \sigma'$ then σ and σ' have the same outermost former: both are occurrences $I(\vec{u})$, $I(\vec{u}')$ of the same datatype I with $u_h \simeq u'_h$ for all h , or both are $\Pi x:\rho.\tau$ -types whose domains and codomains are respectively convertible, or both are $\varphi \rightarrow \tau$ -types with convertible components, or both are refinement types with convertible components. The analogous statement holds for propositions.*

Proof. No root rule of Definition 3.10 applies at the root of a set type or a proposition — the root rules rewrite terms and proofs only — so every $\rightarrow$ -step inside a type or proposition happens strictly below its outermost former: inside an index term of a datatype occurrence, inside a component type or proposition, or inside an embedded term or proof. The claim follows by induction on the length of the conversion path, observing that a single step preserves the outermost former and relates components (index terms included) pointwise. $\square$

Lemma 3.12 (Substitution). *All judgements admit substitution: if $\Gamma, x:\sigma, \Gamma' \vdash \mathcal{J}$ and $\Gamma \vdash a : \sigma$, then $\Gamma, \Gamma'[a/x] \vdash \mathcal{J}[a/x]$, and analogously for proof variables and predicate variables.*

Proof. Routine simultaneous induction on the second derivation, exactly as in [Ext26]: $\simeq$ is closed under substitution (each root rule is, including the new case and **scase** rules, whose instantiated proofs and branches substitute componentwise), and substitution into predicate atoms $X(\vec{a})[P/X] = P \cdot \vec{a}$

is given by predicate application, which terminates because predicate expressions cannot mention the variable being substituted after α -renaming. The new rules (constructor, case, **scase**, **discr**, **ind_I**) are homomorphic in all their components; for **discr** note that substitution preserves the constructor-headedness of both equation sides, and for **scase**(; ;) that θ commutes with the ambient substitution because the clause data $(\vec{u}^0, j)$ are closed declaration components while $\vec{u}$ substitutes pointwise. $\square$

3.5 The shallow fragment

Definition 3.13 (Shallow fragment $\mathfrak{S}$). A proposition is *shallow* if it contains no subformula $\forall X:K.\varphi$ and no atom $X(\vec{a})$ with X a predicate variable; a set type is shallow if all propositions occurring in it (refinement payloads and premises) are shallow. $\mathfrak{S}$ is the class of shallow propositions and set types.

Index terms of datatype occurrences are terms and are always shallow material; only genuine predicate quantification is excluded. As in the companion paper, the translation is defined on $\mathfrak{S}$; proofs inside the environment may use the full calculus, but a lemma whose statement leaves $\mathfrak{S}$ contributes no premise axiom, a definition whose type leaves $\mathfrak{S}$ contributes no specification axiom, and goals must be shallow and closed. We require the declarations of $\mathbb{E}$ (datatype index types, constructor argument types, premises and index terms; inductive-predicate components) to lie in $\mathfrak{S}$.

3.6 Programming with forced case analysis

The index equations $\vec{q}_i$ in branches are what makes dependently typed idioms expressible with *first-order* motives. We record the two standard examples and the derivability of injectivity; both are used in Section 5 as running translation examples.

Lemma 3.14 (Derivable injectivity). *Let c be a constructor of a datatype I whose j -th argument type A_j is closed — it mentions no earlier telescope variables; it may be a non-recursive type or a recursive occurrence $I(\vec{v})$ at closed indices (for a non-indexed datatype, in particular, any recursive position qualifies). Then from $\Gamma \vdash \pi : c(\vec{a}; \vec{\pi}_1) =_\sigma c(\vec{b}; \vec{\pi}_2)$ one can derive $\Gamma \vdash \text{inj}_j(\pi) : a_j =_{A_j} b_j$ for a proof term $\text{inj}_j(\pi)$ definable uniformly from the derivation.*

Proof. Since $\Gamma \vdash c(\vec{a}; \vec{\pi}_1) : \sigma$ and the constructor rule produces the type $I(\vec{t}[\vec{a}/\vec{y}])$, by Lemma 3.11 $\sigma \simeq I(\vec{u})$ for some $\vec{u}$; fix such an occurrence. Define the projection

$$F := \lambda w:I(\vec{u}). \text{case}(w; \vec{z}.z.A_j; \{c(\vec{y}; \vec{p}; \vec{q}) \Rightarrow y_j \mid d(\dots) \Rightarrow a_j \ (d \neq c)\}).$$

The motive is the constant type A_j , well formed in the motive context since A_j is closed (this includes the recursive case $A_j = I(\vec{v})$ with $\vec{v}$ closed: a legal occurrence like any other); the c -branch is well typed because $y_j : A_j$ (again by closedness, no instantiation of earlier telescope variables occurs) and the branch obligation is A_j itself; the default branches use a_j , well typed over Γ by weakening (its type $A_j[\vec{a}_{<j}/\vec{y}_{<j}] = A_j$ by closedness). Then F is well typed, $F\ c(\vec{a}; \vec{\pi}_1) \simeq a_j$ and $F\ c(\vec{b}; \vec{\pi}_2) \simeq b_j$ by a β - and a case-step. Now

$$\text{inj}_j(\pi) := \text{rew}(\pi; z. (a_j =_{A_j} F\ z); \text{refl}_{A_j}(a_j)),$$

where the base proof is well typed at $(a_j =_{A_j} F\ c(\vec{a}; \vec{\pi}_1)) \simeq (a_j =_{A_j} a_j)$ by the conversion rule, and the conclusion converts to $a_j =_{A_j} b_j$. $\square$

Example 3.15 (Head and tail). With the declarations of Example 3.4:

```

vhead :=  $\lambda n:\text{nat}. \lambda v:\text{vec}(\mathbb{S}\ n).$ 
  case( $v; \vec{z}.z.A; \{ \text{vnil}(\;; q) \Rightarrow \text{efq}(\text{discr}(q); A) \mid \text{vcons}(k, a, w; \;; q) \Rightarrow a \}$ )
vtail :=  $\lambda n:\text{nat}. \lambda v:\text{vec}(\mathbb{S}\ n).$ 
  case( $v; \vec{z}.z.\text{vec}(n); \{ \text{vnil}(\;; q) \Rightarrow \text{efq}(\text{discr}(q); \text{vec}(n)) \mid \quad \quad \quad \}$ 
     $\text{vcons}(k, a, w; \;; q) \Rightarrow$ 
       $\text{cast}(\text{sym}(\text{inj}_1(q)); z.\text{vec}(z); w) \}$ 

```

where $\text{sym}(\pi) := \text{rew}(\pi; z. (z =_{\text{nat}} n); \text{refl}_{\text{nat}}(n))$ derives $k =_{\text{nat}} n$ from $\text{inj}_1(q) : n =_{\text{nat}} k$, and the vcons branch of vtail is, written in full, $\text{cast}(\text{sym}(\text{inj}_1(q)); z.\text{vec}(z); w)$. In the vnil branches, $q : \mathbb{S}\ n =_{\text{nat}} \mathbb{O}$ and $\text{discr}(q) : \perp$ discharges the impossible branch — with a *constant* motive; no large elimination, no unit-valued return-type trick. In the vcons branch of vtail , $q : \mathbb{S}\ n =_{\text{nat}} \mathbb{S}\ k$ gives $\text{inj}_1(q) : n =_{\text{nat}} k$ by Lemma 3.14, whence $\text{sym}(\text{inj}_1(q)) : k =_{\text{nat}} n$, and the cast transports $w : \text{vec}(k)$ to $\text{vec}(n)$. Both definitions are well typed in CIC_{ix}^- ; their translations are computed in Example 5.6.

3.7 Relation to CIC

CIC_{ix}^- is designed as the image of a first-order, monomorphic slice of CIC under analyses that are part of the CoqHammer translation pipeline or of program extraction. The discussion of stratification, monomorphisation, refinement classification, structural fixpoints and the propositional connectives in [Ext26, §2.6] applies verbatim; we record only what is new for indexed families.

- *Parameters and indices.* CIC distinguishes uniform parameters from indices; the distinction matters for elimination (a parameter is fixed across a match, an index is generalized). After monomorphisation, *type*

parameters are ground and disappear into the datatype name (`vec` above is really `vecA`). *Term* and *proposition* parameters become indices (`eqσ`'s left argument) or premises (`rflφ₀`'s *P*), and the forced-variable analysis of Definition 3.6 recovers, per occurrence, exactly the specialization that CIC's parameter mechanism provided — this is the content of Remark 3.9. Non-uniform parameters are indices here, as they are in CIC's kernel.

- *Dependent case analysis.* CIC's `match a in I ↦ return τ with ...` has branches typed at each constructor's own indices, with no equational premises; impossible branches are discharged by large-eliminating motives. The forced case of CIC_{ix}^- instead hands each branch the index equations. The two styles are intertranslatable in CIC extended with UIP — this is precisely the Goguen–McBride–McKinna elaboration [GMM06] — and the front end is expected to produce forced branches by the same analysis Coq's `inversion` performs [CT96, Mon10]: generalize the indices, introduce equations, and (optionally) simplify them. Branches the front end can discharge by rigid-index conflict arrive as `efq(discr(·); ·)` as in Example 3.15; branches it cannot discharge (green slime) arrive with their equations intact, which is the point (Remark 5.16). Large elimination remains excluded from CIC_{ix}^- : fording removes its main *programming* use (impossible branches), and its *logical* uses (proving discrimination) are covered by the primitive `discr`, which is CIC-derivable by exactly such a large elimination.
- *Propositional index positions.* Index types in CIC_{ix}^- are set types; CIC families indexed by *propositions* or *proofs* have no direct image. The front end drops proof-sorted index positions before entering the fragment; this is sound for the same reason the model is proof-irrelevant (all proofs of a proposition denote one element — Section 6), and is the translation-level counterpart of the rule that erased data may appear in indices only where nothing relevant observes them [Atk18, GCST19]. Proof *subterms* of index terms, by contrast, are in the fragment and are erased structurally (Remark 4.6).
- *Recursion.* The `fix` former captures structural recursion on a family element with a designated guard argument. Recursive calls may be at arbitrary index instantiations (the call `f k w` inside `vmap`-style definitions instantiates the index argument with the pattern variable *k*); the guard condition constrains only the guard position. Well-founded (Acc-based) recursion remains excluded, for the reasons analysed in [Ext26, §10.3]: its unfolding equations are sound only with their propositional premises retained, and its guardedness is semantic rather than syntactic [GCST19]. Relatedly, predicate clauses with recursive premises under

quantifiers (Acc’s constructor) are excluded from the declaration format.

4 The intermediate language and program extraction

The first half of the translation is Letouzey-style program extraction: an erasure of CIC_{ix}^- terms into an untyped functional calculus $\lambda_{\square}$. The calculus is *the same* as in the companion paper — indexed families change constructor *arities* (a constructor keeps only its informative arguments) but not the shape of the target language — and the erasure function extends that of [Ext26] with clauses for the new formers. Two design points carry over: since the target of the overall translation is an equational theory rather than a programming language with an evaluation order, erased arguments are *dropped* rather than replaced by dummies; and the singleton eliminations collapse to their branches structurally. A third point is new: erasure is *index-blind*. Neither the motive indices of a case, nor the index equations of its branches, nor the index terms of types leave any trace; index terms occurring as *constructor arguments* (the length argument of vcons) are informative and are kept, exactly as in Letouzey’s extraction [Let03, Let08], where redundant stored indices are likewise kept. The re-materialisation of index information is the job of specification extraction (Section 5.3), not of the program.

4.1 The calculus $\lambda_{\square}$

Definition 4.1 (Syntax of $\lambda_{\square}$). Fix the datatype constructors c of a CIC_{ix}^- environment $\mathbb{E}$, each with arity r equal to its number of *informative* arguments (premises do not count), the term-definition constants c of $\mathbb{E}$, and a distinguished constant $\square$ (“dead code”). The terms of $\lambda_{\square}$ are

$$e ::= x \mid c \mid \square \mid \lambda x. e \mid e e' \mid c(e_1, \dots, e_r) \mid \text{case}(e; \{c_i(\vec{x}_i) \Rightarrow e_i\}_i) \mid \text{fix } f. e$$

where constructors are fully applied and a case expression carries one branch for each constructor of the datatype of its constructors.

Definition 4.2 (Reduction of $\lambda_{\square}$). The relation $\rightarrow_{\square}$ is the contextual closure of:

$$\begin{aligned} (\lambda x. e) e' &\rightarrow_{\square} e[e'/x] & (\beta) \\ \text{case}(c_i(\vec{e}); \{c_j(\vec{x}_j) \Rightarrow e_j\}_j) &\rightarrow_{\square} e_i[\vec{e}/\vec{x}_i] & (\iota) \\ \text{fix } f. e &\rightarrow_{\square} e[\text{fix } f. e/f] & (\mu) \\ c &\rightarrow_{\square} \mathcal{E}(t) \quad \text{if } (c := t : \sigma) \in \mathbb{E} & (\delta) \end{aligned}$$

where $\mathcal{E}$ is the erasure of Definition 4.5 (the δ -rule is well defined because the body of c mentions only constants declared before c). We write $=_{\square}$ for the least congruence containing $\rightarrow_{\square}$.

The four rule families have pairwise distinct, non-overlapping, left-linear redex shapes with rigid heads; $\lambda_\square$ is an orthogonal higher-order rewrite system, and its confluence is independent of the constructor arities:

Theorem 4.3 (Confluence of $\lambda_\square$). $\rightarrow_\square$ is confluent. Consequently, $e_1 =_\square e_2$ if and only if e_1 and e_2 have a common $\rightarrow_\square^*$ -reduct.

Proof. By the Tait–Martin–Löf parallel-reduction method in Takahashi’s formulation [Tak95]; the full proof is in Appendix A. Alternatively, $\lambda_\square$ is an orthogonal combinatory reduction system, so confluence follows from Klop’s theorem [Klo80]. $\square$

Corollary 4.4 (Constructor discrimination and injectivity). Let c, d be constructors.

1. If $c(\vec{e}) =_\square d(\vec{e}')$ then $c = d$ and $e_i =_\square e'_i$ for all i .
2. An application $e_1 e_2$ in which e_1 is normal and not a λ -abstraction (for instance $e_1 = c(\vec{e})$ with $\vec{e}$ normal) is not convertible to any constructor form $d(\vec{e}')$.

Proof. (1) Constructor-headed terms only reduce internally: every reduct of $c(\vec{e})$ has the form $c(\vec{e}'')$ with $e_i \rightarrow_\square^* e''_i$. By Theorem 4.3, the two sides have a common reduct, which is then simultaneously c -headed and d -headed, forcing $c = d$; its arguments are common reducts of the respective e_i, e'_i . (2) Such an application is $\rightarrow_\square$ -normal apart from reductions inside e_1 and e_2 , which preserve the application shape, so a common reduct with a constructor form is impossible as in (1). $\square$

4.2 Erasure

Definition 4.5 (Erasure). The function $\mathcal{E}$ from CIC_{ix}^- terms to $\lambda_\square$ terms is defined by structural recursion:

$$\begin{aligned}
\mathcal{E}(x) &= x & \mathcal{E}(c) &= c \\
\mathcal{E}(t a) &= \mathcal{E}(t) \mathcal{E}(a) & \mathcal{E}(t \pi) &= \mathcal{E}(t) \\
\mathcal{E}(\lambda x:\sigma. t) &= \lambda x. \mathcal{E}(t) & \mathcal{E}(\lambda p:\varphi. t) &= \mathcal{E}(t) \\
\mathcal{E}(c(\vec{a}; \vec{\pi})) &= c(\mathcal{E}(\vec{a})) & \mathcal{E}(\text{fix } f:T. t) &= \text{fix } f. \mathcal{E}(t) \\
\mathcal{E}(\text{case}(a; \vec{z}.z.\tau; \{c_i(\vec{y}_i; \vec{p}_i; \vec{q}_i) \Rightarrow t_i\}_i)) &= \text{case}(\mathcal{E}(a); \{c_i(\vec{y}_i) \Rightarrow \mathcal{E}(t_i)\}_i) \\
\mathcal{E}(\text{refine}(t, \pi)) &= \mathcal{E}(t) & \mathcal{E}(\text{val}(t)) &= \mathcal{E}(t) \\
\mathcal{E}(\text{split}(\pi; p_1 p_2. t)) &= \mathcal{E}(t) & \mathcal{E}(\text{cast}(\pi; z.\tau; t)) &= \mathcal{E}(t) \\
\mathcal{E}(\text{scase}_J(\pi; \vec{z}.\tau; \vec{p}.t)) &= \mathcal{E}(t) & \mathcal{E}(\text{efq}(\pi; \sigma)) &= \square
\end{aligned}$$

The clauses for the new formers say: constructor terms keep their informative arguments and lose their premise proofs; a case on a family keeps its

scrutinee and branch bodies and loses the motive, the premise binders and the index-equation binders; the singleton elimination $\mathbf{scase}(\cdot; \cdot)$ collapses to its branch like $\mathbf{split}(\cdot)$ and $\mathbf{cast}(\cdot; \cdot)$. Proofs are never inspected — $\mathcal{E}$ does not recurse into the category of proofs at all — which is the syntactic form of proof irrelevance.

Remark 4.6 (Proofs inside index terms). An index term may contain proof subterms — $\mathbf{vec}(f \ x \ \pi)$ for $f : \Pi x:\mathbf{nat}. \varphi \rightarrow \mathbf{nat}$ — and these vanish under erasure: $\mathcal{E}(f \ x \ \pi) = f \ x$. Consequently the translated index (Definition 5.7) and its interpretation (Definition 6.2) are independent of the proof, for *arbitrary* f , not only parametric ones. This is the translation-level form of interpreting all erased data by one canonical element, with three precedents: Letouzey’s and MetaCoq’s $\square$ (operational) [Let03, SBF⁺20], Atkey’s $!_0x = !_0y$ canonical erased realiser (algebraic) [Atk18], and the canonical proof element of set-theoretic proof-irrelevant models (denotational) [LW11]. Lemma 6.7(2) is the precise statement in our model.

Lemma 4.7 (Erasure commutes with substitution). $\mathcal{E}(t[a/x]) = \mathcal{E}(t)[\mathcal{E}(a)/x]$; $\mathcal{E}(t[\pi/p]) = \mathcal{E}(t)$; $\mathcal{E}(t[P/X]) = \mathcal{E}(t)$.

Proof. Induction on t . For the first equation the variable case is immediate, and all other cases follow from the induction hypothesis because $\mathcal{E}$ is homomorphic in exactly the informative positions and substitution acts pointwise on them; positions discarded by $\mathcal{E}$ (proofs, motives, annotations, premise and equation binders) absorb the substitution without residue. For the second equation, observe that p can occur in t only inside proof positions or inside types, propositions and annotations, all of which $\mathcal{E}$ discards; similarly for the third. $\square$

Lemma 4.8 (Erasure simulates reduction). *If $t \rightarrow u$ then $\mathcal{E}(t) \rightarrow_{\square} \mathcal{E}(u)$ or $\mathcal{E}(t) = \mathcal{E}(u)$. Consequently $t \simeq u$ implies $\mathcal{E}(t) =_{\square} \mathcal{E}(u)$.*

Proof. By induction on the derivation of $t \rightarrow u$. For root steps:

- $(\lambda x:\sigma.t) a \rightarrow t[a/x]$: the erasure β -reduces to $\mathcal{E}(t[a/x])$ by Lemma 4.7.
- $(\lambda p:\varphi.t) \pi \rightarrow t[\pi/p]$: both sides erase to $\mathcal{E}(t)$; zero steps.
- Case on a constructor:

$$\begin{aligned} \mathcal{E}(\mathbf{case}(\mathbf{c}_i(\vec{a}; \vec{\pi}); \vec{z}.z.\tau; \dots)) &= \mathbf{case}(\mathbf{c}_i(\mathcal{E}(\vec{a})); \{\mathbf{c}_j(\vec{y}_j) \Rightarrow \mathcal{E}(t_j)\}_j) \\ &\rightarrow_{\square} \mathcal{E}(t_i)[\mathcal{E}(\vec{a})/\vec{y}_i], \end{aligned}$$

one ι -step, and by Lemma 4.7 the target equals $\mathcal{E}(t_i[\vec{a}/\vec{y}_i, \vec{\pi}/\vec{p}_i, \overrightarrow{\mathbf{refl}}/\vec{q}_i])$: the informative substitution is simulated pointwise and the proof substitutions (for $\vec{p}_i$ and $\vec{q}_i$) are invisible.

- μ (fixpoint unfolding) and δ : erase to the corresponding $\lambda_\square$ steps via Lemma 4.7.
- The singleton rules for $\text{val}()/\text{refine}()$, $\text{cast}();$, $\text{split}();$ and $\text{scase}_J();$: both sides erase to $\mathcal{E}(t)$ (using Lemma 4.7 for the proof substitutions); zero steps.

For congruence steps: a step inside an informative position maps to a step (or equality) in the corresponding position of the erasure by the induction hypothesis; a step inside a proof, type, proposition, motive, index or annotation position leaves the erasure unchanged. $\square$

Remark 4.9. Lemma 4.8 is the equational shadow of extraction correctness [Let04, §2.3]; the semantic half — the extracted term *realizes* its type — becomes the fundamental lemma of Section 7. Note where the indexed extension is invisible: the $\lambda_\square$ image of vhead is $\lambda n v. \text{case}(v; \{\text{vnil} \Rightarrow \square \mid \text{vcons}(k, a, w) \Rightarrow a\})$ — the impossible branch has become dead code, the possible branch a projection — and nothing in $\lambda_\square$ records that v was drawn from an indexed family. Soundness of the translation cannot, therefore, be an invariant of the program half alone; it is recovered in Section 6 by interpreting the *types*.

5 The translation

This section defines the emitted first-order theory. Throughout, fix a well-formed definitional environment $\mathbb{E}$ whose declarations lie in $\mathfrak{S}$ and a closed shallow goal proposition G . The target logic is classical first-order logic with equality $\approx$; $\vdash_{\text{FOL}}$ is provability in any standard sound and complete calculus [Sho67]; the propositional constants $\top, \perp$ are assumed available.

5.1 Signature

Definition 5.1 (Signature $\Sigma_{\mathbb{E}}$). The signature $\Sigma_{\mathbb{E}}$ contains:

- a $(1+m)$ -ary predicate symbol $\hat{I}$ (the *guard predicate*) for each datatype $I : (\sigma_1, \dots, \sigma_m) \rightarrow \mathbf{Set}$ of $\mathbb{E}$; the first argument place holds the guarded term, the remaining m the indices;
- a k -ary predicate symbol $\hat{J}$ for each inductive predicate $J : (\sigma_1, \dots, \sigma_k) \rightarrow \mathbf{Prop}$ of $\mathbb{E}$;
- an r -ary function symbol $\hat{c}$ for each constructor c of a datatype of $\mathbb{E}$, where r is the number of informative arguments of c (its $\lambda_\square$ arity);
- a function symbol $\hat{c}$ of arity a_c , and if $a_c > 0$ additionally a constant $\hat{c}^0$ (the *pointer* of c), for each term definition $c := t : \tau$ of $\mathbb{E}$, where the *erasure arity* a_c is determined from $\mathcal{E}(t)$ in Definition 5.3;

- a binary function symbol $@$ (infix, associating left), and a constant $\odot$ (the image of dead code $\square$);
- countably many *auxiliary* function symbols, produced by the compilation procedure below. As in [Cza18, §4] we assume injective naming functions $\Lambda(\cdot)$, $X(\cdot)$, $\Phi(\cdot)$ from $\lambda_\square$ terms (up to renaming of free variables) to auxiliary symbols (hash-consing).

Source term variables double as first-order variables.

Two negative observations, both load-bearing. First, there is no binary inhabitation predicate and no encoding of *types* as first-order terms: the indices of a family are *term* arguments of its guard predicate, so the type $\text{vec}(Sn)$ contributes the atom $\widehat{\text{vec}}(w, \widehat{S}(n))$ — native first-order structure — rather than a term $\text{vec}(S(n))$ inside a `HasType` atom. Second, because types are not terms, no formula of the signature can even *state* injectivity of the type former I itself; Remark 8.16 explains why this inexpressibility is a feature.

5.2 Compilation of erased programs

The compilation of $\lambda_\square$ programs into defining equations is that of the companion paper, reproduced here for self-containedness; indexed families require no change to it whatsoever, because $\lambda_\square$ is unchanged. It is a recursive procedure that *emits* first-order axioms and, for every symbol f it introduces, records a *witness*: a $\lambda_\square$ term e_f with free variables among the closure parameters of f , used only by the model construction of Section 6.

Compilation carries a finite map θ assigning to certain $\lambda_\square$ variables (fix-bound recursion variables) a *template* $(f, \vec{w}, m)$: an $(|\vec{w}| + m)$ -ary symbol f , a list of first-order terms $\vec{w}$, and an argument count m .

Definition 5.2 (Term translation $\mathcal{C}_\theta$). For a $\lambda_\square$ term e whose free variables are first-order variables or θ -bound, the first-order term $\mathcal{C}_\theta(e)$ is defined by recursion on e :

- $\mathcal{C}_\theta(x) = x$ for an ordinary variable; $\mathcal{C}_\theta(\square) = \odot$; $\mathcal{C}_\theta(c(\vec{e})) = \widehat{c}(\mathcal{C}_\theta(\vec{e}))$.
- For an application spine $e = h e_1 \cdots e_j$ ($j \geq 0$, h not an application, e itself not of the previous forms):
 - if $h = g$ with $\theta(g) = (f, \vec{w}, m)$: by the guard condition $j \geq m$, and $\mathcal{C}_\theta(e) = f(\vec{w}, \mathcal{C}_\theta(e_1), \dots, \mathcal{C}_\theta(e_m)) @ \mathcal{C}_\theta(e_{m+1}) @ \cdots @ \mathcal{C}_\theta(e_j)$;
 - if $h = c$ is a term-definition constant: if $j \geq a_c$ then $\mathcal{C}_\theta(e) = \widehat{c}(\mathcal{C}_\theta(e_1), \dots, \mathcal{C}_\theta(e_{a_c})) @ \cdots @ \mathcal{C}_\theta(e_j)$, and otherwise $\mathcal{C}_\theta(e) = \widehat{c}^0 @ \mathcal{C}_\theta(e_1) @ \cdots @ \mathcal{C}_\theta(e_j)$;
 - if $h = x$ is a variable: $\mathcal{C}_\theta(e) = x @ \mathcal{C}_\theta(e_1) @ \cdots @ \mathcal{C}_\theta(e_j)$;

- if h is a λ -abstraction, case or fix (a *complex head*): let u be its *value translation* defined below; then $\mathcal{C}_\theta(e) = u @ \mathcal{C}_\theta(e_1) @ \dots @ \mathcal{C}_\theta(e_j)$.
- Value translations of complex heads h with $\vec{w} = \text{FV}(h)$ (ordered):
 - $h = \lambda x. e_0$: let $f = \Lambda(h)$, an $|\vec{w}|$ -ary symbol with witness $e_f = h$; run $B_\theta(f(\vec{w}); h)$; the value is $f(\vec{w})$.
 - $h = \text{case}(s; \vec{b}r)$: let $g = X(\lambda z. \text{case}(z; \vec{b}r))$ for fresh z , of arity $1 + |\vec{w}'|$ with $\vec{w}' = \text{FV}(\vec{b}r)$, with witness $e_g = \lambda z. \text{case}(z; \vec{b}r)$; run $B_\theta(g(z, \vec{w}'); \text{case}(z; \vec{b}r))$; the value is $g(\mathcal{C}_\theta(s), \vec{w}')$.
 - $h = \text{fix } g_0. \lambda z_1 \dots \lambda z_m. e_0$ (maximal prefix): let $f = \Phi(h)$ of arity $|\vec{w}| + m$ with witness $e_f = h$, and $f^0 = \Phi^0(h)$ of arity $|\vec{w}|$ with witness $e_{f^0} = h$; emit the pointer axiom $\forall \vec{w} \vec{z}. f^0(\vec{w}) @ z_1 @ \dots @ z_m \approx f(\vec{w}, \vec{z})$; run $B_{\theta'}(f(\vec{w}, \vec{z}); e_0)$ where $\theta' = \theta[g_0 \mapsto (f, \vec{w}, m)]$; the value is $f^0(\vec{w})$.

Definition 5.3 (Body compilation B). $B_\theta(L; e)$ takes a first-order term L (the *left-hand side*, linear in its variables, whose variables include $\text{FV}(e)$ minus the θ -bound ones) and a $\lambda_\square$ term e , and emits axioms:

1. If $e = \lambda x. e'$: run $B_\theta(L @ x; e')$.
2. If $e = \text{case}(x; \{c_i(\vec{z}_i) \Rightarrow b_i\}_i)$ with x a variable of L : for each i , with $\vec{z}_i$ fresh, run $B_\theta(L[\widehat{c}_i(\vec{z}_i)/x]; b_i[c_i(\vec{z}_i)/x])$.
3. If $e = \text{case}(s; \vec{b}r)$ with s not a variable of L : let $g = X(\lambda z. \text{case}(z; \vec{b}r))$ with closure $\vec{w}' = \text{FV}(\vec{b}r)$ and witness as in Definition 5.2; emit $\forall. L \approx g(\mathcal{C}_\theta(s), \vec{w}')$ (universal closure over the variables of L), and run $B_\theta(g(z, \vec{w}'); \text{case}(z; \vec{b}r))$ for fresh z .
4. If $e = \text{fix } g_0. e'$: compute the value translation $u = f^0(\vec{w})$ of e as in Definition 5.2 (which also compiles e); emit $\forall. L \approx u$.
5. Otherwise: emit $\forall. L \approx \mathcal{C}_\theta(e)$.

For a term definition $c := t : \tau$ of $\mathbb{E}$, compilation is initialised on $e = \mathcal{E}(t)$:

- if $e = \text{fix } g_0. \lambda z_1 \dots \lambda z_m. e_0$ (maximal prefix): set $a_c := m$ and run $B_{[g_0 \mapsto (\widehat{c}, (), m)]}(\widehat{c}(\vec{z}); e_0)$;
- otherwise, with $e = \lambda z_1 \dots \lambda z_m. e_0$ (maximal prefix, possibly $m = 0$): set $a_c := m$ and run $B_\emptyset(\widehat{c}(\vec{z}); e_0)$.

The witness of $\widehat{c}$ and of $\widehat{c}^0$ is the $\lambda_\square$ constant c ; if $a_c > 0$, the pointer axiom $\forall \vec{z}. \widehat{c}^0 @ z_1 @ \dots @ z_{a_c} \approx \widehat{c}(\vec{z})$ is emitted.

Lemma 5.4 (Compilation terminates). *The procedure of Definitions 5.2 and 5.3 terminates on every input and, with the naming functions of Definition 5.1, emits a finite set of axioms for a given environment and goal.*

Proof. As in [Ext26]: order calls by the pair (number of case/ λ /fix nodes, node count) lexicographically. Rule (2) consumes the outer case node and passes to a branch with a variable-for-pattern replacement that creates no case, λ or fix nodes; rule (3) replaces a non-variable scrutinee by a variable; rules (1), (4), (5) and all $\mathcal{C}$ recursions strictly decrease the pair. Finiteness follows since each emitted axiom is attached to a symbol, each symbol to a $\lambda_{\square}$ term in the finite compilation tree, and hash-consing merges repeated liftings. $\square$

Remark 5.5 (Shape of the equations). All axioms emitted by compilation are universally quantified equations $L \approx R$ whose left-hand sides are linear terms $f(\vec{\pi})@x_1@ \dots @x_k$ with $\vec{\pi}$ constructor patterns and variables; grouped by head symbol they are pairwise non-overlapping, so oriented left to right they form an orthogonal rewrite system [BN98]. Nothing about indexed families disturbs this: the patterns are patterns of *informative* constructor arguments, and index information never reaches $\lambda_{\square}$.

Example 5.6 (Compilation over families). For the definitions of Example 3.15 (writing hatted symbols without hats):

$$\mathcal{E}(\text{vhead}) = \lambda n v. \text{case}(v; \{\text{vnil} \Rightarrow \square \mid \text{vcons}(k, a, w) \Rightarrow a\}),$$

so $a_{\text{vhead}} = 2$ and rule (2) emits

$$\forall n. \text{vhead}(n, \text{vnil}) \approx \odot, \quad \forall n k a w. \text{vhead}(n, \text{vcons}(k, a, w)) \approx a,$$

and likewise $\text{vtail}(n, \text{vnil}) \approx \odot$ and $\forall n k a w. \text{vtail}(n, \text{vcons}(k, a, w)) \approx w$. Note three things. (i) The equations are unguarded and unconditional: no premise relates n to k . (ii) The first equation of each pair describes a *statically impossible* branch (vnil cannot inhabit $\text{vec}(\mathbf{S} n)$); it is nevertheless *valid* in the model (Lemma 6.12), because the $\lambda_{\square}$ case fires on any constructor form regardless of typability — the branch body is the image of the $\perp$ -elimination, dead code. Section 8.3 analyses the index-premised and pruned variants. (iii) For a recursive example, $\text{vmap}_g := \text{fix}f: (\Pi n: \text{nat}. \text{vec}(n) \rightarrow \text{vec}(n)). \lambda n v. \text{case}(v; \dots; \dots)$ (mapping a previously defined $g : A \rightarrow A$ over a vector) compiles to

$$\begin{aligned} \forall n. \text{vmap}_g(n, \text{vnil}) &\approx \text{vnil}, \\ \forall n k a w. \text{vmap}_g(n, \text{vcons}(k, a, w)) &\approx \text{vcons}(k, g(a), \text{vmap}_g(k, w)), \end{aligned}$$

orientable rewrite rules in which the recursive call's index argument is the pattern variable k — the equations a human would state.

5.3 Specification extraction

Definition 5.7 (Guard formulas and proposition translation). For every shallow set type σ and first-order term u , the *guard formula* $\mathcal{G}_\sigma(u)$, and for every shallow proposition φ , the first-order formula $\mathcal{F}(\varphi)$, are defined by mutual structural recursion:

$$\begin{aligned}
\mathcal{G}_{I(\vec{a})}(u) &= \widehat{I}(u, \mathcal{C}(\mathcal{E}(a_1)), \dots, \mathcal{C}(\mathcal{E}(a_m))) \\
\mathcal{G}_{\Pi x:\sigma.\tau}(u) &= \forall x. \mathcal{G}_\sigma(x) \rightarrow \mathcal{G}_\tau(u @ x) \\
\mathcal{G}_{\varphi \rightarrow \tau}(u) &= \mathcal{F}(\varphi) \rightarrow \mathcal{G}_\tau(u) \\
\mathcal{G}_{\{x:\sigma|\varphi\}}(u) &= \mathcal{G}_\sigma(u) \wedge \mathcal{F}(\varphi)[u/x] \\
\mathcal{F}(\top) &= \top, \quad \mathcal{F}(\perp) = \perp \\
\mathcal{F}(\varphi \circledast \psi) &= \mathcal{F}(\varphi) \circledast \mathcal{F}(\psi) \quad (\circledast \in \{\wedge, \vee, \rightarrow\}) \\
\mathcal{F}(\forall x:\sigma.\varphi) &= \forall x. \mathcal{G}_\sigma(x) \rightarrow \mathcal{F}(\varphi) \\
\mathcal{F}(\exists x:\sigma.\varphi) &= \exists x. \mathcal{G}_\sigma(x) \wedge \mathcal{F}(\varphi) \\
\mathcal{F}(a =_\sigma b) &= \mathcal{C}(\mathcal{E}(a)) \approx \mathcal{C}(\mathcal{E}(b)) \\
\mathcal{F}(J(\vec{a})) &= \widehat{J}(\mathcal{C}(\mathcal{E}(a_1)), \dots, \mathcal{C}(\mathcal{E}(a_k)))
\end{aligned}$$

where in the binder clauses the bound source variable is reused as the bound first-order variable, and $\mathcal{C} = \mathcal{C}_\emptyset$ is the term translation of Definition 5.2, whose complex-head clauses may emit further axioms. $\mathcal{G}$ and $\mathcal{F}$ are total on $\mathfrak{S}$ and structurally recursive; no normalisation of the source is involved.

The new clause is the first: the guard of a family occurrence carries the *translated index terms* as arguments of the guard atom. Index terms are ordinary terms — they may contain defined functions ($\text{vec}(\widehat{\text{add}}(m, n))$ arises from $\text{vec}(\text{add } m \ n)$), matches (lifted by $\mathcal{C}$'s complex-head clauses), and proof subterms (erased, Remark 4.6). Nothing is solved, nothing is normalised, nothing is encoded: the constraint material of the family arrives at the prover as first-order terms in predicate arguments, available to superposition and to nothing else.

Definition 5.8 (Specification of a constant). For a term definition $c := t : \tau$ with $\tau \in \mathfrak{S}$, the *specification axiom* $\mathcal{S}(c)$ is $S(u_0; \tau)$ where $u_0 = \widehat{c}$ if $a_c = 0$ and otherwise $u_0 = \widehat{c}()$ is the empty accumulator, and

$$\begin{aligned}
S(u; \Pi x:\sigma.\tau') &= \forall x. \mathcal{G}_\sigma(x) \rightarrow S(u \cdot x; \tau') \\
S(u; \varphi \rightarrow \tau') &= \mathcal{F}(\varphi) \rightarrow S(u; \tau') \\
S(u; \tau') &= \mathcal{G}_{\tau'}(\bar{u}) \quad \text{if } \tau' \text{ is a datatype occurrence or refinement type,}
\end{aligned}$$

where $u \cdot x$ appends x to the accumulator if fewer than a_c arguments have been accumulated and otherwise applies with $@$, and $\bar{u}$ closes the accumulator (an under-applied accumulator is rendered as the pointer chain, as in [Ext26]).

Example 5.9. $\mathcal{S}(\text{vtail})$ is

$$\forall n. \widehat{\text{nat}}(n) \rightarrow \forall v. \widehat{\text{vec}}(v, \widehat{\mathbf{S}}(n)) \rightarrow \widehat{\text{vec}}(\widehat{\text{vtail}}(n, v), n),$$

the statement of vtail 's type as a first-order formula: the index arithmetic of the family has become term structure inside guard atoms. For vmap_g the axiom is $\forall n. \widehat{\text{nat}}(n) \rightarrow \forall v. \widehat{\text{vec}}(v, n) \rightarrow \widehat{\text{vec}}(\widehat{\text{vmap}}_g(n, v), n)$ — index preservation, for free, from the type.

5.4 Structural axioms: the ford of a family

Definition 5.10 (Emitted theory). The first-order theory $\Delta(\mathbb{E}, G)$ consists of the following axioms.

1. For each datatype $I : (\vec{\sigma}) \rightarrow \text{Set}$ with constructors $\mathbf{c}_i : \forall \vec{y} : \vec{A}_i. \varphi_{i1} \Rightarrow \dots \Rightarrow \varphi_{i\ell_i} \Rightarrow I(t_i)$:

- (*constructor typing*) for each i ,

$$\forall \vec{y}. \bigwedge_j \mathcal{G}_{A_{ij}}(y_j) \rightarrow \bigwedge_l \mathcal{F}(\varphi_{il}) \rightarrow \widehat{I}(\widehat{\mathbf{c}}_i(\vec{y}), \mathcal{C}(\mathcal{E}(t_{i1})), \dots, \mathcal{C}(\mathcal{E}(t_{im})));$$

- (*inversion, forded*)

$$\forall w \vec{z}. \widehat{I}(w, \vec{z}) \rightarrow \bigvee_i \exists \vec{y}. \bigwedge_j \mathcal{G}_{A_{ij}}(y_j) \wedge \bigwedge_l \mathcal{F}(\varphi_{il}) \wedge w \approx \widehat{\mathbf{c}}_i(\vec{y}) \wedge \bigwedge_{h=1}^m z_h \approx \mathcal{C}(\mathcal{E}(t_{ih}));$$

- (*injectivity*) $\forall \vec{y} \vec{y}'. \widehat{\mathbf{c}}_i(\vec{y}) \approx \widehat{\mathbf{c}}_i(\vec{y}') \rightarrow \bigwedge_j y_j \approx y'_j$ for each i with $r_i > 0$;
- (*discrimination*) $\forall \vec{y} \vec{y}'. \neg(\widehat{\mathbf{c}}_i(\vec{y}) \approx \widehat{\mathbf{c}}_j(\vec{y}'))$ for all $i \neq j$.

2. For each inductive predicate J with clauses $\mathbf{k}_j : \forall \vec{x} : \vec{\tau}_j. \varphi_{j1} \rightarrow \dots \rightarrow \varphi_{j\ell_j} \rightarrow J(\vec{u}_j)$:

- (*clause*) $\forall \vec{x}. \bigwedge_i \mathcal{G}_{\tau_{ji}}(x_i) \rightarrow \mathcal{F}(\varphi_{j1}) \rightarrow \dots \rightarrow \mathcal{F}(\varphi_{j\ell_j}) \rightarrow \widehat{J}(\mathcal{C}(\mathcal{E}(\vec{u}_j)))$;
- (*case analysis*) $\forall \vec{z}. \widehat{J}(\vec{z}) \rightarrow \bigvee_j \exists \vec{x}. \bigwedge_i \mathcal{G}_{\tau_{ji}}(x_i) \wedge \bigwedge_i \mathcal{F}(\varphi_{ji}) \wedge \bigwedge_h z_h \approx \mathcal{C}(\mathcal{E}(u_{jh}))$.

3. For each term definition $c := t : \tau$: the equations, pointer axioms and auxiliary axioms emitted by compiling $\mathcal{E}(t)$ (Definition 5.3), and the specification axiom $\mathcal{S}(c)$ if $\tau \in \mathfrak{S}$.
4. For each lemma $c := \pi : \varphi$ with $\varphi \in \mathfrak{S}$: the *premise axiom* $\mathcal{F}(\varphi)$.
5. The auxiliary axioms emitted by the computations of $\mathcal{F}(G)$ and of the guard formulas and $\mathcal{F}$ -images of declarations (via the complex-head clauses of $\mathcal{C}$ on embedded terms).

The *translation of the problem* $(\mathbb{E}, G)$ is the pair $(\Delta(\mathbb{E}, G), \mathcal{F}(G))$.

The forded inversion axiom is the first-order image of the forded declaration: it quantifies over an *arbitrary* index tuple $\vec{z}$ — the whole aperture, in McBride’s phrase — and each disjunct carries, beside the constructor equation and payload, the index equations $z_h \approx \mathcal{C}(\mathcal{E}(t_{ih}))$ that recover specificity. The predicate case-analysis axiom (present already in the companion paper) is the same shape one level down — predicates have no element, so the constructor equation is absent; this is why indexed inductive *predicates* required no new mechanism there, while indexed *datatypes* do. Note also the division of labour inherited from the companion paper: exhaustiveness lives *only* in the guarded inversion axiom; the defining equations of item 3 make no such claim.

Example 5.11 (The running families, translated). For `vec` (constructor typing, then inversion):

$$\begin{aligned} \widehat{\text{vec}}(\widehat{\text{vnil}}, \widehat{\text{O}}), \quad & \forall n a v. \widehat{\text{nat}}(n) \rightarrow \mathcal{G}_A(a) \rightarrow \widehat{\text{vec}}(v, n) \rightarrow \widehat{\text{vec}}(\widehat{\text{vcons}}(n, a, v), \widehat{\text{S}}(n)), \\ & \forall w z. \widehat{\text{vec}}(w, z) \rightarrow (w \approx \widehat{\text{vnil}} \wedge z \approx \widehat{\text{O}}) \vee \\ & \exists n a v. (\widehat{\text{nat}}(n) \wedge \mathcal{G}_A(a) \wedge \widehat{\text{vec}}(v, n) \wedge w \approx \widehat{\text{vcons}}(n, a, v) \wedge z \approx \widehat{\text{S}}(n)), \end{aligned}$$

plus injectivity for `vcons` and discrimination $\widehat{\text{vnil}} \not\approx \widehat{\text{vcons}}(n, a, v)$. For rfl_{φ_0} :

$$\mathcal{F}(\varphi_0) \rightarrow \widehat{\text{rfl}}(\widehat{\text{RT}}, \widehat{\text{true}}), \quad \mathcal{F}(\neg\varphi_0) \rightarrow \widehat{\text{rfl}}(\widehat{\text{RF}}, \widehat{\text{false}}),$$

$$\begin{aligned} \forall w b. \widehat{\text{rfl}}(w, b) \rightarrow (\mathcal{F}(\varphi_0) \wedge w \approx \widehat{\text{RT}} \wedge b \approx \widehat{\text{true}}) \vee \\ (\mathcal{F}(\neg\varphi_0) \wedge w \approx \widehat{\text{RF}} \wedge b \approx \widehat{\text{false}}), \end{aligned}$$

with $\widehat{\text{RT}} \not\approx \widehat{\text{RF}}$: the propositional payloads and the index equations sit side by side in the disjuncts — “index equations are just more propositional payload.” For `bnd`, inversion is $\forall w z. \widehat{\text{bnd}}(w, z) \rightarrow \exists n k. \widehat{\text{nat}}(n) \wedge \widehat{\text{nat}}(k) \wedge \widehat{\text{lt}}(k, n) \wedge w \approx \widehat{\text{B}}(n, k) \wedge z \approx n$, which Lemma 5.14 simplifies by solving $n := z$. For the predicate `le`, the case-analysis axiom is

$$\begin{aligned} \forall z_1 z_2. \widehat{\text{le}}(z_1, z_2) \rightarrow \exists n. (\widehat{\text{nat}}(n) \wedge z_1 \approx n \wedge z_2 \approx n) \vee \\ \exists n m. (\widehat{\text{nat}}(n) \wedge \widehat{\text{nat}}(m) \wedge \widehat{\text{le}}(n, m) \wedge z_1 \approx n \wedge z_2 \approx \widehat{\text{S}}(m)), \end{aligned}$$

the classical inversion principle.

Example 5.12 (Empty instances refute for free). No axiom asserts any instance inhabited; empty instances are refutable. From $\widehat{\text{fin}}(w, \widehat{\text{O}})$, the inversion axiom for `fin` yields a disjunction whose every disjunct contains an equation $\widehat{\text{O}} \approx \widehat{\text{S}}(n)$, refuted by discrimination for `nat`; hence $\Delta(\mathbb{E}, G) \vdash_{\text{FOL}}$

$\neg \exists w. \widehat{\text{fin}}(w, \widehat{\text{O}})$, and a goal $\mathcal{F}(\forall x:\text{fin}(\text{O}). \varphi) = \forall w. \widehat{\text{fin}}(w, \widehat{\text{O}}) \rightarrow \mathcal{F}(\varphi)$ is provable outright. The same one-step pattern refutes $\widehat{\text{vec}}(\text{vnil-form}, \widehat{\text{S}}(n))$ -style hypotheses. Contrast the hazard class of selector/rank axioms on degenerate instances (Verus #1366 [The24b]): the theory of Definition 5.10 contains no axiom whose *hypothesis-free* content mentions an arbitrary instance, so empty instances can only make guards false, never derive new equations.

Example 5.13 (A user-declared equality). For eq_σ of Example 3.4, the emitted axioms are the clause $\forall x. \mathcal{G}_\sigma(x) \rightarrow \widehat{\text{eq}}_\sigma(x, x)$ and the case analysis $\forall z_1 z_2. \widehat{\text{eq}}_\sigma(z_1, z_2) \rightarrow \exists x. \mathcal{G}_\sigma(x) \wedge z_1 \approx x \wedge z_2 \approx x$, which Lemma 5.14 simplifies to $\forall z_1 z_2. \widehat{\text{eq}}_\sigma(z_1, z_2) \rightarrow \mathcal{G}_\sigma(z_1) \wedge z_2 \approx z_1$. Thus the declared equality implies the primitive $\approx$ and membership in σ , while $\mathcal{F}(a =_\sigma b) = \mathcal{C}(\mathcal{E}(a)) \approx \mathcal{C}(\mathcal{E}(b))$ is bare untyped equality; both directions of the correspondence are first-order derivable under the guard of σ . A user-written transport through eq_σ — the `scase`(; ;) former — erases (Definition 4.5), so a definition $c := \lambda x y:\sigma. \lambda e:\text{eq}_\sigma(x, y). \lambda w:\tau[x]. \text{scase}_{\text{eq}_\sigma}(e; \vec{z}.\tau[z_2]; w)$ compiles to the single unguarded equation $\forall x y w. \widehat{c}(x, y, w) \approx w$ together with its specification axiom

$$\forall x y. \mathcal{G}_\sigma(x) \rightarrow \mathcal{G}_\sigma(y) \rightarrow \forall w. \widehat{\text{eq}}_\sigma(x, y) \rightarrow \mathcal{G}_{\tau[x]}(w) \rightarrow \mathcal{G}_{\tau[y]}(\widehat{c}(x, y, w)).$$

This closes, in the idealised setting, the implementation gap in which a user-written match on an equality proof receives no definitional axiom at all while a fixed list of library transport constants is special-cased (Section 1.4); the soundness of the unguarded equation is exactly the business of Section 8.5.

5.5 Forced-argument simplification

The inversion and case-analysis axioms frequently contain existentially quantified variables that some index equation solves outright — the image of McBride’s coalescence [McB99] and of the forced-argument analysis of Brady et al. [BMM04, SM19]. Eliminating them is a first-order equivalence, not a soundness matter:

Lemma 5.14 (Solving forced equations). *Let $\Phi(\vec{y})$ be a first-order formula, y_j one of the variables, and z a term not containing any y . Then*

$$\vdash_{\text{FOL}} \left(\exists \vec{y}. z \approx y_j \wedge \Phi(\vec{y}) \right) \leftrightarrow \exists \vec{y} \setminus y_j. \Phi(\vec{y})[z/y_j],$$

and the same holds with the equation oriented $y_j \approx z$, and under any prefix of universal quantifiers and implications. Consequently, replacing any disjunct of an inversion or case-analysis axiom by its solved form yields a logically equivalent axiom, and Theorem 7.9 is unaffected by performing any number of such replacements.

Proof. Standard: left to right, instantiate the conjunct and rewrite; right to left, witness $y_j := z$. Both directions are derivations in pure first-order logic with equality, so the replacement preserves logical equivalence of the whole axiom (replacement of equivalents), and a theory with an axiom replaced by an equivalent proves the same formulas. $\square$

Remark 5.15. After solving, the `bnd` inversion of Example 5.11 becomes $\forall w z. \widehat{\text{bnd}}(w, z) \rightarrow \exists k. \widehat{\text{nat}}(z) \wedge \widehat{\text{nat}}(k) \wedge \widehat{\text{lt}}(k, z) \wedge w \approx \widehat{\text{B}}(z, k)$ — the refinement-style unfolding with no existential residue on the index. Which equations are solvable is exactly the forced-argument analysis; the translation may apply Lemma 5.14 greedily (each application removes one quantifier) and the model-side soundness statement is untouched. Unsolved equations — non-variable patterns like $z \approx \widehat{\text{S}}(n)$, non-linear occurrences, defined-function images like $z \approx \widehat{\text{add}}(m, n)$ — simply remain, as they must (Remark 5.16).

Remark 5.16 (Green slime, stated not solved). Nothing in Definition 5.10 requires index terms to be constructor-headed. A family `stree` with `node :  $\forall (x y : \text{nat}) (l : \text{stree}(x)) (r : \text{stree}(y)). \text{stree}(\text{S}(\text{add } x \ y))$` contributes the inversion disjunct $\dots \wedge z \approx \widehat{\text{S}}(\widehat{\text{add}}(x, y))$, next to which the compilation of `add` has already placed its two defining equations. An ATP confronted with a hypothesis `$\text{stree}(w, \widehat{\text{S}}(\widehat{\text{O}}))$` derives $\widehat{\text{S}}(\widehat{\text{add}}(x, y)) \approx \widehat{\text{S}}(\widehat{\text{O}})$, then $\widehat{\text{add}}(x, y) \approx \widehat{\text{O}}$ by injectivity of $\widehat{\text{S}}$, and can close or constrain the branch by superposition with `add`'s equations and `nat`'s inversion — reasoning that elaboration-time unification, for which `add` is not a rigid symbol, cannot perform [McB14, GMM06]. The translation performs *specialization by unification* nowhere; the prover performs it as ordinary equational inference, and the soundness proof (Theorem 7.8) is entirely indifferent to the shape of index terms. No decidability claim is made or needed: with defined symbols in the signature the first-order theory of the term algebra is Π_1^1 -complete [KRV17, Thm. 2], and the translation targets refutational usefulness, not decision.

Remark 5.17 (Relation to the implementation). Each axiom family has a direct site in the CoqHammer pipeline: item 1's constructor typing and inversion generalize the existing declaration-level axioms from inductive predicates to indexed datatypes (the current translation emits them only behind an inhabitation atom); item 3's equations are the per-constructor case equations of the extraction branch, now specified to be emitted for indexed scrutinees unchanged; the guard clause of Definition 5.7 is the removal of the classifier's indexed-family gate, with the index terms flowing into the guard atom's argument places. The per-occurrence *expansion* of guard atoms — the implementation's enum/subset/singleton classifications — is Section 8.1.

6 Semantics

We now build the structures against which soundness is proved: a proof-irrelevant, classical, syntactic semantics of the source, and a first-order structure $\mathfrak{M}$ over the same individuals interpreting $\Sigma_{\mathbb{E}}$. Throughout, $\mathbb{E}$ is a fixed well-formed definitional environment with $\mathfrak{S}$ declarations. The construction follows [Ext26]; the new material is the interpretation of indexed families as sets of (index tuple, element) pairs, the *index reflection* lemma — the semantic content of fording — and element ranks.

6.1 The domain

Definition 6.1 (Domain). Let $\mathcal{D}$ be the set of closed $\lambda_{\square}$ terms modulo $=_{\square}$, written $[e]$ for classes. $\mathcal{D}$ is non-empty ($[\square] \in \mathcal{D}$) and carries the application $d \cdot d' := [e e']$ for representatives, well defined since $=_{\square}$ is a congruence.

$\mathcal{D}$ deliberately contains *junk*: classes such as $[\mathbf{O} \mathbf{O}]$ or $[\text{fix } f.f]$ that are the value of no source term of any type, and — new in the indexed setting — *well-formed constructor forms at the wrong index*, such as $[\text{vnil}]$ considered at index $[\mathbf{S}(\mathbf{O})]$. By Corollary 4.4, distinct constructor forms lie in distinct classes and junk like $[\mathbf{O} \mathbf{O}]$ lies outside every constructor form. Junk is what makes the guard analysis of Section 8 non-trivial; keeping it in the model is what lets the model arbitrate which axioms need guards, and the wrong-index phenomenon is what makes the *index* arguments of guards non-negotiable (Proposition 8.9).

Definition 6.2 (Valuations and term denotation). A *valuation* ρ assigns to finitely many term variables elements of $\mathcal{D}$ and to finitely many predicate variables X (of kind $(\sigma_1, \dots, \sigma_n) \rightarrow \text{Prop}$) functions $\mathcal{D}^n \rightarrow \{0, 1\}$. Proof variables are not assigned anything. For a CIC_{ix}^- term t whose free term variables are assigned by ρ ,

$$\llbracket t \rrbracket_{\rho} := [\mathcal{E}(t)^{\rho}] \in \mathcal{D},$$

where $\mathcal{E}(t)^{\rho}$ substitutes a representative of $\rho(x)$ for each free variable x of $\mathcal{E}(t)$; this is well defined exactly as in [Ext26] (free variables of $\mathcal{E}(t)$ are free term variables of t ; classes are representative-independent). For an index tuple we write $\llbracket \vec{u} \rrbracket_{\rho} = (\llbracket u_1 \rrbracket_{\rho}, \dots, \llbracket u_m \rrbracket_{\rho}) \in \mathcal{D}^m$.

6.2 Interpretation of types, propositions and predicates

Definition 6.3 (Interpretations). By recursion on the position of declarations in $\mathbb{E}$ and, within each declaration, on the structure of phrases, we define for every set type σ , proposition φ and predicate expression P over a prefix of $\mathbb{E}$, and every valuation ρ covering their free variables: a set $\llbracket \sigma \rrbracket_{\rho} \subseteq \mathcal{D}$; a truth value $\langle \varphi \rangle_{\rho} \in \{0, 1\}$; a function $\langle P \rangle_{\rho} : \mathcal{D}^n \rightarrow \{0, 1\}$.

Indexed datatypes. For $I : (\sigma_1, \dots, \sigma_m) \rightarrow \mathbf{Set}$ with constructors $c_i : \forall \vec{y}:\vec{A}_i. \varphi_{i1} \Rightarrow \dots \Rightarrow \varphi_{i\ell_i} \Rightarrow I(\vec{t}_i)$ (r_i informative arguments), define stages $\llbracket I \rrbracket^s \subseteq \mathcal{D}^m \times \mathcal{D}$ by $\llbracket I \rrbracket^0 = \emptyset$ and

$$\begin{aligned} \llbracket I \rrbracket^{s+1} = \Big\{ & (\llbracket \vec{t}_i \rrbracket_{\rho_0}, [c_i(e_1, \dots, e_{r_i})]) \mid i \leq M, \rho_0 = [\vec{y}_i \mapsto \vec{e}], \\ & [e_j] \in \llbracket A_{ij} \rrbracket_{\rho_0} \text{ for non-recursive } j, \\ & (\llbracket \vec{v}_{ij} \rrbracket_{\rho_0}, [e_j]) \in \llbracket I \rrbracket^s \text{ for recursive } j \text{ with } A_{ij} = I(\vec{v}_{ij}), \\ & \langle \varphi_{il} \rangle_{\rho_0} = 1 \text{ for all } l \Big\}, \end{aligned}$$

which is well defined: non-recursive argument types and all premises are I -free phrases over the environment prefix and the telescope, so their interpretations and truth values are supplied by the outer recursion, and index-term denotations $\llbracket \cdot \rrbracket_{\rho_0}$ are erasure classes, defined independently of any type interpretation. The stages are increasing in s (induction on s , the defining condition being monotone in the stage at recursive positions). Set $\llbracket I \rrbracket = \bigcup_s \llbracket I \rrbracket^s$, and for an occurrence:

$$\llbracket I(\vec{u}) \rrbracket_\rho = \{ d \in \mathcal{D} \mid (\llbracket \vec{u} \rrbracket_\rho, d) \in \llbracket I \rrbracket \}.$$

Other set types (as in [Ext26]):

$$\begin{aligned} \llbracket \Pi x:\sigma. \tau \rrbracket_\rho &= \{ d \in \mathcal{D} \mid \forall e \in \llbracket \sigma \rrbracket_\rho. d \cdot e \in \llbracket \tau \rrbracket_{\rho[x \mapsto e]} \} \\ \llbracket \varphi \rightarrow \tau \rrbracket_\rho &= \{ d \in \mathcal{D} \mid \langle \varphi \rangle_\rho = 1 \implies d \in \llbracket \tau \rrbracket_\rho \} \\ \llbracket \{x:\sigma \mid \varphi\} \rrbracket_\rho &= \{ d \in \llbracket \sigma \rrbracket_\rho \mid \langle \varphi \rangle_{\rho[x \mapsto d]} = 1 \} \end{aligned}$$

Inductive predicates. For $J : (\sigma_1, \dots, \sigma_k) \rightarrow \mathbf{Prop}$ with clauses $k_j : \forall \vec{x}:\vec{\tau}_j. \varphi_{j1} \rightarrow \dots \rightarrow \varphi_{j\ell_j} \rightarrow J(\vec{u}_j)$, stages $\langle J \rangle^0 = \emptyset \subseteq \mathcal{D}^k$ and

$$\begin{aligned} \langle J \rangle^{s+1} = \Big\{ & \llbracket \vec{u}_j \rrbracket_{\rho_0} \mid j, \rho_0 = [\vec{x} \mapsto \vec{d}], d_i \in \llbracket \tau_{ji} \rrbracket_{\rho_0}, \\ & \langle \varphi \rangle_{\rho_0} = 1 \text{ for non-recursive premises,} \\ & \llbracket \vec{v} \rrbracket_{\rho_0} \in \langle J \rangle^s \text{ for recursive premises } J(\vec{v}) \Big\}, \end{aligned}$$

with $\langle J \rangle = \bigcup_s \langle J \rangle^s$.

Propositions and predicate expressions: exactly the clauses of [Ext26]: classical truth tables for the connectives, min / max over $\llbracket \sigma \rrbracket_\rho$ for quantifiers (with $\min \emptyset = 1$, $\max \emptyset = 0$), min over all $Q : \mathcal{D}^n \rightarrow \{0, 1\}$ for predicate quantification, $\langle a =_\sigma b \rangle_\rho = 1$ iff $\llbracket a \rrbracket_\rho = \llbracket b \rrbracket_\rho$, $\langle J(\vec{a}) \rangle_\rho = 1$ iff $\llbracket \vec{a} \rrbracket_\rho \in \langle J \rangle$, $\langle X(\vec{a}) \rangle_\rho = \rho(X)(\llbracket \vec{a} \rrbracket_\rho)$, $\langle \lambda(\vec{x}:\vec{\sigma}). \varphi \rangle_\rho(\vec{d}) = \langle \varphi \rangle_{\rho[\vec{x} \mapsto \vec{d}]}$.

The recursion is well founded exactly as before: stage constructions refer only to earlier declarations and structurally smaller phrases of their own declaration; all other clauses recurse on proper sub-phrases. The semantics

is classical and proof-irrelevant by construction: truth values do not depend on proofs, and $\llbracket \cdot \rrbracket$ ignores proof subterms. The interpretation of equality is untyped (class equality, not mentioning σ), matching the translation clause $\mathcal{F}(a =_\sigma b) = \mathcal{C}(\mathcal{E}(a)) \approx \mathcal{C}(\mathcal{E}(b))$; and the interpretation of a family is *index-aware* — membership is a relation between an index tuple and an element — matching the $(1+m)$ -ary guard predicate.

Definition 6.4 (Element rank). For d with $(\vec{d}, d) \in \llbracket I \rrbracket$ for some $\vec{d}$, let $\text{rk}_I(d)$ be the least s such that $(\vec{d}, d) \in \llbracket I \rrbracket^s$ for some $\vec{d}$. (By Lemma 6.5(3) the tuple is in fact unique, but the definition does not need this.) For $\langle J \rangle$, ranks of tuples are defined as in [Ext26]: the least s with $\vec{d} \in \langle J \rangle^s$.

Lemma 6.5 (Index reflection). *Let I be a datatype as above and $(\vec{d}, d) \in \llbracket I \rrbracket$. Then:*

1. (Decomposition.) *There are a constructor index i , closed $\lambda_\square$ terms $\vec{e} = e_1, \dots, e_{r_i}$ and the valuation $\rho_0 = [\vec{y}_i \mapsto [\vec{e}]]$ such that $d = [\mathbf{c}_i(\vec{e})]$; $[e_j] \in \llbracket A_{ij} \rrbracket_{\rho_0}$ for every non-recursive j ; $(\llbracket \vec{v}_{ij} \rrbracket_{\rho_0}, [e_j]) \in \llbracket I \rrbracket$ for every recursive j ; $\langle \varphi_{il} \rangle_{\rho_0} = 1$ for every l ; and*

$$\vec{d} = \llbracket \vec{t}_i \rrbracket_{\rho_0}.$$

2. (Uniqueness.) *The index i and the classes $[e_1], \dots, [e_{r_i}]$ (hence ρ_0 , hence all data of (1)) are uniquely determined by d .*
3. (Index uniqueness.) *If also $(\vec{d}', d) \in \llbracket I \rrbracket$ then $\vec{d}' = \vec{d}$.*
4. (Rank descent.) *$\text{rk}_I(d)$ is defined, and for every recursive j in the decomposition of (1), $\text{rk}_I([e_j]) < \text{rk}_I(d)$.*
5. (Completeness of stages.) *Conversely, any data as in (1) — an i , classes $[\vec{e}]$ with the memberships, premises and recursive conditions displayed there — yield $(\llbracket \vec{t}_i \rrbracket_{\rho_0}, [\mathbf{c}_i(\vec{e})]) \in \llbracket I \rrbracket$.*

Proof. (1) $(\vec{d}, d) \in \llbracket I \rrbracket^{s+1}$ for some s ; unfolding Definition 6.3 gives exactly the displayed data, with the recursive conditions in $\llbracket I \rrbracket^s \subseteq \llbracket I \rrbracket$.

(2) Suppose $d = [\mathbf{c}_i(\vec{e})] = [\mathbf{c}_{i'}(\vec{e}')]$ arise from two decompositions. By Corollary 4.4(1), $i = i'$ and $[e_j] = [e'_j]$ for all j . Since all conditions of (1) are conditions on the classes (interpretations, truth values and denotations are class functions), the two decompositions coincide.

(3) Apply (1) to $(\vec{d}, d)$ and to $(\vec{d}', d)$; by (2) both decompositions have the same i and ρ_0 , so $\vec{d} = \llbracket \vec{t}_i \rrbracket_{\rho_0} = \vec{d}'$.

(4) Definedness is immediate. Let $s_0 = \text{rk}_I(d)$, witnessed by $(\vec{d}_0, d) \in \llbracket I \rrbracket^{s_0}$; necessarily $s_0 = s + 1$ for some s , and unfolding the stage gives a decomposition whose recursive components satisfy $(\llbracket \vec{v}_{ij} \rrbracket_{\rho_0}, [e_j]) \in \llbracket I \rrbracket^s$, whence $\text{rk}_I([e_j]) \leq s < s_0$. By (2) this decomposition is *the* decomposition of (1).

(5) Each recursive condition $(\llbracket \vec{v}_{ij} \rrbracket_{\rho_0}, [e_j]) \in \llbracket I \rrbracket$ holds at some finite stage; by monotonicity all hold at their maximum s , and the defining clause puts the conclusion pair in $\llbracket I \rrbracket^{s+1}$. $\square$

Part (1) is the *inversion* direction of fording (used for hypothesis-position content: the forded inversion axiom, the case rule); part (5) is the *introduction* direction (constructor typing, goal-position content); part (3) is the semantic fact that in the term model every element carries its indices — Proposition 8.15 derives its first-order shadow; and parts (1)+(3) together are what uniqueness of identity proofs buys in type theory, spent here by the `scase`(; ;) and `cast`(; ;) cases of Lemma 7.6 (see Section 8.5).

Lemma 6.6 (Elementary properties). *1. $\llbracket \sigma \rrbracket_\rho, \langle \varphi \rangle_\rho, \llbracket P \rrbracket_\rho$ depend only on the restriction of ρ to the free variables of the phrase, and are unchanged by extending ρ on fresh variables.*

2. Every $(\vec{d}, d) \in \llbracket I \rrbracket$ has $(\vec{d}, d) \in \llbracket I \rrbracket^s$ for all $s > \text{rk}_I(d)$; similarly for $\langle J \rangle$.

Proof. (1) routine induction on the phrase (index terms depend on ρ only through their free variables). (2) monotonicity of stages and Lemma 6.5(3): the witnessing tuple at the minimal stage is the unique tuple. $\square$

Lemma 6.7 (Semantic substitution). *Let a be a term, π a proof and P a predicate expression, with ρ covering the relevant free variables.*

- 1. $\llbracket t[a/x] \rrbracket_\rho = \llbracket t \rrbracket_{\rho[x \mapsto \llbracket a \rrbracket_\rho]}$, and analogously for substituted types $(\llbracket \sigma[a/x] \rrbracket_\rho = \llbracket \sigma \rrbracket_{\rho[x \mapsto \llbracket a \rrbracket_\rho]})$, propositions and predicate expressions.*
- 2. Substitution of proofs is invisible: $\llbracket t[\pi/p] \rrbracket_\rho = \llbracket t \rrbracket_\rho$, $\llbracket \sigma[\pi/p] \rrbracket_\rho = \llbracket \sigma \rrbracket_\rho$, $\langle \varphi[\pi/p] \rangle_\rho = \langle \varphi \rangle_\rho$.*
- 3. $\langle \varphi[P/X] \rangle_\rho = \langle \varphi \rangle_{\rho[X \mapsto \llbracket P \rrbracket_\rho]}$, and analogously for types and terms (invisible).*

Proof. (1) For terms, by Lemma 4.7 as in [Ext26]. For types, propositions and predicates: induction on the phrase; the new case $\llbracket I(\vec{u}) \rrbracket_\rho$ follows from the term statement applied to each index term, since the clause depends on $\vec{u}$ only through $\llbracket \vec{u} \rrbracket_\rho$. (2) Erasure discards the substituted proof (Lemma 4.7); proofs do not occur in any semantic clause — in particular proofs inside *index terms* are invisible, which is the model-theoretic form of Remark 4.6. (3) As in (1). $\square$

Lemma 6.8 (Conversion invariance). *If $t \simeq t'$ then $\llbracket t \rrbracket_\rho = \llbracket t' \rrbracket_\rho$. If $\sigma \simeq \sigma'$ (both well formed) then $\llbracket \sigma \rrbracket_\rho = \llbracket \sigma' \rrbracket_\rho$; if $\varphi \simeq \varphi'$ then $\langle \varphi \rangle_\rho = \langle \varphi' \rangle_\rho$.*

Proof. For terms: $t \simeq t'$ implies $\mathcal{E}(t) =_\square \mathcal{E}(t')$ (Lemma 4.8), and $=_\square$ is substitutive. For types and propositions, consider a single step and proceed by induction on the structure of the phrase, using Lemma 3.11: the step

occurs inside an embedded term (an index term of a datatype occurrence, an equation side, a predicate-atom argument) — covered by the term statement, since the semantic clauses depend on embedded terms only through $\llbracket \cdot \rrbracket_\rho$ — or inside a component type or proposition, covered by the induction hypothesis at every extension of ρ . The general claim follows by induction on the conversion path. $\square$

6.3 The first-order structure $\mathfrak{M}$

Definition 6.9 (The structure $\mathfrak{M}$). $\mathfrak{M}$ is the first-order $\Sigma_{\mathbb{E}}$ -structure with universe $\mathcal{D}$ and:

- $\approx$ interpreted as identity of classes; $@$ as $\cdot$; $\odot$ as $[\Box]$;
- $\hat{I}^{\mathfrak{M}} = \{(d, \vec{d}) \mid (\vec{d}, d) \in \llbracket I \rrbracket\}$ for each datatype I (subject first, matching Definition 5.1), and $\hat{J}^{\mathfrak{M}} = \langle J \rangle$ for each inductive predicate J ;
- $\hat{\mathcal{C}}^{\mathfrak{M}}(d_1, \dots, d_r) = [c(e_1, \dots, e_r)]$ for representatives $e_i \in d_i$;
- for every symbol f introduced by compilation with closure arity s , extension arity m and witness e_f (free variables $\vec{w}$): $f^{\mathfrak{M}}(\vec{d}, \vec{d}') = [e_f[\vec{e}/\vec{w}] e'_1 \cdots e'_m]$ for representatives; in particular $\hat{\mathcal{C}}^{\mathfrak{M}}(\vec{d}) = [c \vec{e}]$ and $(\hat{\mathcal{C}}^0)^{\mathfrak{M}} = [c]$.

All interpretations are well defined on classes by congruence of $=_{\Box}$. We write $u^{\mathfrak{M}, \rho}$ for the denotation of a first-order term u under an assignment ρ , and $\mathfrak{M}, \rho \models \phi$ for Tarskian satisfaction. Valuations restricted to term variables serve as first-order assignments.

6.4 Adequacy of the term translation and of compilation

The two lemmas of this subsection are stated and proved exactly as in the companion paper; we reproduce them because the soundness proof cites their content, and because their *independence* of the indexed extension is itself the point: compilation concerns only the program half, which is index-blind.

Definition 6.10 (Coherent template maps). For a template map θ and a substitution σ_θ assigning closed $\lambda_{\Box}$ terms to $\text{dom}(\theta)$, say (θ, σ_θ) is *coherent* for an assignment ρ if for every g with $\theta(g) = (f, \vec{w}, m)$ and all $d_1, \dots, d_m \in \mathcal{D}$ with representatives e_i : $f^{\mathfrak{M}}(\vec{w}^{\mathfrak{M}, \rho}, \vec{d}) = [\sigma_\theta(g) e_1 \cdots e_m]$.

Lemma 6.11 (Adequacy of $\mathcal{C}$). *Let e be a $\lambda_{\Box}$ term whose free variables are first-order variables assigned by ρ or bound by θ , and let (θ, σ_θ) be coherent for ρ . Then $\mathcal{C}_\theta(e)^{\mathfrak{M}, \rho} = [e^{\rho, \sigma_\theta}]$, where e^{ρ, σ_θ} substitutes representatives of $\rho(x)$ for the first-order variables and $\sigma_\theta(g)$ for the θ -bound variables. In particular $\mathcal{C}(\mathcal{E}(t))^{\mathfrak{M}, \rho} = \llbracket t \rrbracket_\rho$ for every CIC_{ix}^- term t with $\theta = \emptyset$.*

Proof. By induction on e following the clauses of Definition 5.2, exactly as in [Ext26, §5.4]. Variables, $\square$ and constructors are immediate ($\widehat{c}^{\mathfrak{M}}$ applies the constructor to the argument classes). For spines: a θ -bound head uses coherence and the induction hypothesis on the arguments; a constant head uses $\widehat{c}^{\mathfrak{M}}(\vec{d}) = [c \vec{e}]$ and the pointer interpretation; a variable head is immediate; a complex head uses the interpretation of the lifted symbol via its witness, plus one β -step for case lifts. The final claim is the special case $e = \mathcal{E}(t)$, using Definition 6.2. $\square$

Lemma 6.12 (Soundness of compilation). *Consider any invocation of the compilation procedure on the environment or on a term embedded in the goal or in a declaration, and let $B_\theta(L; e)$ range over the body-compilation calls it performs. Then:*

1. *for every such call there is a substitution σ_θ such that for every assignment ρ of the variables of L , the pair (θ, σ_θ) is coherent for ρ and $L^{\mathfrak{M}, \rho} = [e^{\rho, \sigma_\theta}]$;*
2. *every axiom emitted by the procedure holds in $\mathfrak{M}$ — under every assignment of its universally quantified variables, junk included.*

Proof. By induction on the tree of procedure calls, exactly as in [Ext26, §5.4]; we reproduce the argument in compact form since Section 8 leans on part (2).

Roots. For $c := t : \tau$ with $\mathcal{E}(t) = \lambda \vec{z}. e_0$: $\widehat{c}(\vec{z})^{\mathfrak{M}, \rho} = [c \vec{e}] = [\mathcal{E}(t) \vec{e}]$ (one δ -step) $= [e_0^\rho]$ (β -steps), with $\theta = \emptyset$. For $\mathcal{E}(t) = \text{fix } g_0. \lambda \vec{z}. e_0$: set $\sigma_\theta(g_0) = \mathcal{E}(t)$; coherence is the δ -step, and the invariant follows by a μ -step and β -steps. For λ -, case- and fix-lifts the invariant holds by the definition of the lifted symbol's interpretation via its witness (for case lifts with one β -step; for fix lifts, extend σ_θ with the closed instance of the witness at the representatives of $\vec{w}^{\mathfrak{M}, \rho}$).

Rule (1): one β -step. *Rule (2):* an assignment of the refined left-hand side induces the assignment of L mapping x to $[c_i(\vec{e})]$; a head ι -step converts the representative — note the ι -step of $\lambda_\square$ fires on *any* constructor form, with no index side condition, which is why the emitted equations hold at all elements. *Rule (3):* both sides of the emitted axiom denote $[\text{case}(s; \vec{br})^{\rho, \sigma_\theta}]$ by the invariant, Lemma 6.11 and one β -step. *Rule (4):* both sides denote $[e^{\rho, \sigma_\theta}]$ via the fix-lift witness. *Rule (5):* the invariant and Lemma 6.11. *Pointer axioms:* both sides denote $[c \vec{e}]$ (resp. the fix-lift value applied). $\square$

6.5 Coincidence: the translation reflects the semantics

Lemma 6.13 (Coincidence). *Let σ be a shallow set type and φ a shallow proposition over $\mathbb{E}$ and a context of term variables, and let ρ be a valuation of their free variables. Then:*

1. for every first-order term u over the variables of ρ : $\mathfrak{M}, \rho \models \mathcal{G}_\sigma(u) \iff u^{\mathfrak{M}, \rho} \in \llbracket \sigma \rrbracket_\rho$;
2. $\mathfrak{M}, \rho \models \mathcal{F}(\varphi) \iff \langle \varphi \rangle_\rho = 1$.

Proof. Mutual induction on the structure of σ and φ , following Definition 5.7.

- $\sigma = I(\vec{a})$: $\mathfrak{M}, \rho \models \hat{I}(u, \mathcal{C}(\mathcal{E}(\vec{a})))$ iff $(u^{\mathfrak{M}, \rho}, \mathcal{C}(\mathcal{E}(\vec{a}))^{\mathfrak{M}, \rho}) \in \hat{I}^{\mathfrak{M}}$ iff $(\llbracket \vec{a} \rrbracket_\rho, u^{\mathfrak{M}, \rho}) \in \llbracket I \rrbracket$ — using Lemma 6.11 to identify $\mathcal{C}(\mathcal{E}(a_h))^{\mathfrak{M}, \rho} = \llbracket a_h \rrbracket_\rho$ — iff $u^{\mathfrak{M}, \rho} \in \llbracket I(\vec{a}) \rrbracket_\rho$ by Definition 6.3.
- $\sigma = \Pi x:\sigma'.\tau$, $\sigma = \varphi' \rightarrow \tau$, $\sigma = \{x:\sigma' \mid \varphi'\}$: as in [Ext26], by the induction hypotheses and, for the refinement clause, the first-order substitution lemma.
- Connectives, $\top$, $\perp$, quantifiers: the classical satisfaction clauses match the truth-value clauses literally, with induction hypothesis (1) carving out exactly $\llbracket \sigma' \rrbracket_\rho$ under each guard.
- Equations and J -atoms: by Lemma 6.11, $\mathcal{C}(\mathcal{E}(a))^{\mathfrak{M}, \rho} = \llbracket a \rrbracket_\rho$, and the model interprets $\approx$ as class equality and $\hat{J}$ as $\langle J \rangle$.

Shallowness is used exactly here: predicate quantifiers and predicate-variable atoms have no translation; all remaining formers are covered. $\square$

The new clause deserves a comment in the direction of Section 2.4: the coincidence for family guards is an equivalence *at every index tuple*, junk tuples included — both sides are false at a wrong-index pair such as $([\text{vnil}], [\text{S}(\text{O})])$ by Lemma 6.5(3). It is this two-sided, index-aware coincidence that licenses using guard atoms in *both* polarities of the translation, and its failure under index-forgetting that Proposition 8.9 exhibits.

7 Soundness

This section proves the main results: every axiom of $\Delta(\mathbb{E}, G)$ holds in $\mathfrak{M}$, and consequently first-order provability of $\mathcal{F}(G)$ entails truth of G in the semantics of Section 6.

7.1 Guard disciplines

Definition 7.1 (Context satisfaction). A valuation ρ *satisfies* a context Γ , written $\rho \models \Gamma$, if in telescope order: $\rho(x) \in \llbracket \sigma \rrbracket_\rho$ for $x : \sigma$; $\rho(X) : \mathcal{D}^n \rightarrow \{0, 1\}$ for $X : (\vec{\sigma}) \rightarrow \text{Prop}$; and $\langle \varphi \rangle_\rho = 1$ for $p : \varphi$.

The treatment of fix interpolates between “ $\rho(f)$ is anything” and “ $\rho(f)$ inhabits its type” by a rank-restricted membership; the only change against [Ext26] is that the rank of the guard argument is its *element rank* in the guard family (Definition 6.4), which Lemma 6.5 makes well defined without reference to the indices at which the argument is used.

Lemma 7.2 (Flattening of telescopes). *Let $T = \Pi x_1:\sigma_1 \dots \Pi x_n:\sigma_n.\tau$. Then $F \in \llbracket T \rrbracket_\rho$ iff for all $d_1, \dots, d_n$ chosen successively with $d_i \in \llbracket \sigma_i \rrbracket_{\rho[x_1 \mapsto d_1, \dots, x_{i-1} \mapsto d_{i-1}]}$ we have $F \cdot d_1 \dots d_n \in \llbracket \tau \rrbracket_{\rho[\vec{x} \mapsto \vec{d}]}$.*

Proof. Unfold Definition 6.3 n times. $\square$

Definition 7.3 (Rank-restricted interpretation). Let $T = \Pi x_1:\sigma_1 \dots \Pi x_n:\sigma_n.\tau$ be a fix type with guard position k , $\sigma_k = I(\vec{v})$, and let $m \in \mathbb{N}$. Define $\llbracket T \rrbracket_\rho^{k, < m}$ as the set of $F \in \mathcal{D}$ such that for all successively chosen $d_1, \dots, d_n$ as in Lemma 7.2 with additionally $\text{rk}_I(d_k) < m$:

$$F \cdot d_1 \dots d_n \in \llbracket \tau \rrbracket_{\rho[\vec{x} \mapsto \vec{d}]}.$$

(The rank is defined: the chain condition places d_k in $\llbracket I(\vec{v}) \rrbracket_{\rho[\dots]}$, so $(\cdot, d_k) \in \llbracket I \rrbracket_\rho$.) By Lemma 7.2 and finiteness of ranks, $\llbracket T \rrbracket_\rho = \bigcap_m \llbracket T \rrbracket_\rho^{k, < m}$, and $\llbracket T \rrbracket_\rho^{k, < m}$ is antitone in m .

Definition 7.4 (Guard discipline). A *guard discipline* $\mathcal{D}$ over Γ is a finite set of triples (f, V_f, m_f) where the f are pairwise distinct term variables declared in Γ with fix types T_f (guard position k_f , guard family I_f), V_f is a set of term variables declared in Γ whose types are occurrences of I_f , and $m_f \in \mathbb{N}$. A phrase t *respects* $\mathcal{D}$ if for each $(f, V_f, m_f) \in \mathcal{D}$, every occurrence of f in t lies at the head of an application spine $f a_1 \dots a_{n_f}$ occurring as a subterm of t , with $a_{k_f} \in V_f$. A valuation ρ satisfies Γ *under* $\mathcal{D}$, written $\rho \models_{\mathcal{D}} \Gamma$, if ρ satisfies all declarations of Γ as in Definition 7.1 except that for each $(f, V_f, m_f) \in \mathcal{D}$ only $\rho(f) \in \llbracket T_f \rrbracket_\rho^{k_f, < m_f}$ is required, and additionally $\text{rk}_{I_f}(\rho(y)) < m_f$ for every $y \in V_f$.

Lemma 7.5 (Locality of respect). *Let t respect $\mathcal{D}$ and let s be a subterm of t occurring in a non-applied position (any position other than the function part of an application). Then s respects $\mathcal{D}$.*

Proof. As in [Ext26]: if the witnessing spine of an occurrence of f in s were not contained in s , then s would be a proper prefix of that spine and hence occur applied, contradicting the assumption. $\square$

7.2 The fundamental lemma

Throughout this subsection we assume the *environment hypothesis* for the fixed environment $\mathbb{E}$ (discharged in Theorem 7.8):

(H) for every term definition $c := t : \tau_c$ in $\mathbb{E}$, $[c] \in \llbracket \tau_c \rrbracket$; and for every lemma $c := \pi : \varphi_c$ in $\mathbb{E}$, $\langle \varphi_c \rangle = 1$.

We use the simultaneous form of Lemma 6.7 throughout: for a simultaneous substitution of terms $\vec{a}$ for distinct variables $\vec{x}$, $\llbracket \sigma[\vec{a}/\vec{x}] \rrbracket_\rho = \llbracket \sigma \rrbracket_{\rho[\vec{x} \mapsto \llbracket \vec{a} \rrbracket_\rho]}$, obtained from the single-variable form by α -renaming to fresh variables and iterating; likewise for terms and propositions.

Lemma 7.6 (Fundamental lemma). *Assume (H). Let $\mathcal{D}$ be a guard discipline over Γ and $\rho \models_{\mathcal{D}} \Gamma$.*

1. *If $\Gamma \vdash t : \sigma$ and t respects $\mathcal{D}$, then $\llbracket t \rrbracket_\rho \in \llbracket \sigma \rrbracket_\rho$.*
2. *If $\Gamma \vdash \pi : \varphi$ and π respects $\mathcal{D}$, then $\langle \varphi \rangle_\rho = 1$.*

Proof. By induction on the size of the typing derivation, mutually for (1) and (2). We freely use Lemma 6.6 (weakening), Lemma 6.7 (semantic substitution, in simultaneous form), Lemma 6.8 (conversion invariance, which handles the two conversion rules), and Lemma 7.5 to propagate the discipline to subphrases; types, propositions and predicate expressions contain no applied occurrences of disciplined variables in well-formed judgements, by the same locality argument. When extending ρ at bound variables we α -rename so that disciplined variables, V_f -variables and context variables are not shadowed.

Variables, constants. If $t = x$ is not disciplined, $\rho(x) \in \llbracket \sigma \rrbracket_\rho$ by $\rho \models_{\mathcal{D}} \Gamma$; a bare disciplined f does not respect $\mathcal{D}$, so this case cannot arise for it. For a term constant, $\llbracket c \rrbracket_\rho = [c] \in \llbracket \tau_c \rrbracket$ by (H), the δ -rule and weakening; for a lemma constant in (2), directly by (H).

Disciplined spines. Suppose $t = f a_1 \cdots a_j$ with $(f, V_f, m_f) \in \mathcal{D}$ and $j \geq n_f$ (respect makes $f a_1 \cdots a_{n_f}$ a spine of t), and let $T_f = \Pi x_1 : \sigma_1 \dots \Pi x_{n_f} : \sigma_{n_f} . \tau$. The typing derivation decomposes into j applications; by induction hypothesis (1) applied to each argument (which respects $\mathcal{D}$ by Lemma 7.5), $d_i := \llbracket a_i \rrbracket_\rho \in \llbracket \sigma_i \rrbracket_{\rho[x_1 \mapsto d_1, \dots]}$ for $i \leq n_f$, the intermediate assignments being handled by semantic substitution as in the application case below. By respect, $a_{k_f} \in V_f$, so $\text{rk}_{I_f}(d_{k_f}) = \text{rk}_{I_f}(\rho(a_{k_f})) < m_f$; since $\rho(f) \in \llbracket T_f \rrbracket_\rho^{k_f, < m_f}$,

$$\llbracket f a_1 \cdots a_{n_f} \rrbracket_\rho = \rho(f) \cdot d_1 \cdots d_{n_f} \in \llbracket \tau \rrbracket_{\rho[\vec{x} \mapsto \vec{d}]} = \llbracket \tau[\vec{a}/\vec{x}] \rrbracket_\rho,$$

the last step by semantic substitution; remaining arguments $a_{n_f+1}, \dots, a_j$ are absorbed by the application case.

Applications and abstractions. For $t a$ with head not disciplined: the function part respects $\mathcal{D}$ (its disciplined occurrences have their spines within it, by the argument of Lemma 7.5 applied to the maximal spine decomposition), so by the induction hypotheses $\llbracket t \rrbracket_\rho \in \llbracket \Pi x : \sigma . \tau \rrbracket_\rho$ and $\llbracket a \rrbracket_\rho \in \llbracket \sigma \rrbracket_\rho$, whence $\llbracket t a \rrbracket_\rho = \llbracket t \rrbracket_\rho \cdot \llbracket a \rrbracket_\rho \in \llbracket \tau \rrbracket_{\rho[x \mapsto \llbracket a \rrbracket_\rho]} = \llbracket \tau[a/x] \rrbracket_\rho$. For $t \pi$: $\mathcal{E}(t \pi) = \mathcal{E}(t)$;

induction hypothesis (2) gives $\langle \varphi \rangle_\rho = 1$, so induction hypothesis (1) and the clause for $\llbracket \varphi \rightarrow \tau \rrbracket_\rho$ conclude. For $\lambda x:\sigma.t$ and $e \in \llbracket \sigma \rrbracket_\rho$: $\llbracket \lambda x:\sigma.t \rrbracket_\rho \cdot e = [(\lambda x.\mathcal{E}(t)^\rho) \hat{e}] = \llbracket t \rrbracket_{\rho[x \mapsto e]}$ (one β -step), which lies in $\llbracket \tau \rrbracket_{\rho[x \mapsto e]}$ by the induction hypothesis. For $\lambda p:\varphi.t$: if $\langle \varphi \rangle_\rho = 1$ then $\rho \models_{\mathcal{D}} \Gamma, p:\varphi$ and $\llbracket \lambda p:\varphi.t \rrbracket_\rho = \llbracket t \rrbracket_\rho \in \llbracket \tau \rrbracket_\rho$ by the induction hypothesis, as the clause for $\varphi \rightarrow \tau$ requires.

Constructors. Let $t = c_i(\vec{a}; \vec{\pi})$ with the typing premises of Figure 2, so the displayed type is $I(\vec{t}_i[\vec{a}/\vec{y}])$. Let $\rho_0 := [\vec{y} \mapsto \llbracket \vec{a} \rrbracket_\rho]$. By induction hypothesis (1) on each a_j and semantic substitution, $\llbracket a_j \rrbracket_\rho \in \llbracket A_{ij}[\vec{a}_{<j}/\vec{y}_{<j}] \rrbracket_\rho = \llbracket A_{ij} \rrbracket_{\rho_0}$; for recursive j with $A_{ij} = I(\vec{v}_{ij})$ this reads $(\llbracket \vec{v}_{ij} \rrbracket_{\rho_0}, \llbracket a_j \rrbracket_\rho) \in \llbracket I \rrbracket$. By induction hypothesis (2) on each π_l and semantic substitution, $\langle \varphi_{il} \rangle_{\rho_0} = \langle \varphi_{il}[\vec{a}/\vec{y}] \rangle_\rho = 1$. By Lemma 6.5(5),

$$(\llbracket \vec{t}_i \rrbracket_{\rho_0}, [c_i(\vec{e})]) \in \llbracket I \rrbracket, \quad \text{where } [c_i(\vec{e})] = \llbracket t \rrbracket_\rho,$$

and $\llbracket \vec{t}_i \rrbracket_{\rho_0} = \llbracket \vec{t}_i[\vec{a}/\vec{y}] \rrbracket_\rho$ by semantic substitution, so $\llbracket t \rrbracket_\rho \in \llbracket I(\vec{t}_i[\vec{a}/\vec{y}]) \rrbracket_\rho$.

Forded case. Let $t = \text{case}(a; \vec{z}.z.\tau; \{c_i(\vec{y}_i; \vec{p}_i; \vec{q}_i) \Rightarrow t_i\}_i)$ with $\Gamma \vdash a : I(\vec{u})$, and displayed type $\tau[\vec{u}/\vec{z}, a/z]$. By induction hypothesis (1), $(\llbracket \vec{u} \rrbracket_\rho, \llbracket a \rrbracket_\rho) \in \llbracket I \rrbracket$. Lemma 6.5(1) yields the unique decomposition: an i , classes $[\vec{e}]$ with $\llbracket a \rrbracket_\rho = [c_i(\vec{e})]$, the memberships and premise truths at $\rho_0 = [\vec{y}_i \mapsto [\vec{e}]]$, and

$$\llbracket \vec{u} \rrbracket_\rho = \llbracket \vec{t}_i \rrbracket_{\rho_0}. \quad (*)$$

Let $\rho' := \rho[\vec{y}_i \mapsto [\vec{e}]]$ ($\vec{y}_i$ fresh). Then $\rho' \models_{\mathcal{D}} \Gamma, \vec{y}_i:\vec{A}_i, \vec{p}_i:\vec{\varphi}_i, \vec{q}_i:(\vec{u} =_{\vec{\sigma}} \vec{t}_i)$: the Γ -part by weakening; the memberships and premises by the decomposition (their free variables are among earlier declarations and $\vec{y}_i$, on which ρ' agrees with ρ_0); and for the index equations, $\langle u_h =_{\sigma_h} t_{ih} \rangle_{\rho'} = 1$ because $\llbracket u_h \rrbracket_{\rho'} = \llbracket u_h \rrbracket_\rho$ ($\vec{y}_i$ fresh for u_h) = $\llbracket t_{ih} \rrbracket_{\rho_0} = \llbracket t_{ih} \rrbracket_{\rho'}$ by (*). On erasures,

$$\mathcal{E}(t)^\rho = \text{case}(\mathcal{E}(a)^\rho; \dots) =_{\square} \text{case}(c_i(\vec{e}); \dots) \rightarrow_{\square} \mathcal{E}(t_i)^{\rho'},$$

so $\llbracket t \rrbracket_\rho = \llbracket t_i \rrbracket_{\rho'}$. By the induction hypothesis on the branch derivation (t_i respects $\mathcal{D}$ by Lemma 7.5),

$$\llbracket t_i \rrbracket_{\rho'} \in \llbracket \tau[\vec{t}_i/\vec{z}, c_i(\vec{y}_i; \vec{p}_i)/z] \rrbracket_{\rho'} = \llbracket \tau \rrbracket_{\rho'[\vec{z} \mapsto \llbracket \vec{t}_i \rrbracket_{\rho'}, z \mapsto [c_i(\vec{y}_i; \vec{p}_i)]]_{\rho'}}$$

by semantic substitution. Now $\llbracket \vec{t}_i \rrbracket_{\rho'} = \llbracket \vec{t}_i \rrbracket_{\rho_0} = \llbracket \vec{u} \rrbracket_\rho$ by (*), and $\llbracket c_i(\vec{y}_i; \vec{p}_i) \rrbracket_{\rho'} = [c_i(\vec{e})] = \llbracket a \rrbracket_\rho$; by weakening on the remaining free variables of τ and semantic substitution backwards, the displayed set equals $\llbracket \tau[\vec{u}/\vec{z}, a/z] \rrbracket_\rho$, as required. This is the place where the index equations of the model — Lemma 6.5(1) — convert the branch's type, stated at the constructor's indices, to the case's type, stated at the occurrence's indices; in CIC the same conversion is performed by index unification, and in the forded reading by rewriting along $\vec{q}_i$.

Fix. Let $t = \text{fix } f:T. \lambda \vec{x}. \text{case}(x_k; \vec{z}.z.\tau_0; \{c_i(\vec{y}_i; \vec{p}_i; \vec{q}_i) \Rightarrow b_i\}_i)$ with $T = \Pi x_1:\sigma_1 \dots \Pi x_n:\sigma_n. \tau$, $\sigma_k = I(\vec{v})$, $\tau = \tau_0[\vec{v}/\vec{z}, x_k/z]$, and let $F := \llbracket t \rrbracket_\rho$. Since

$\llbracket T \rrbracket_\rho = \bigcap_m \llbracket T \rrbracket_\rho^{k, < m}$, it suffices to prove $F \in \llbracket T \rrbracket_\rho^{k, < m}$ for all m , by strong induction on m (the *rank induction*, nested inside the main induction).

Let m be given with $F \in \llbracket T \rrbracket_\rho^{k, < m'}$ for all $m' < m$, and let $d_1, \dots, d_n$ be a chain as in Definition 7.3 with $m_0 := \text{rk}_I(d_k) < m$. Write $\rho_c := \rho[x_1 \mapsto d_1, \dots, x_{k-1} \mapsto d_{k-1}]$; the chain condition gives $(\llbracket \vec{v} \rrbracket_{\rho_c}, d_k) \in \llbracket I \rrbracket$. By Lemma 6.5(1) there are i and $[\vec{e}]$ with $d_k = [\mathbf{c}_i(\vec{e})]$, the memberships and premises at $\rho_0 = [\vec{y}_i \mapsto [\vec{e}]]$, and $\llbracket \vec{v} \rrbracket_{\rho_c} = \llbracket \vec{t}_i \rrbracket_{\rho_0}$; by Lemma 6.5(4), $\text{rk}_I([e_j]) < m_0$ for recursive j . Unfolding one μ -step, n β -steps and one ι -step on a representative of $F \cdot d_1 \cdots d_n$:

$$F \cdot d_1 \cdots d_n = \llbracket b_i \rrbracket_{\rho''}, \quad \rho'' := \rho[f \mapsto F, \vec{x} \mapsto \vec{d}, \vec{y}_i \mapsto [\vec{e}]].$$

Apply the *main* induction hypothesis to the branch derivation

$$\Gamma, f:T, \vec{x}:\vec{\sigma}, \vec{y}_i:\vec{A}_i, \vec{p}_i:\vec{\varphi}_i, \vec{q}_i:(\vec{v} =_{\vec{\sigma}'} \vec{t}_i) \vdash b_i : \tau_0[\vec{t}_i/\vec{z}, \mathbf{c}_i(\vec{y}_i; \vec{p}_i)/z]$$

with discipline $\mathcal{D}' = \mathcal{D} \cup \{(f, \vec{y}_i^{\text{ec}}, m_0)\}$ and valuation ρ'' . The premises hold: b_i respects $\mathcal{D}'$ (for f by the guard condition grd_k ; for the outer disciplines by Lemma 7.5); $\rho''(f) = F \in \llbracket T \rrbracket_\rho^{k, < m_0}$ by the rank induction hypothesis; $\text{rk}_I(\rho''(y)) < m_0$ for $y \in \vec{y}_i^{\text{ec}}$ as noted; the $\vec{x}$ - and $\vec{y}_i$ -declarations are satisfied by the chain condition and the decomposition; the premise proofs $\vec{p}_i$ by the decomposition; and the index equations $\vec{q}_i$ by $\llbracket \vec{v} \rrbracket_{\rho''} = \llbracket \vec{v} \rrbracket_{\rho_c} = \llbracket \vec{t}_i \rrbracket_{\rho_0} = \llbracket \vec{t}_i \rrbracket_{\rho''}$ ($\vec{v}$ mentions only $x_1, \dots, x_{k-1}$). The conclusion gives $\llbracket b_i \rrbracket_{\rho''} \in \llbracket \tau_0[\vec{t}_i/\vec{z}, \mathbf{c}_i(\vec{y}_i; \vec{p}_i)/z] \rrbracket_{\rho''}$; by semantic substitution this set is $\llbracket \tau_0 \rrbracket_{\rho''[\vec{z} \mapsto \llbracket \vec{t}_i \rrbracket_{\rho''}, \vec{z} \mapsto d_k]} = \llbracket \tau_0 \rrbracket_{\rho''[\vec{z} \mapsto \llbracket \vec{v} \rrbracket_{\rho''}, \vec{z} \mapsto \llbracket x_k \rrbracket_{\rho''}]} = \llbracket \tau_0[\vec{v}/\vec{z}, x_k/z] \rrbracket_{\rho''} = \llbracket \tau \rrbracket_{\rho[\vec{x} \mapsto \vec{d}]}$ (weakening: $f, \vec{y}_i$ do not occur in τ). This closes the rank induction, hence the fix case.

Refinement formers. **refine**(t, π): by the induction hypotheses, $\llbracket t \rrbracket_\rho \in \llbracket \sigma \rrbracket_\rho$ and $\langle \varphi[t/x] \rangle_\rho = 1$, i.e. $\langle \varphi \rangle_{\rho[x \mapsto \llbracket t \rrbracket_\rho]} = 1$ by semantic substitution; since $\mathcal{E}(\text{refine}(t, \pi)) = \mathcal{E}(t)$, this is membership in $\llbracket \{x:\sigma \mid \varphi\} \rrbracket_\rho$. **val**(t): membership in the refinement interpretation includes membership in $\llbracket \sigma \rrbracket_\rho$, and $\mathcal{E}(\text{val}(t)) = \mathcal{E}(t)$. For **valid**(t) in (2): the payload component of the same membership, rewritten by semantic substitution using $\llbracket \text{val}(t) \rrbracket_\rho = \llbracket t \rrbracket_\rho$.

Propositional singleton eliminations. **split**($\pi; p_1 p_2. t$): induction hypothesis (2) gives both conjuncts, so $\rho \models_{\mathcal{D}} \Gamma, p_1:\varphi_1, p_2:\varphi_2$; conclude by induction hypothesis (1) on t , the erasures agreeing. **cast**($\pi; z.\tau; t$): induction hypothesis (2) gives $\llbracket a \rrbracket_\rho = \llbracket b \rrbracket_\rho$, hence $\llbracket \tau[a/z] \rrbracket_\rho = \llbracket \tau \rrbracket_{\rho[z \mapsto \llbracket a \rrbracket_\rho]} = \llbracket \tau[b/z] \rrbracket_\rho$; conclude by induction hypothesis (1) on t . **efq**($\pi; \sigma$) (and **efq**($\pi; \varphi$) in (2)): vacuous — induction hypothesis (2) would give $\langle \perp \rangle_\rho = 1$.

Singleton elimination. Let $t = \text{scase}_J(\pi; \vec{z}.\tau; \vec{p}.t_0)$ with J a forced singleton (clause $k : \forall \vec{x}:\vec{\tau}. \varphi_1 \rightarrow \cdots \rightarrow \varphi_\ell \rightarrow J(\vec{u}^0)$, solution j), $\Gamma \vdash \pi : J(\vec{u})$, $\theta = [u_{j(i)}/x_i]_i$, and displayed type $\tau[\vec{u}/\vec{z}]$. By induction hypothesis (2), $\llbracket \vec{u} \rrbracket_\rho \in \langle J \rangle$; unfolding the least stage containing the tuple (the predicate analogue of Lemma 6.5(1), immediate from Definition 6.3 since J has a single clause): there is $\rho_0 = [\vec{x} \mapsto \vec{d}]$ with $d_i \in \llbracket \tau_i \rrbracket_{\rho_0}$, $\langle \varphi_l \rangle_{\rho_0} = 1$ for every l

(for a recursive premise $J(\vec{v})$, membership of $\llbracket \vec{v} \rrbracket_{\rho_0}$ in an earlier stage gives exactly $\langle J(\vec{v}) \rangle_{\rho_0} = 1$), and $\llbracket \vec{u} \rrbracket_{\rho} = \llbracket \vec{u}^0 \rrbracket_{\rho_0}$. By forcedness, for each i :

$$\llbracket \theta(x_i) \rrbracket_{\rho} = \llbracket u_{j(i)} \rrbracket_{\rho} = (\llbracket \vec{u}^0 \rrbracket_{\rho_0})_{j(i)} = \llbracket u_{j(i)}^0 \rrbracket_{\rho_0} = \llbracket x_i \rrbracket_{\rho_0} = d_i.$$

Hence by semantic substitution $\langle \varphi_l[\theta] \rangle_{\rho} = \langle \varphi_l \rangle_{\rho[\vec{x} \mapsto \vec{d}]} = \langle \varphi_l \rangle_{\rho_0} = 1$, so $\rho \models_{\mathcal{D}} \Gamma, \vec{p} : \theta(\vec{\varphi})$, and the induction hypothesis (1) on t_0 gives $\llbracket t_0 \rrbracket_{\rho} \in \llbracket \tau[\theta(\vec{u}^0)/\vec{z}] \rrbracket_{\rho} = \llbracket \tau \rrbracket_{\rho[\vec{z} \mapsto \llbracket \theta(\vec{u}^0) \rrbracket_{\rho}]}$. Again by the displayed computation, $\llbracket \theta(u_h^0) \rrbracket_{\rho} = \llbracket u_h^0 \rrbracket_{\rho_0} = \llbracket u_h \rrbracket_{\rho}$ for every h , so this set is $\llbracket \tau[\vec{u}/\vec{z}] \rrbracket_{\rho}$; and $\llbracket t \rrbracket_{\rho} = \llbracket t_0 \rrbracket_{\rho}$ since $\mathcal{E}(t) = \mathcal{E}(t_0)$.

Discrimination. For $\text{discr}(\pi)$ with $\Gamma \vdash \pi : c(\vec{a}; \cdot) =_{\sigma} d(\vec{b}; \cdot)$, $c \neq d$: induction hypothesis (2) would give $\llbracket c(\mathcal{E}(\vec{a})^{\rho}) \rrbracket = \llbracket d(\mathcal{E}(\vec{b})^{\rho}) \rrbracket$, contradicting Corollary 4.4(1); the case is vacuous — no $\rho \models_{\mathcal{D}} \Gamma$ reaches it.

Proof formers. The introduction and elimination rules for $\rightarrow$, $\wedge$, $\vee$, $\forall$, $\exists$, $\top$ and predicate quantification are sound for the truth-value semantics by direct computation: each rule matches the corresponding min/max clause of Definition 6.3; the instantiation rules use semantic substitution (parts (1) and (3) of Lemma 6.7); and the witnesses in **ecase** and predicate-quantifier elimination are semantic (an element $d \in \llbracket \sigma \rrbracket_{\rho}$, a function $\llbracket P \rrbracket_{\rho}$), requiring no syntactic counterpart. $\text{refl}_{\sigma}(a)$ is reflexivity of class equality; $\text{rew}(\pi; z.\psi; \pi')$ uses $\llbracket a \rrbracket_{\rho} = \llbracket b \rrbracket_{\rho}$ and semantic substitution exactly as $\text{cast}(\cdot; \cdot)$.

Predicate-constructor introduction. For $k(\vec{a}; \vec{\pi})$: by induction hypothesis (1) and iterated semantic substitution along the dependent telescope, each $\llbracket a_i \rrbracket_{\rho}$ lies in $\llbracket \tau_i \rrbracket_{\rho_0}$ where $\rho_0 = [\vec{x} \mapsto \llbracket \vec{a} \rrbracket_{\rho}]$; by induction hypothesis (2) the non-recursive premises are true at ρ_0 and each recursive premise $J(\vec{v})$ yields $\llbracket \vec{v} \rrbracket_{\rho_0} \in \langle J \rangle$, hence in some finite stage. The stage construction (at the maximum of those stages) puts $\llbracket \vec{u} \rrbracket_{\rho_0}$ in $\langle J \rangle$, and semantic substitution identifies it with $\llbracket \vec{u}[\vec{a}/\vec{x}] \rrbracket_{\rho}$.

Predicate induction. For $\text{ind}_J(\vec{z}.\psi; \vec{\pi}; \pi_0)$ with $\pi_0 : J(\vec{a})$: by induction hypothesis (2), $\llbracket \vec{a} \rrbracket_{\rho} \in \langle J \rangle$ and $\langle \Xi_j(\vec{z}.\psi) \rangle_{\rho} = 1$ for every clause j . We show by strong induction on s that every $\vec{d} \in \langle J \rangle^s$ satisfies $\langle \psi \rangle_{\rho[\vec{z} \mapsto \vec{d}]} = 1$: a tuple enters $\langle J \rangle^s$ (with $s = s' + 1$) through some clause j with witnesses $\vec{d}' \in \llbracket \vec{\tau}_j \rrbracket$ -telescope, true non-recursive premises, and recursive-premise tuples in $\langle J \rangle^{s'}$; instantiating the semantic universal quantifiers and implications of $\langle \Xi_j \rangle_{\rho} = 1$ at these witnesses — the inductive-hypothesis premises supplied by the inner induction at s' — yields $\langle \psi[\vec{u}_j/\vec{z}] \rangle_{\rho[\vec{x} \mapsto \vec{d}]} = 1$, which by semantic substitution is $\langle \psi \rangle_{\rho[\vec{z} \mapsto \vec{d}]} = 1$. Conclude at $\llbracket \vec{a} \rrbracket_{\rho}$.

Datatype induction. Let $\pi = \text{ind}_I(\vec{z}.z.\psi; (\vec{y}_i \vec{p}_i \vec{h}_i.\pi_i)_i; a)$ with $\Gamma \vdash a : I(\vec{u})$ and the clause derivations of Figure 5. By induction hypothesis (1), $(\llbracket \vec{u} \rrbracket_{\rho}, \llbracket a \rrbracket_{\rho}) \in \llbracket I \rrbracket$. We show by strong induction on $\text{rk}_I(d)$ that every $(\vec{d}, d) \in \llbracket I \rrbracket$ satisfies $\langle \psi \rangle_{\rho[\vec{z} \mapsto \vec{d}, z \mapsto d]} = 1$. Decompose $(\vec{d}, d)$ by Lemma 6.5(1) into $i, [\vec{e}], \rho_0$; for each recursive j , $(\llbracket \vec{v}_{ij} \rrbracket_{\rho_0}, [e_j]) \in \llbracket I \rrbracket$ with $\text{rk}_I([e_j]) < \text{rk}_I(d)$

(Lemma 6.5(4)), so the inner induction gives $\langle \psi \rangle_{\rho[\vec{z} \mapsto \llbracket \vec{v}_{ij} \rrbracket_{\rho_0}, z \mapsto [e_j]]} = 1$, i.e., by semantic substitution and weakening, $\langle \psi[\vec{v}_{ij}/\vec{z}, y_{ij}/z] \rangle_{\rho'} = 1$ at $\rho' := \rho[\vec{y}_i \mapsto [\vec{e}]]$. Together with the memberships and premise truths of the decomposition, $\rho' \models_{\mathcal{D}}$ the clause context, so the main induction hypothesis (2) applied to π_i gives $\langle \psi[\vec{t}_i/\vec{z}, c_i(\vec{y}_i; \vec{p}_i)/z] \rangle_{\rho'} = 1$, which by semantic substitution is $\langle \psi \rangle_{\rho[\vec{z} \mapsto \vec{d}, z \mapsto d]} = 1$ (using $\llbracket \vec{t}_i \rrbracket_{\rho_0} = \vec{d}$ and $\llbracket c_i(\vec{y}_i; \vec{p}_i) \rrbracket_{\rho'} = d$). Conclude at $(\llbracket \vec{u} \rrbracket_{\rho}, \llbracket a \rrbracket_{\rho})$. $\square$

7.3 Validity of the emitted theory

Lemma 7.7 (Validity of specification axioms). *Let $c := t : \tau$ be a term definition with $\tau \in \mathfrak{S}$, and suppose $[c] \in \llbracket \tau \rrbracket$. Then $\mathfrak{M} \models \mathcal{S}(c)$.*

Proof. By induction on the type τ' in the recursion of Definition 5.8, with the invariant: if ρ assigns the accumulated variables, $v := [c \hat{e}_1 \cdots \hat{e}_j]$ for representatives of the accumulated values, and $v \in \llbracket \tau' \rrbracket_{\rho}$, then $\mathfrak{M}, \rho \models S(u; \tau')$. The Π -case extends ρ under a guard, which Lemma 6.13 converts to a membership; the premise case uses Lemma 6.13 for $\mathcal{F}(\varphi)$ and leaves the accumulator unchanged; the base case (datatype occurrence or refinement type) is Lemma 6.13(1) — for a family occurrence this includes the index arguments of the final guard atom, whose denotations are the interpretations of the index terms by Lemma 6.11. $\square$

Theorem 7.8 (Environment validity). *Let $\mathbb{E}$ be a well-formed definitional environment with $\mathfrak{S}$ declarations, and G a closed shallow goal. Then:*

1. (H) holds;
2. $\mathfrak{M} \models \Delta(\mathbb{E}, G)$.

Proof. (1) By induction on the position of the definition in $\mathbb{E}$: the body of $c := t : \tau_c$ is typed over the strict prefix, for which (H) holds by induction; Lemma 7.6 with empty context, discipline and valuation gives $\llbracket t \rrbracket \in \llbracket \tau_c \rrbracket$, and $[c] = \llbracket t \rrbracket$ by the δ -rule. Lemmas identically via part (2) of the fundamental lemma.

(2) We check each axiom family of Definition 5.10.

Compilation, pointer and auxiliary axioms: Lemma 6.12.

Specification axioms: Lemma 7.7, using (1).

Premise axioms: (1) and Lemma 6.13 at the empty valuation.

Constructor typing (datatypes): let $\vec{d}$ be an assignment of $\vec{y}$ with $\mathfrak{M} \models \bigwedge_j \mathcal{G}_{A_{ij}}(y_j) \wedge \bigwedge_l \mathcal{F}(\varphi_{il})$ at $\vec{d}$. By Lemma 6.13, $d_j \in \llbracket A_{ij} \rrbracket_{\rho_0}$ for every j and $\langle \varphi_{il} \rangle_{\rho_0} = 1$ for every l , where $\rho_0 = [\vec{y} \mapsto \vec{d}]$ (for recursive j the guard is the atom $\widehat{I}(y_j, \mathcal{C}(\mathcal{E}(\vec{v}_{ij})))$ and coincidence yields $(\llbracket \vec{v}_{ij} \rrbracket_{\rho_0}, d_j) \in \llbracket I \rrbracket$). By Lemma 6.5(5), $(\llbracket \vec{t}_i \rrbracket_{\rho_0}, \widehat{\mathcal{C}}_i^{\mathfrak{M}}(\vec{d})) \in \llbracket I \rrbracket$, and by Lemma 6.11 the index arguments of the conclusion atom denote exactly $\llbracket \vec{t}_i \rrbracket_{\rho_0}$; hence the atom holds.

Inversion (datatypes): let $(d, \vec{d}) \in \hat{I}^{\mathfrak{M}}$, i.e. $(\vec{d}, d) \in \llbracket I \rrbracket$. Lemma 6.5(1) yields i , $[\vec{e}]$, ρ_0 ; instantiate the existentials of the i -th disjunct with $[\vec{e}]$. The guards and premise conjuncts hold by Lemma 6.13 (memberships and truths of the decomposition); $w \approx \hat{c}_i(\vec{y})$ holds by the interpretation of $\hat{c}_i$; and each index equation $z_h \approx \mathcal{C}(\mathcal{E}(t_{ih}))$ holds because $d_h = (\llbracket \vec{t}_i \rrbracket_{\rho_0})_h$ and Lemma 6.11 identifies the right-hand side's denotation at the chosen witnesses with $\llbracket t_{ih} \rrbracket_{\rho_0}$.

Injectivity and discrimination: Corollary 4.4(1) — at all elements of $\mathcal{D}$, with no guard.

Predicate clause and case-analysis axioms: as constructor typing and inversion one level down, exactly as in [Ext26], using the stage construction of $\langle J \rangle$, Lemma 6.13 and Lemma 6.11. $\square$

7.4 Main theorem and corollaries

Theorem 7.9 (Soundness). *Let $\mathbb{E}$ be a well-formed definitional environment with $\mathfrak{S}$ declarations and G a closed shallow proposition over $\mathbb{E}$. If $\Delta(\mathbb{E}, G) \vdash_{\text{FOL}} \mathcal{F}(G)$ then $\langle G \rangle = 1$: G is true in the proof-irrelevant classical semantics of $\mathbb{E}$.*

Proof. $\mathfrak{M} \models \Delta(\mathbb{E}, G)$ (Theorem 7.8); by soundness of classical first-order logic, $\mathfrak{M} \models \mathcal{F}(G)$; by Lemma 6.13 at the empty valuation, $\langle G \rangle = 1$. $\square$

Corollary 7.10 (Consistency). *$\Delta(\mathbb{E}, G)$ is consistent, for every well-formed definitional $\mathbb{E}$ and goal G . In particular no combination of the emitted axiom families — across all declared families, instances and definitions — can derive $\perp$.*

Proof. $\mathfrak{M} \models \Delta(\mathbb{E}, G)$ and $\mathfrak{M} \not\models \perp$. No consistency assumption on the source is needed: a definitional environment always has the model $\mathfrak{M}$. $\square$

Corollary 7.11 (Refutable goals are not provable). *If $\vdash \pi : \neg G$ for some closed proof over $\mathbb{E}$, then $\Delta(\mathbb{E}, G) \not\vdash_{\text{FOL}} \mathcal{F}(G)$; more generally $\Delta(\mathbb{E}, G)$ never proves both $\mathcal{F}(G)$ and $\mathcal{F}(\neg G)$.*

Proof. By Lemma 7.6 (via Theorem 7.8(1)), $\langle \neg G \rangle = 1$, so $\langle G \rangle = 0$ and Theorem 7.9 applies contrapositively; for the second statement, $\mathfrak{M}$ satisfies every consequence of $\Delta(\mathbb{E}, G)$ and $\mathcal{F}(\neg G) = \mathcal{F}(G) \rightarrow \perp$. $\square$

Proposition 7.12 (Standardness on purely structural families). *Call a datatype I purely structural if its constructors have no premises and every non-recursive constructor argument type is an occurrence of a purely structural datatype declared earlier. Then:*

1. *For every purely structural I and $(\vec{d}, d) \in \llbracket I \rrbracket$, d is the class of a closed term built exclusively from constructors of purely structural datatypes; and two such closed constructor terms have the same class only if they*

carry the same constructor at the root and componentwise equal-class arguments, so each class determines its constructor tree over classes uniquely.

2. For $\mathbf{vec}$ over a purely structural element type A , the map

$$(a_1, \dots, a_n) \mapsto \left([S^n(\mathbf{O})], [\mathbf{vcons}(S^{n-1}(\mathbf{O}), \hat{a}_1, \mathbf{vcons}(S^{n-2}(\mathbf{O}), \hat{a}_2, \dots, \mathbf{vnil} \dots))] \right)$$

(for representatives $\hat{a}_i$) is a bijection from the set of finite sequences over $\llbracket A \rrbracket$ to $\llbracket \mathbf{vec} \rrbracket$; in particular $\llbracket \mathbf{vec}(S^n(\mathbf{O})) \rrbracket$ is in bijection with $\llbracket A \rrbracket^n$, and $n \mapsto [S^n(\mathbf{O})]$ is a bijection $\mathbb{N} \rightarrow \llbracket \mathbf{nat} \rrbracket$.

Consequently, for propositions whose quantifiers range over purely structural datatypes, $\langle \cdot \rangle$ is satisfaction in the corresponding standard structure of index-respecting finite trees.

Proof. (1) By induction on declaration order and, within a datatype, on $\text{rk}_I(d)$: Lemma 6.5(1) decomposes d as $[c_i(\vec{e})]$ with non-recursive components in interpretations of earlier purely structural datatypes (induction on declaration order) and recursive components of smaller rank (induction on rank); premises are absent. Distinctness is Corollary 4.4(1) by induction on the terms.

(2) Write Φ for the displayed map. *Well-definedness and injectivity:* the image element is a constructor term; injectivity of Φ follows from Corollary 4.4(1) by induction on sequence length, together with the injectivity of $n \mapsto [S^n(\mathbf{O})]$ (also Corollary 4.4(1)). *Image in $\llbracket \mathbf{vec} \rrbracket$:* by induction on n using Lemma 6.5(5) and, for the index computation, $\llbracket S n \rrbracket_{[n \mapsto [S^{n-1}(\mathbf{O})]]} = [S^n(\mathbf{O})]$.

Surjectivity: let $(\vec{d}, d) \in \llbracket \mathbf{vec} \rrbracket$ and induct on $\text{rk}_{\mathbf{vec}}(d)$. If the decomposition is by $\mathbf{vnil}$: $\vec{d} = ([\mathbf{O}]) = ([\mathbf{O}])$ and $d = [\mathbf{vnil}] = \Phi(\varepsilon)$. If by $\mathbf{vcons}$ with components $[e_n], [e_a], [e_w]$: the recursive condition gives $([e_n], [e_w]) \in \llbracket \mathbf{vec} \rrbracket$ (the index tuple of the recursive argument is $(\llbracket n \rrbracket_{\rho_0}) = ([e_n])$), so by the inner induction $[e_w] = \Phi(a_2, \dots, a_n)$ -image with $[e_n] = [S^{n-1}(\mathbf{O})]$; moreover $[e_a] \in \llbracket A \rrbracket$, and $\vec{d} = (\llbracket S n \rrbracket_{\rho_0}) = ([S^n(\mathbf{O})])$; hence $(\vec{d}, d) = \Phi([e_a], a_2, \dots, a_n)$. The bijection $\mathbb{N} \cong \llbracket \mathbf{nat} \rrbracket$ is the $m = 0$ instance, as in [Ext26]. The final claim unfolds Definition 6.3 clause by clause under the bijections. $\square$

Remark 7.13 (Postulated axioms). If $\mathbb{E}$ additionally contains postulates $c : \varphi$ (or $c : \sigma$) without bodies, all results relativise as in [Ext26]: Theorem 7.8 goes through provided each postulate is true in the semantics ($\langle \varphi \rangle = 1$; for a postulated constant, $[c] \in \llbracket \sigma \rrbracket$ under the evident extension of $\mathfrak{M}$). Excluded middle and proof irrelevance are true; functional extensionality is *not* (Remark 8.19); a postulated inhabitant of an empty instance ($c : \text{fin}(\mathbf{O})$) is not, and voids the guarantee — which is the correct account of the corresponding SMT-datatype hazard (Example 5.12): the danger is never the empty instance itself but an axiom asserting it inhabited.

Remark 7.14 (What is not claimed). Theorem 7.9 does not claim G is provable in CIC_{ix}^- : the semantics validates excluded middle, proof irrelevance and (as spent in the `scase`(; ;) case) uniqueness of identity proofs, so $\mathcal{F}(G)$ may be provable for G true-but-unprovable. This matches the design of the implemented tool, which runs classical provers and relies on reconstruction. Section 8.5 localises the UIP debt precisely.

8 Fording at the level of formulas: design questions settled

The model $\mathfrak{M}$ was built once; this section spends it. Each subsection takes a design question documented in implementations of related translations — where may the forced inversion disjunction be inlined? which guards are load-bearing? do match equations need index premises? what may be said about indices without saying it about type formers? — and settles it as a theorem about $\mathfrak{M}$ or about the theory $\Delta(\mathbb{E}, G)$. Throughout, $\mathbb{E}$ is well formed and definitional with $\mathfrak{S}$ declarations.

8.1 Per-occurrence expansion of forced guards

The translation of Section 5 keeps guard atoms atomic and states the ford once per family, in the inversion axiom. A shallower variant — the analogue, for families, of the inline expansion of refinement payloads — replaces a guard atom, at any chosen occurrence, by the corresponding disjunction of constructor payloads and index equations. This is sound, in both polarities, and changes nothing logically:

Definition 8.1 (One-level expansion). For a datatype I as in Definition 5.10, define the first-order formula (with free variables $w, \vec{z}$)

$$\mathcal{X}_I(w, \vec{z}) := \bigvee_i \exists \vec{y}. \bigwedge_j \mathcal{G}_{A_{ij}}(y_j) \wedge \bigwedge_l \mathcal{F}(\varphi_{il}) \wedge w \approx \widehat{c}_i(\vec{y}) \wedge \bigwedge_h z_h \approx \mathcal{C}(\mathcal{E}(t_{ih})),$$

i.e. the right-hand side of the inversion axiom; for an inductive predicate J , let $\mathcal{X}_J(\vec{z})$ be the right-hand side of its case-analysis axiom. An *expansion variant* of the translation is any pair $(\mathcal{G}', \mathcal{F}')$ obtained from Definition 5.7 by replacing, at any chosen set of occurrences in the structural recursion, the clause value $\widehat{I}(u, \mathcal{C}(\mathcal{E}(\vec{a})))$ by $\mathcal{X}_I(u, \mathcal{C}(\mathcal{E}(\vec{a})))$ and/or atoms $\widehat{J}(\mathcal{C}(\mathcal{E}(\vec{a})))$ by $\mathcal{X}_J(\mathcal{C}(\mathcal{E}(\vec{a})))$ (arguments substituted for $w, \vec{z}$). The variant theory $\Delta'(\mathbb{E}, G)$ is Definition 5.10 with items 3–5 (specification, premise and goal formulas and their auxiliary axioms) computed by $(\mathcal{G}', \mathcal{F}')$; items 1–2 (structural axioms) are unchanged.

Note that the expansion is *one level*: the guards $\mathcal{G}_{A_{ij}}(y_j)$ inside $\mathcal{X}_I$ are the atomic guards of the base translation, so the definition is not recursive

and expanded formulas are finite regardless of recursion in I . Constraint discharge remains pay-as-you-go, one constructor layer per expansion, exactly as in the source calculus (Section 2.2).

Lemma 8.2 (Expansion is valid). $\mathfrak{M} \models \forall w \vec{z}. \widehat{I}(w, \vec{z}) \leftrightarrow \mathcal{X}_I(w, \vec{z})$, and $\mathfrak{M} \models \forall \vec{z}. \widehat{J}(\vec{z}) \leftrightarrow \mathcal{X}_J(\vec{z})$.

Proof. Left to right is the validity of inversion resp. case analysis (Theorem 7.8). Right to left, for I : given a disjunct's witnesses $\vec{d}$ satisfying its conjuncts, Lemma 6.13 converts the guard and premise conjuncts into memberships and truths at $\rho_0 = [\vec{y} \mapsto \vec{d}]$; Lemma 6.5(5) gives $([\vec{t}_i]_{\rho_0}, \widehat{c}_i^{\mathfrak{M}}(\vec{d})) \in \llbracket I \rrbracket$; and the equation conjuncts identify $w^{\mathfrak{M}}$ with the constructor value and $\vec{z}^{\mathfrak{M}}$ with the index tuple (using Lemma 6.11 for the right-hand sides). For J : the stage construction closes $\langle J \rangle$ under exactly the displayed conditions. $\square$

Theorem 8.3 (Soundness of expansion variants). *Let $(\mathcal{G}', \mathcal{F}')$ be any expansion variant. Then Lemma 6.13 holds verbatim for $(\mathcal{G}', \mathcal{F}')$; consequently $\mathfrak{M} \models \Delta'(\mathbb{E}, G)$, and if $\Delta'(\mathbb{E}, G) \vdash_{\text{FOL}} \mathcal{F}'(G)$ then $\langle G \rangle = 1$. All corollaries of Section 7.4 hold for the variant.*

Proof. The coincidence proof is by structural induction on types and propositions; the only change is at a replaced occurrence, where $\mathfrak{M}, \rho \models \mathcal{X}_I(u, \mathcal{C}(\mathcal{E}(\vec{a})))$ iff $\mathfrak{M}, \rho \models \widehat{I}(u, \mathcal{C}(\mathcal{E}(\vec{a})))$ by Lemma 8.2 (a universally valid equivalence may be instantiated at the argument terms), and the base-translation coincidence handles the atom. The auxiliary axioms emitted by computing the $\mathcal{C}$ -translations inside expanded occurrences are compilation axioms, valid by Lemma 6.12. With coincidence in place, the proofs of Lemma 7.7, Theorem 7.8 and Theorem 7.9 go through unchanged for items 3–5, and items 1–2 are shared with $\Delta(\mathbb{E}, G)$. $\square$

Proposition 8.4 (Expansion changes nothing logically). $\Delta(\mathbb{E}, G) \vdash_{\text{FOL}} \forall w \vec{z}. \widehat{I}(w, \vec{z}) \leftrightarrow \mathcal{X}_I(w, \vec{z})$ (from the constructor-typing and inversion axioms of I), and likewise for J from clause and case-analysis axioms. Consequently $\Delta(\mathbb{E}, G)$ and any $\Delta'(\mathbb{E}, G)$ are mutually derivable, and the choice among expansion variants is purely operational.

Proof. Left to right is the inversion axiom. Right to left: in the i -th disjunct, the witnesses' guards and premises feed the constructor-typing axiom of c_i , yielding $\widehat{I}(\widehat{c}_i(\vec{y}), \mathcal{C}(\mathcal{E}(\vec{t}_i)))$; rewriting with $w \approx \widehat{c}_i(\vec{y})$ and $z_h \approx \mathcal{C}(\mathcal{E}(t_{ih}))$ gives $\widehat{I}(w, \vec{z})$. Mutual derivability of the theories: each axiom of Δ' differs from the corresponding axiom of $\Delta(\mathbb{E}, G)$ by replacements of subformulas by $\Delta(\mathbb{E}, G)$ -provably equivalent ones (both theories contain items 1–2), so each direction follows by the replacement theorem of classical first-order logic. $\square$

Example 8.5 (The reflect payoff). For rfl_{φ_0} (Example 5.11), a hypothesis occurrence $r : \text{rfl}_{\varphi_0}(b)$ inside a premise or goal expands to

$$(\mathcal{F}(\varphi_0) \wedge r \approx \widehat{\text{RT}} \wedge b \approx \widehat{\text{true}}) \vee (\mathcal{F}(\neg\varphi_0) \wedge r \approx \widehat{\text{RF}} \wedge b \approx \widehat{\text{false}}).$$

Both directions of the view are now local: from $b \approx \widehat{\text{true}}$ the second disjunct dies by discrimination on `bool`, leaving $\mathcal{F}(\varphi_0)$ (elimination is no longer too weak); and in a positive position, establishing the instance at tag `RF` together with $b \approx \widehat{\text{true}}$ is impossible for the same reason (introduction cannot cheat the index). In the atomic variant the same two inferences route through the declaration-level inversion axiom — one resolution step more, contingent on premise selection. Which variant is operationally better is an empirical question; the content of Theorem 8.3 and Proposition 8.4 is that it is *only* an operational question.

Remark 8.6 (Existential residue). Expansion introduces $\exists \vec{y}$ in guard positions; in negative positions these become universals after clausification. For enumeration-like and subset-like families the residue is empty or a single carrier variable after Lemma 5.14 (`rfl`: none; `bnd`: one), which is exactly the class of families for which implementations enable expansion; for recursive families like `vec` the atomic guard is the reasonable default. Nothing in the soundness statement forces either choice.

8.2 Which guards are load-bearing

The companion paper proved that for non-indexed datatypes the guard of the inversion axiom is semantically mandatory (false at junk) while the per-constructor equations need no guards at all. For families both halves sharpen.

Proposition 8.7 (Unguarded inversion is inconsistent). *Let I be a datatype some constructor c_i of which has a constructor-headed index term: $t_{ih} = c'(\dots)$ for some position h , where c' belongs to a datatype I' that has a second constructor $d \neq c'$. Let Inv_I^- be the inversion axiom of I with its guard atom $\widehat{I}(w, \vec{z})$ deleted. Then $\{\text{Inv}_I^-\}$ together with the discrimination axioms of I and of I' — axioms the translation must emit in any case — is inconsistent. In particular, for `vec` (Example 5.11) the theory $\{\text{Inv}_{\text{vec}}^-\} \cup \{\text{discrimination for nat, vec}\}$ is inconsistent.*

Proof. Instantiate the universally quantified variables of Inv_I^- by $w := \widehat{c}_i(\vec{x})$, $z_h := \widehat{d}(\vec{x}')$, the remaining z 's by fresh variables ($\vec{x}, \vec{x}'$ fresh). In the resulting disjunction, every disjunct $i' \neq i$ contains the equation $\widehat{c}_i(\vec{x}) \approx \widehat{c}_{i'}(\vec{y})$, refuted by discrimination for I ; the disjunct i contains $\widehat{d}(\vec{x}') \approx \mathcal{C}(\mathcal{E}(t_{ih}))[\vec{y}]$, whose right-hand side is $\widehat{c}'$ -headed ($\mathcal{C}(\mathcal{E}(c'(\dots)))$ is an application of c' by Definition 5.2), refuted by discrimination for I' . All disjuncts refuted, the theory proves $\perp$. For `vec`, concretely: $w := \widehat{\text{vnil}}$, $z := \widehat{S}(x')$ — the `vnil` disjunct dies on $\widehat{S}(x') \approx \widehat{O}$, the `vcons` disjunct on $\widehat{\text{vnil}} \approx \widehat{\text{vcons}}(n, a, v)$. $\square$

Remark 8.8. The hypothesis delimits the blast radius precisely. Rigid index patterns (the proposition) cover `vec`, `fin` and every family whose constructors specialise an index. Enumeration-shaped families with bare-variable or

absent index constraints are caught by the *cardinality* mechanism instead: for rfl_{φ_0} , the unguarded inversion entails $\forall w. w \approx \widehat{\text{RT}} \vee w \approx \widehat{\text{RF}}$ (each disjunct contains the corresponding tag equation as a conjunct), a two-element bound on the whole untyped universe; in any environment that also declares nat — or any datatype with three provably distinct closed terms — instantiating w at $\widehat{\text{O}}$, $\widehat{\text{S}}(\widehat{\text{O}})$ and $\widehat{\text{S}}(\widehat{\text{S}}(\widehat{\text{O}}))$ — pairwise distinct by discrimination and injectivity of nat — forces two of them equal by pigeonhole, hence $\perp$. This is Meng and Paulson’s two-element universe [MP08, §2.3]; exhaustiveness sentences are exactly the sentences whose guards can never be dropped [BBPS16]. For a fully generic family (every index a bare variable, constructors carrying arguments) the unguarded axiom is false in $\mathfrak{M}$ — any junk instantiation of w refutes it there — but need not be inconsistent; soundness fails all the same, via Lemma 6.13. Note the strengthening against the non-indexed case of [Ext26]: there, guard omission made the disjunctive axiom false in $\mathfrak{M}$, and inconsistency required an auxiliary single-constructor datatype; here the family’s own index equations or tag equations detonate it. For the implementation this means: an indexed inversion axiom accidentally emitted unguarded is typically not a weak axiom but a refutation-generator, poisoning every problem it enters (cf. Corollary 7.10).

Proposition 8.9 (Index arguments of guards are mandatory). *Consider the index-forgetting variant of the translation: guard predicates $\widehat{I}^-$ of arity 1, guard clause $\mathcal{G}_{I(\vec{a})}^-(u) = \widehat{I}^-(u)$, and the axioms of Definition 5.10 recomputed accordingly (index equations deleted from inversion, index arguments deleted from constructor-typing conclusions). This variant is unsound: for $\mathbb{E}$ declaring nat and vec (over $A = \text{nat}$) there is a closed shallow goal G with $\langle G \rangle = 0$ whose translation is derivable,*

$$G := \exists v : \text{vec}(\text{O}). \neg(v =_{\text{vec}(\text{O})} \text{vnil}),$$

$$\Delta^-(\mathbb{E}, G) \vdash_{\text{FOL}} \exists v. \widehat{\text{vec}}^-(v) \wedge \neg(v \approx \widehat{\text{vnil}}).$$

Proof. $\langle G \rangle = 0$: by Lemma 6.5(1) and Corollary 4.4, every $([\text{O}], d) \in \llbracket \text{vec} \rrbracket$ decomposes by vnil — the vcons index tuple is a class $[\text{S}(e)] \neq [\text{O}]$ — so $\llbracket \text{vec}(\text{O}) \rrbracket = \{\{\text{vnil}\}\}$ and the matrix is false at the only element.

Derivability: the index-forgetting constructor-typing axioms are $\widehat{\text{vec}}^-(\widehat{\text{vnil}})$ and $\forall n a v. \widehat{\text{nat}}(n) \rightarrow \widehat{\text{nat}}(a) \rightarrow \widehat{\text{vec}}^-(v) \rightarrow \widehat{\text{vec}}^-(\widehat{\text{vcons}}(n, a, v))$. From $\widehat{\text{nat}}(\widehat{\text{O}})$ (constructor typing for nat) instantiate $n = a := \widehat{\text{O}}$, $v := \widehat{\text{vnil}}$ to get $\widehat{\text{vec}}^-(\widehat{\text{vcons}}(\widehat{\text{O}}, \widehat{\text{O}}, \widehat{\text{vnil}}))$; discrimination for vec gives $\neg(\widehat{\text{vcons}}(\dots) \approx \widehat{\text{vnil}})$; the displayed existential follows with witness $\widehat{\text{vcons}}(\widehat{\text{O}}, \widehat{\text{O}}, \widehat{\text{vnil}})$. $\square$

Remark 8.10 (The F^* #1542 shape). Proposition 8.9 is the indexed-family form of a documented unsoundness. In F^* #1542 [Ase18], the type-indexed extensional equality $\text{feq} \# a$ — “equal pointwise at domain a ” — was encoded through the SMT solver’s untyped $=$; an equality whose *meaning depends on*

an *index* was represented by an atom that forgot the index, and facts established at index `nat` were consumed at index `int`, proving `False`. Here, family membership — whose meaning depends on the index — is represented in the variant by an atom $\widehat{\text{vec}}^-(v)$ that forgets it, and membership established at index `S(O)` is consumed at index `O`. The polarity analysis explains where the damage occurs: forgetting indices in *negative* (premise) positions merely strengthens hypotheses and weakens the translation; in *positive* (conclusion, existential-witness) positions it lets the prover manufacture members of the wrong instance, as in the derivation above. Since one guard clause serves both polarities, the indices must stay. The translation of Section 5 is immune for a reason worth stating precisely: it is not that its untyped $\approx$ is somehow safer than F^* 's — it is that *no index-dependent notion of the source is represented by an index-blind notion of the target*. Source equality $a =_\sigma b$ is index-blind (σ is not consulted) because its *semantics* in $\mathfrak{M}$ is index-blind class equality (Definition 6.3), and family membership is index-aware because its semantics is (Lemma 6.13); each atom is exactly as discriminating as the semantic notion it names. The residual caveat is the intensionality of that semantics: see Remark 8.19.

8.3 Match equations for indexed scrutinees

Proposition 8.11 (Unguarded equations, impossible branches included). *Every equation emitted by the compilation of a definition matching on a family — with no guard on the matched term, no guards on the constructor-argument variables, and no index premises, and including the equations of branches whose index patterns clash with the scrutinee occurrence's indices — holds in $\mathfrak{M}$ under every assignment. In particular (Example 5.6):*

$$\begin{aligned}\mathfrak{M} \models \forall n. \widehat{\text{vhead}}(n, \widehat{\text{vnil}}) &\approx \odot, \\ \mathfrak{M} \models \forall n k a w. \widehat{\text{vhead}}(n, \widehat{\text{vcons}}(k, a, w)) &\approx a.\end{aligned}$$

Proof. An instance of Lemma 6.12(2). The point of displaying it: the ι -rule of $\lambda_\square$ fires on any constructor form with no index side condition, so validity of the equations is a fact about the *program*, blind to the typing discipline that made the branch impossible; the branch body is the erasure of the $\perp$ -elimination, namely $\square$. $\square$

Proposition 8.12 (Index-premised equations are strictly weaker). *Fix the definition `vhead` and let the premised variant of its `vnil`-equation be*

$$\text{Eq}_{\text{vnil}}^\pi := \forall n. \widehat{S}(n) \approx \widehat{O} \rightarrow \widehat{\text{vhead}}(n, \widehat{\text{vnil}}) \approx \odot,$$

and analogously $\text{Eq}_{\text{vcons}}^\pi = \forall n k a w. \widehat{S}(n) \approx \widehat{S}(k) \rightarrow \widehat{\text{vhead}}(n, \widehat{\text{vcons}}(k, a, w)) \approx a$ (premises: occurrence indices $\approx$ constructor index pattern). Then:

1. each premised equation is derivable from its unguarded form, hence valid in $\mathfrak{M}$;
2. the unguarded **vnil**-equation is not derivable from the theory obtained from $\Delta(\mathbb{E}, G)$ by replacing **vhead**'s equations with their premised forms;
3. deleting the **vnil**-equation altogether (rigid-clash pruning) preserves Theorem 7.8, hence Theorem 7.9 and Corollary 7.10.

Proof. (1) Weakening. (3) Validity of a subset of a valid axiom set.

(2) Let $\mathbb{E}_0$ declare **nat**, $A := \text{nat}$, **vec** and **vhead** only, and let Δ^π be the resulting theory with **vhead**'s two equations replaced by their premised forms. We build $\mathfrak{M}' \models \Delta^\pi$ falsifying the unguarded **vnil**-equation. A pointwise reinterpretation of **vhead** is not available: the pointer axiom $\forall z_1 z_2. \widehat{\text{vhead}}^0 @ z_1 @ z_2 \approx \widehat{\text{vhead}}(z_1, z_2)$ ties the symbol to the applicative structure of $\mathcal{D}$, which is fixed. Instead, reinterpret both symbols through an *alternative witness*: let

$$\kappa := \lambda n v. \text{case}(v; \{\text{vnil} \Rightarrow \text{O} \mid \text{vcons}(k, a, w) \Rightarrow a\})$$

— the erasure of **vhead** with the dead code of the impossible branch replaced by **O** — and let $\mathfrak{M}'$ agree with $\mathfrak{M}$ on all symbols except $(\widehat{\text{vhead}}^0)^{\mathfrak{M}'} := [\kappa]$ and $\widehat{\text{vhead}}^{\mathfrak{M}'}(d_1, d_2) := [\kappa e_1 e_2]$ (representatives $e_i \in d_i$). Then: the pointer axiom holds by construction (**@** is application); the premised **vnil**-equation holds vacuously, its premise $\widehat{\text{S}}(n) \approx \widehat{\text{O}}$ being false (Corollary 4.4; constructor interpretations are unchanged); the premised **vcons**-equation holds because $\kappa e_1 \text{vcons}(\vec{e}) \rightarrow_{\square}^* e_a$, and this is where κ agrees with $\mathcal{E}(\text{vhead})$; the specification axiom $\forall n. \widehat{\text{nat}}(n) \rightarrow \forall v. \widehat{\text{vec}}(v, \widehat{\text{S}}(n)) \rightarrow \widehat{\text{nat}}(\widehat{\text{vhead}}(n, v))$ constrains the symbol only at $v \in \llbracket \text{vec}(\text{S } n) \rrbracket$ -instances, which by Lemma 6.5(1,3) are **vcons**-forms (**[vnil]**'s unique index tuple is **[O]**), where $\mathfrak{M}'$ computes the same values as $\mathfrak{M}$, so the axiom transfers from Theorem 7.8; and all axioms not mentioning **vhead**'s symbols hold as in $\mathfrak{M}$. Finally the unguarded **vnil**-equation fails: $\widehat{\text{vhead}}^{\mathfrak{M}'}(d, [\text{vnil}]) = [\kappa e \text{vnil}] = [\text{O}] \neq [\square] = \odot^{\mathfrak{M}'}$, the last two being distinct $\rightarrow_{\square}$ -normal forms (Theorem 4.3).

The moral generalises: the premised theory cannot distinguish the extracted program from any *repair of its dead code*, because every axiom that could see the dead branch is either premised on a refutable index equation or guarded by an uninhabited instance. $\square$

Proposition 8.13 (Pruned branches are semantically dead). *Say two terms rigidly clash if both are constructor-headed with distinct constructors, or both are applications of the same constructor and some pair of corresponding informative arguments rigidly clashes; extend to index tuples by “some position clashes”. If, in a definition $c := \lambda \vec{x}. \text{case}(x_k; \dots; \dots)$ (or the fix form) with*

$x_k : I(\vec{v})$, the occurrence indices $\vec{v}$ rigidly clash with the index pattern $\vec{t}_i$ of constructor c_i , then for every valuation ρ of $\vec{x}$ with $\rho(x_k) \in \llbracket I(\vec{v}) \rrbracket_\rho$, the value $\rho(x_k)$ is not of the form $[c_i(\vec{e})]$. Consequently the pruned equation constrains $\hat{c}$ only at argument tuples excluded by the guards of c 's specification axiom.

Proof. First, rigid clash implies semantic distinctness at all valuations: by induction on the definition, if u and t rigidly clash then $\llbracket u \rrbracket_{\rho_1} \neq \llbracket t \rrbracket_{\rho_2}$ for all ρ_1, ρ_2 — distinct constructor heads give distinct classes (Corollary 4.4(1)), and for equal heads, componentwise equality of classes would contradict the inner clash by the induction hypothesis (erasure preserves constructor heads of informative arguments). Now suppose $\rho(x_k) = [c_i(\vec{e})] \in \llbracket I(\vec{v}) \rrbracket_\rho$. By Lemma 6.5(1,3) the unique index tuple of this element is $\llbracket \vec{t}_i \rrbracket_{\rho_0}$, so $\llbracket \vec{v} \rrbracket_\rho = \llbracket \vec{t}_i \rrbracket_{\rho_0}$, contradicting semantic distinctness at the clashing position. $\square$

Remark 8.14 (Choice of equation form). Propositions 8.11 to 8.13 bound the design space: the unguarded equations are the strongest sound form and are what the factored architecture naturally produces; index premises are sound but strictly weaker, buying nothing semantically (their one virtue is to keep the equation silent at wrong-index instantiations, a reconstruction-signal nicety); pruning is sound and removes only equations that no guarded reasoning can reach, at the price that we do *not* claim conservativity — a first-order derivation of a goal may in principle route through junk instances of a pruned equation, so pruning can only be justified operationally, not by the model. All three verdicts are instances of the companion paper's division of labour: equations carry computation, valid everywhere; typing content lives in guarded specification and inversion axioms.

8.4 Indices, elements, and injectivity of type formers

Proposition 8.15 (Index uniqueness is a first-order consequence). *For every datatype I ,*

$$\Delta(\mathbb{E}, G) \vdash_{\text{FOL}} \forall w \vec{z} \vec{z}'. \hat{I}(w, \vec{z}) \rightarrow \hat{I}(w, \vec{z}') \rightarrow \bigwedge_h z_h \approx z'_h,$$

and the formula holds in $\mathfrak{M}$.

Proof. Model validity is Lemma 6.5(3). Derivability: apply inversion to both hypotheses, obtaining disjuncts i (witnesses $\vec{y}$) and i' (witnesses $\vec{y}'$) with $w \approx \hat{c}_i(\vec{y}) \approx \hat{c}_{i'}(\vec{y}')$. If $i \neq i'$, discrimination refutes the branch. If $i = i'$, injectivity gives $y_j \approx y'_j$ for all j ; each index equation pair $z_h \approx \mathcal{C}(\mathcal{E}(t_{ih}))[\vec{y}]$, $z'_h \approx \mathcal{C}(\mathcal{E}(t_{ih}))[\vec{y}']$ then yields $z_h \approx z'_h$ by congruence of $\approx$ (the two right-hand sides are the same first-order term at componentwise-equal arguments). For $r_i = 0$ the two right-hand sides are identical closed terms. $\square$

Remark 8.16 (Injectivity of type formers is neither expressible nor wanted). Index uniqueness is the sound first-order shadow of the type-theoretic intuition “ $I(\vec{u})$ and $I(\vec{u}')$ are disjoint unless $\vec{u} = \vec{u}'$ ”: it speaks about *elements*. The stronger, type-level statement — injectivity of the type former, $I\vec{u} = I\vec{u}' \rightarrow \vec{u} = \vec{u}'$ — is not even a formula of $\Sigma_{\mathbb{E}}$: types are not terms in this translation (Definition 5.1), so there is no equality between type images to invert. This inexpressibility is a feature. In type theory, injectivity of inductive type constructors is *anti-classical*: Hur’s construction [Hur10] refutes excluded middle in Agda with injective type constructors, by a Cantor-style diagonal through an injection $(\text{Set} \rightarrow \text{Set}) \rightarrow \text{Set}$ — so a translation validating classical logic, as any ATP-facing translation must, cannot emit any image of it. Encodings that do represent type expressions as first-order terms (types-as-terms with a `HasType` predicate) must therefore be careful that no axiom makes the type-term constructor injective-and-surjective in a way that imports the paradox; in the present design the question cannot arise. The same discipline governs constructor discrimination: the axioms of Definition 5.10 concern fully applied constructor images only — there are no partially applied constructor symbols in $\Sigma_{\mathbb{E}}$ at all, constructors being r -ary function symbols — which is exactly the boundary drawn by Counterexample 35 of [CD18]: conflict between *under-applied* constructors is refutable under functional extensionality (Agda issue #1497) and must not be asserted by a translation whose source may consistently assume it.

8.5 Singleton collapse and uniqueness of identity proofs

The `scase`(; ;) and `cast`(; ;) cases of Lemma 7.6, and the unconditional identity equations that user-written transports compile to (Example 5.13), are the exact locations where the model’s proof irrelevance is spent as *uniqueness of identity proofs*. We make the accounting precise.

Fact 8.17 (The semantic UIP surrogate). *Let J be a forced singleton with clause $k : \forall \vec{x}:\vec{\tau}. \vec{\varphi} \rightarrow J(\vec{u}^0)$, solution j , and let $J(\vec{u})$ be an occurrence with solution substitution θ . For every valuation ρ with $\langle J(\vec{u}) \rangle_{\rho} = 1$:*

$$\llbracket \theta(\vec{u}^0) \rrbracket_{\rho} = \llbracket \vec{u} \rrbracket_{\rho} \quad \text{and} \quad \langle \varphi_l[\theta] \rangle_{\rho} = 1 \text{ for all } l.$$

Proof. Extracted from the `scase`(; ;) case of Lemma 7.6, which proves exactly this. $\square$

In type-theoretic terms, Fact 8.17 says: the mere *truth* of the singleton family at $\vec{u}$ collapses the motive instance at the generic (constructor) indices with the motive instance at $\vec{u}$, and recovers the premises — without inspecting any proof. Gilbert, Cockx, Sozeau and Tabareau prove that this collapse, availed as a definitional principle for an equality-like family under proof irrelevance, *is* UIP [GCST19, §2.2]; and the without-K analysis of pattern matching isolates the same strength in the unification steps of deletion

and of injectivity at non-linear or non-self-unifiable indices [CDP14, CD18]. Our model validates all of them at once, because $\langle \cdot \rangle$ is proof-irrelevant and Lemma 6.5 holds; consequently:

- *Deletion* costs nothing: a match on $\text{eq}_\sigma(a, a)$ (or a $\text{cast}(\cdot; \cdot)$ along it) is validated by Fact 8.17 with $\theta(\vec{u}^0)$ and $\vec{u}$ literally equal.
- *Non-linear index patterns* cost nothing: nothing in Definition 3.6 or in Lemma 6.5 restricts repeated variables or self-unifiability; the without-K side conditions exist to avoid UIP, which the model has.
- *Heterogeneity* costs nothing at the target: the index equations of Definition 5.10 are pointwise untyped equations; the formation problem that motivates John Major equality [McB99] does not exist in $\Sigma_{\mathbb{E}}$.

What the strength *costs* is reconstruction, not soundness: a first-order proof that uses a transport-collapse equation at a type without decidable equality may need UIP (or an axiom-free inversion argument where one exists [Mon10, CD18]) to replay in CIC. This is the same debt the companion paper documents for $\text{cast}(\cdot; \cdot)$ -erasure, unchanged in kind; the premise-carrying variant below eliminates it at the price of weaker axioms.

Proposition 8.18 (Werner’s guarded collapse as an emission variant). *Let $\mathbb{E}_0$ declare `bool`, the forced singleton eq_{bool} (Example 3.4), and*

$$c := \lambda x y:\text{bool}. \lambda e:\text{eq}_{\text{bool}}(x, y). \lambda w:\text{bool}. \\ \text{scase}_{\text{eq}_{\text{bool}}}(e; \vec{z}.\text{bool}; \cdot.w),$$

with emitted equation $\forall x y w. \widehat{c}(x, y, w) \approx w$ (Example 5.13, at the constant motive). The premised variant

$$\text{Eq}_c^\pi := \forall x y w. y \approx x \rightarrow \widehat{c}(x, y, w) \approx w$$

(premise: the non-trivial index equation of the ford, $\vec{u} \approx \theta(\vec{u}^0)$) is derivable from the unguarded equation, hence $\mathfrak{M}$ -valid; and the unguarded equation is not derivable from the theory Δ^π obtained from $\Delta(\mathbb{E}_0, \top)$ by replacing it with Eq_c^π .

Proof. Derivability is weakening. For non-derivability, a witness-repair modification of $\mathfrak{M}$ in the style of Proposition 8.12 is not available: the pointer axiom forces the behaviour of $\widehat{c}$ to be the applicative behaviour of some element of $\mathcal{D}$, and the premised equation constrains that behaviour on *every* diagonal triple (d, d, w) , junk included — where a $\lambda_\square$ program that inspects its arguments is stuck; rather than pursue a definability analysis of $\mathcal{D}$, we exhibit a finite countermodel. Let $\mathfrak{N}$ have universe $\{0, 1\}$ with: $\approx$ interpreted as identity; $\widehat{\text{true}} := 0$, $\widehat{\text{false}} := 1$; $\widehat{\text{bool}} := \{0, 1\}$; $\widehat{\text{eq}_{\text{bool}}} := \{(0, 0), (1, 1)\}$;

$\odot := 0$; $u @ v := u \oplus v$ (exclusive or); $\widehat{c}^0 := 0$; and $\widehat{c}(x, y, w) := x \oplus y \oplus w$. Every axiom of Δ^π holds: constructor typing and inversion for **bool** (the universe is $\{0, 1\}$), discrimination ($0 \neq 1$), the clause and case-analysis axioms of eq_{bool} (its interpretation is the diagonal of $\widehat{\text{bool}}$), the specification axiom of c (its conclusion $\widehat{\text{bool}}(\widehat{c}(x, y, w))$ is unconditionally true), the pointer axiom ($((\widehat{c}^0 \oplus x) \oplus y) \oplus w = x \oplus y \oplus w$), and the premised equation ($y = x$ gives $x \oplus x \oplus w = w$). The unguarded equation fails: $\widehat{c}(0, 1, 0) = 1 \neq 0$. $\square$

This is precisely the situation Werner analysed for proof-irrelevant type theories [Wer08]: the unconditional singleton reduction $(\text{Eq_rec } A \text{ } P \text{ } a \text{ } b \text{ } p \text{ } e) \triangleright p$ “easily breaks the subject reduction property in incoherent contexts,” and his repair keeps the index condition as a guard: reduce only *if* $a =_{\beta_\varepsilon} b$ (generalized to arbitrary singletons: if $\vec{u} =_{\beta_\varepsilon} \vec{a}$). The premised equation is that rule with “provable” in place of “definitionally checked” — a strictly weaker demand, which is why both variants are sound here while Werner needs the check for decidability and subject reduction. An implementation can therefore expose the choice as a dial: unguarded collapse (strongest, UIP-debt at reconstruction) or premised collapse (weaker, replayable by plain inversion), per definition or globally. Brady, McBride and McKinnin’s run-time warning — collapsing “runs the risk of reducing a proof of something which cannot be constructed, such as $5 \leq 4$ ” in open contexts [BMM04] — is the same point seen from the operational side; in the model-theoretic setting the absurd-context hazard is neutralised because equations are consequences of one consistent structure (Corollary 7.10), and what remains of it is exactly the premised-versus-unguarded choice.

Remark 8.19 (Intensionality: the residual caveat). $\mathfrak{M}$ interprets equality as $\lambda_\square$ -convertibility, which is intensional: for the standard recursive **add**, $[\lambda x. \text{add } 0 \text{ } x] \neq [\lambda x. x]$ in $\mathcal{D}$ (the left representative is stuck at the variable) although the two classes agree applicatively on all of $\llbracket \text{nat} \rrbracket$. So functional extensionality is *not* validated: environments whose premise lemmas are proved from a funext postulate step outside Theorem 7.9 (Remark 7.13), though Corollary 7.10 is unaffected since no extensionality axiom is ever emitted. This caveat is the honest residue of the F^* #1542 analysis (Remark 8.10): that incident needed *both* an extensionality axiom *and* an index-blind equality encoding; the present translation emits neither, and each half of the immunity is a theorem — Proposition 8.9 for the index-blindness half, the present remark for the extensionality half.

9 Shallowness

Shallowness is the design constraint that shaped every clause of Section 5: first-order ATPs are heavily optimised for the primitives of first-order logic, so a translation is efficient to the extent that the logical structure of the

source arrives at the prover *as* first-order structure, with no inhabitation predicate and no bridge axioms to instantiate [Cza18, §1.1]. The companion paper articulates this as a checklist S1–S5 and verifies it for the non-indexed translation; we re-verify each item on the extended translation and add the indexed-specific item S6.

- S1 (Native logical structure).** $\mathcal{F}$ maps every connective and quantifier of a shallow proposition to its first-order counterpart and source equality to $\approx$, at translation time (Definition 5.7); nothing indexed changes this. Expansion variants (Definition 8.1) go further in the same direction: the *content* of a family occurrence — payload and index equations — becomes part of the formula in which it occurs.
- S2 (No inhabitation predicate).** $\Sigma_{\mathbb{E}}$ contains no binary has-type predicate and no term encoding of types (Definition 5.1). The typing-flavoured atoms are the per-family guard predicates, now $(1+m)$ -ary; their index argument places carry *terms*, which is first-order structure the prover indexes and unifies natively. This is the point where the present design parts company with the F^* encoding, the one production system emitting inversion axioms with index equations for type-level families [The24a, Agu16]: there, types are terms and typing is a fuel-instrumented binary predicate, so inversion must be throttled by fuel and triggers to avoid matching loops between inversion and elimination axioms; here there is no self-instantiating typing axiom to loop on — the inversion axiom is guarded by an atom that only constructor typing produces, and (in expansion variants) hypothesis-side content is inlined at translation time rather than recovered by E-matching at solve time. The residual caveat is SMT-specific: a trigger discipline for the guarded inversion axioms of recursive families would be needed by SMT backends, which have no analogue of a saturation prover’s fair clause selection.
- S3 (Per-occurrence payload expansion).** Refinement payloads are expanded inline at every occurrence, in both polarities, as before. The forded view extends this uniformly to families: Theorem 8.3 licenses inlining constructor payloads *and index equations* at any chosen occurrence, in both polarities, and Proposition 8.4 shows the choice is operational. “Index equations are just more propositional payload.”
- S4 (Proofs never become terms).** Erasure drops proof arguments, premise proofs of constructors, and the index-equation binders of forded branches (Definition 4.5); specification extraction keeps the term unapplied across propositional premises (Definition 5.8). Proof subterms *inside index terms* vanish too (Remark 4.6), so no proof material reaches the prover even through the new index argument places.

- S5 (Definitions as oriented computation).** Pattern-matching definitions over families compile to unguarded, orientable, non-overlapping rewrite equations (Remark 5.5, Proposition 8.11) — for vmap_g , literally the recursive program. Case-analysis content (a search operation) is segregated into the guarded inversion axioms, emitted once per family; provers pay for splitting only when a proof needs it.
- S6 (Index constraints as equations, stated not solved).** All index information of a family reaches the prover as first-order equations between ordinary terms — in inversion disjuncts, in expanded guards, in constructor-typing conclusions — computed at translation time by fording, never by unification. Three classical obstructions dissolve at this interface, each by a theorem of this paper rather than by fiat: heterogeneous telescopic equations are stated pointwise in the untyped $\approx$ (Definition 5.10; the formation problem of [McB99] does not exist in $\Sigma_{\mathbb{E}}$); defined functions in index positions are handled by superposition against their defining equations (Remark 5.16), where elaboration-time unification is stuck [McB14]; and the K/UIP side conditions of dependent pattern matching are validated by the model (Section 8.5). Dually, the three soundness obligations the interface creates are discharged as theorems: guarded exhaustiveness only (Proposition 8.7), index-carrying guard atoms only (Proposition 8.9), and no structural axioms beyond constructor images (Remark 8.16).

The residue. As in the companion paper, partial applications go through $@$ and pointer constants, and anonymous λ -abstractions, matches and fix-points inside formulas are lifted to fresh symbols — unavoidable in a language without binders, and local to terms. The indexed extension adds one residue class: *induction on families* is not first-order and is not emitted (Section 11.4); the emitted content of a family is exactly its ford — constructor typing, guarded inversion, injectivity, discrimination, and the defining equations of functions over it.

10 Related work

Fording and the elimination of dependent pattern matching. The constraint technique is McBride’s [McB99, §3.5], including coalescence (our Lemma 5.14) and the worked `vtail` example that Example 3.15 replays inside CIC_{ix}^- ; the datatype-level formulation used in Section 2 is stated by Zhang [Zha23]. Goguen, McBride and McKinna [GMM06] eliminate Coquand-style dependent pattern matching [Coq92] into eliminators plus heterogeneous equality and K, via specialization by unification; their five unification transitions (deletion, solution, injectivity, conflict, cycle) are exactly the first-order theory of constructors, and Remark 5.16 is the observation — made there

for elaboration, here as a soundness theorem — that a saturation prover performs these transitions as ordinary equational inference, with defined-function-headed indices handled strictly better. The without-K refinements of Cockx et al. [CDP14, CD18] supply the exact accounting of which transitions require K/UIP; Section 8.5 shows the model validates the unrestricted set, and Remark 8.16 imports their negative results (Counterexample 35; no type-former injectivity, after Hur [Hur10]). Coq’s `inversion` tactic performs fording at use sites [CT96]; Monin’s small inversions [Mon10] show the generated content is derivable with no axioms, which is what makes the reconstruction obligations of our inversion axioms benign; the Equations package [SM19] implements the whole stack inside Coq, with forced-argument analysis (after Brady, McBride and McKinna [BMM04]) as the UIP-avoidance mechanism that Lemma 5.14 repurposes as an optimisation.

Erasure and proof irrelevance. The program half follows Letouzey [Let03, Let04, Let08]: sort-directed, index-blind erasure, with stored indices kept; MetaCoq mechanises it with the `Subsingleton` criterion for case erasure [SBF⁺20]. Our `scase( ; ; )` is the stratified image of CIC’s singleton elimination [The26], extended from “parameters put aside” to forced arguments; that this extension under proof irrelevance is exactly UIP is the analysis of Gilbert, Cockx, Sozeau and Tabareau [GCST19], whose corrected criterion (non-forced arguments propositional, orthogonal index patterns, guardedness) delimits what can be *definitional* — our model needs only what is *true* in a proof-irrelevant semantics, which is why orthogonality is not required here (Definition 3.6). Werner’s guarded singleton reduction [Wer08] is the syntactic ancestor of Proposition 8.18; the proof-irrelevant model tradition [Wer97, MW03, LW11] is where our truth-value semantics lives, and Lee–Werner’s “singleton with a canonical element” is the denotational form of $\odot$, whose algebraic form is Atkey’s $!_0x = !_0y$ in quantitative type theory [Atk18] and whose operational form is Idris 2’s quantity-0 erasure [Bra21]. Brady, McBride and McKinna’s forcing, detagging and collapsing [BMM04] delete at run time what our translation asserts as equations; their warning about collapsing in open contexts is discussed after Proposition 8.18.

Hammers and type encodings. The immediate context is the CoqHammer translation [CK18, CK16], its proof-theoretically sound core [Cza18], and the companion paper [Ext26] whose architecture this paper extends; Section 1.4 and Remark 5.17 give the correspondence. For HOL hammers, the translation stack and its model-with-junk soundness proofs are due to Meng and Paulson [MP08] and Blanchette et al. [BBPS16]; their monotonicity analysis — a type’s guards are erasable iff no positive equality literal bounds it — identifies exhaustiveness axioms as never-erasable, which Proposition 8.7 sharpens for families. HOL systems never face type-level

index unification: inductive families arrive pre-forded, either as predicates whose `.cases` rules are inversion-with-equations shipped as ordinary lemmas, or as subset types (`vec A n` as $\{l \mid \text{length } l = n\}$) — fording is the default in weaker logics, which is indirect evidence for its suitability as a translation target. Sledgehammer’s SMT route [BBP13] and the SMT datatypes theory [BST07, RB17] axiomatise closed-world initial algebras — exhaustiveness, selectors, demand-driven splitting — which is exactly what an indexed family is not; Lean-auto [QCBA25] accordingly ground-monomorphises index instances before shipping `declare-datatypes`. F* [SHK⁺16] is the closest production design: a single-sorted target with a fuel-instrumented typing predicate and per-family inversion axioms carrying index equations [The24a, Agu16]; its documented incidents — #1542 [Ase18] and the partially-applied-inductive #692 — and the Verus rank-axiom incident [The24b] are analysed against this paper’s design in Remark 8.10 and Example 5.12. Aguirre’s soundness analysis of the F* encoding [Agu16] proves a fragment proof-theoretically and observes that HOL-style monotonicity-based guard removal does not apply to indexed inductive types — consistent with our finding that family guards are the load-bearing ones. Liquid Haskell [VSJ⁺14] performs per-occurrence inversion with measure equations in the *checker*, keeping the solver quantifier-free — the most consistent embodiment of occurrence-driven expansion, at the price of checker-driven unfolding depth; Dafny [Lei10] deliberately emits no exhaustiveness axiom and per-case definitional axioms with constructor-shaped triggers, matching the shape of Definition 5.3’s output; Why3’s `eliminate_algebraic` emits projection-form inversion sound only in a many-sorted target [FP13]; the Dedukti/CoqInE route [BB12] turns inversion into rewriting, with index specialization by computation. The first-order metatheory of term algebras — completeness of injectivity/distinctness/domain-closure/acyclicity, the non-finite-axiomatisability of acyclicity, and the Π_1^1 -completeness with uninterpreted symbols — is Kovács, Robillard and Voronkov [KRV17]; the translation emits the finitely axiomatisable guarded fragment and omits acyclicity (Section 11.4).

Semantics of inductive families. Indexed inductive definitions in type theory are Dybjer’s [Dyb94]; the stage-based interpretation of Definition 6.3 is the classical iterated-inductive-definition construction [Acz77], in the indexed form used by set-theoretic models of CIC [Wer97, LW11]. The rank-restricted treatment of fixpoints is the semantic counterpart of guarded/sized recursion [Gim95, BFG⁺04], extended here with element ranks made well defined by Lemma 6.5(3).

11 Limitations and future work

The fragment CIC_{ix}^- was chosen so that every emitted axiom is provably sound while covering the indexed phenomena the fording architecture is about: constructor-constrained indices, forded case analysis, index-determined singletons, recursion over families. We delimit the boundary, separating presentational restrictions from semantic ones, and record the reconstruction obligations.

11.1 Dependent index telescopes

Index types in CIC_{ix}^- are closed (Definition 3.3): a family may not be indexed over a telescope in which a later index type mentions an earlier index ($\text{dvec} : \Pi n : \text{nat. fin } n \rightarrow \text{Set}$). The restriction is inherited from the shape of source-level equations: the forded branch equations $\vec{u} =_{\vec{\sigma}} \vec{t}_i$ are homogeneous, and with dependent telescopes the equation at position h would relate terms whose types agree only after the earlier equations are substituted — McBride’s motivation for heterogeneous equality [McB99], whose eliminator is K . The *target* side has no such problem: untyped pointwise equations state telescopic constraints directly, and the emitted axioms of Definition 5.10 would be unchanged in form. Lifting the restriction therefore requires work only at the source: either heterogeneous equality in the proof layer (with its K -strength absorbed by the model, which validates it), or cast-chained telescopic equations. We expect no semantic obstacle — the stage construction and Lemma 6.5 nowhere use closedness of index types — but the bookkeeping in Figures 2 to 5 is substantial and is left as future work.

11.2 Polymorphism, large indices, large elimination

CIC_{ix}^- is monomorphic: type parameters are ground instances (Section 3.7), so families indexed by *types* — $\text{eq Set } A \ B$, universe-indexed structures — are outside the fragment, consistently with the intended pipeline in which heuristic monomorphisation runs first. Large elimination remains excluded; fording removes its programming role in impossible branches (Example 3.15), the primitive `discr` covers its role in discrimination, and its remaining uses (types computed by recursion) would break the rigidity of set types (Lemma 3.11) on which the structural definitions of $\mathcal{G}$ and $\llbracket \cdot \rrbracket$ rest.

11.3 Well-founded recursion and **Acc**-style predicates

The hard constraint identified in the companion paper stands, and the indexed extension gives it a second face. `fix` captures structural recursion only; **Acc**-based recursion is excluded, because defining equations obtained by recursion through an erased proof are sound only with the erased propositional

premises retained — the unconditional unfolding equation of Euclidean division at divisor 0 is *inconsistent* with $\forall n. n \not\approx S(n)$ -style facts. On the declaration side, `Acc`’s clause has its recursive premise under a quantifier, which Definition 3.3 excludes; semantically, its guardedness is not syntactic — Gilbert et al. observe that definitional proof irrelevance for `Acc` makes conversion undecidable [GCST19] — and our rank machinery, which is the semantic image of syntactic guardedness, does not apply. Both faces point to the same future work: premised unfolding equations with a semantics interpreting the defined symbol by an arbitrary total extension.

11.4 Acyclicity, induction, completeness

The translation emits no acyclicity axioms ($x \not\approx S(x)$ and its relatives) and no induction: acyclicity is not finitely axiomatisable over term algebras and induction is a schema [KRV17]; both are sound in $\mathfrak{M}$ (acyclicity by Corollary 4.4 and rank induction), so *adding* premise-selected instances would be sound, but the translation makes no completeness claim that would force them. Completeness fails further along the axes inherited from the companion paper: the semantics is classical and proof-irrelevant (Remark 7.14); conversion under binders for lifted symbols is not reflected; goals needing induction on families (e.g. $\forall n (v:\text{vec}(n)). \widehat{\text{vmap}}_{\text{id}}(n, v) \approx v$) need the reconstruction backend or user induction; and expansion variants unfold one constructor layer per occurrence, so properties requiring unbounded inversion depth need an inductive lemma — the same trade-off Liquid Haskell documents for checker-driven unfolding [VSJ⁺14]. A completeness theorem for the ground equational fragment over purely structural families (where Proposition 7.12 makes $\mathfrak{M}$ standard) should follow from Remark 5.5 by standard rewriting arguments and is left as future work.

11.5 Functional extensionality

$\mathfrak{M}$ is intensional (Remark 8.19): premise sets containing funext-derived lemmas step outside Theorem 7.9. Repairing this requires a coarser domain (quotient by observational equivalence), which would disturb Corollary 4.4 and the compilation argument; nothing indexed-specific changes the analysis, but Remark 8.10 raises its stakes — the F^* incident shows extensionality axioms and index-blind encodings interacting is how translations of this kind have actually failed.

11.6 Reconstruction

Soundness here is model-theoretic; the implemented pipeline’s final arbiter is proof reconstruction, and the paper’s results localise its obligations. Uses of inversion axioms and expanded guards replay by *inversion*-style reasoning, derivable without axioms [CT96, Mon10] — K-free, since the mo-

tive never depends on the generated equations. Uses of the unconditional singleton-collapse equations, of deletion, and of the extra permissiveness of Definition 3.6 beyond CIC’s parameter mechanism (Remark 3.9) may need UIP-class tactics at types without decidable equality; Proposition 8.18 is the premise-carrying alternative that eliminates this debt. A proof-theoretic soundness theorem in the style of [Cza18] — transforming first-order derivations from $\Delta(\mathbb{E}, G)$ into CIC_{ix}^- proof terms modulo excluded middle, proof irrelevance and UIP — would turn this localisation into an algorithm and is the natural next step, with the lifted-symbol substitution stability of the companion paper’s discussion as the known hard part.

11.7 Other restrictions

Mutual and nested inductive families, deep and mutual recursion, and coinductive families are excluded syntactically, as in the companion paper; each constructor of a mutual block fords independently, so we expect the extension to be bookkeeping, but coinductive families — whose case-analysis content is a greatest-fixpoint inversion and whose equality is bisimilarity — are a genuine further step. Enumeration-, subset- and singleton-shaped families receive no special-cased treatment in the formal development; the classification exists in implementations as a choice of expansion policy, and Theorem 8.3 is deliberately policy-agnostic so that any such classification is covered.

References

- [Acz77] Peter Aczel. An introduction to inductive definitions. In Jon Barwise, editor, *Handbook of Mathematical Logic*, pages 739–782. North-Holland, 1977.
- [Agu16] Alejandro Aguirre. Towards a provably correct encoding from F^* to SMT. MPRI internship report, Inria Paris, 2016.
- [Ase18] Aseem Rastogi and the F^* Development Team. Unsoundness in the SMT encoding of functional extensionality. FStarLang/FStar issue #1542, <https://github.com/FStarLang/FStar/issues/1542>, 2018.
- [Atk18] Robert Atkey. Syntax and semantics of quantitative type theory. In *Proceedings of the 33rd Annual ACM/IEEE Symposium on Logic in Computer Science (LICS 2018)*, pages 56–65. ACM, 2018.
- [BB12] Mathieu Boespflug and Guillaume Burel. CoqInE: Translating the calculus of inductive constructions into the $\lambda\Pi$ -calculus modulo.

- In *Proceedings of the Second International Workshop on Proof Exchange for Theorem Proving (PxTP 2012)*, pages 44–50, 2012.
- [BBP13] Jasmin Christian Blanchette, Sascha Böhme, and Lawrence C. Paulson. Extending Sledgehammer with SMT solvers. *Journal of Automated Reasoning*, 51(1):109–128, 2013.
- [BBPS16] Jasmin Christian Blanchette, Sascha Böhme, Andrei Popescu, and Nicholas Smallbone. Encoding monomorphic and polymorphic types. *Logical Methods in Computer Science*, 12(4), 2016.
- [BFG⁺04] Gilles Barthe, Maria João Frade, Eduardo Giménez, Luís Pinto, and Tarmo Uustalu. Type-based termination of recursive definitions. *Mathematical Structures in Computer Science*, 14(1):97–141, 2004.
- [BKPU16] Jasmin Christian Blanchette, Cezary Kaliszyk, Lawrence C. Paulson, and Josef Urban. Hammering towards QED. *Journal of Formalized Reasoning*, 9(1):101–148, 2016.
- [BMM04] Edwin Brady, Conor McBride, and James McKinna. Inductive families need not store their indices. In *Types for Proofs and Programs (TYPES 2003)*, volume 3085 of *LNCS*, pages 115–129. Springer, 2004.
- [BN98] Franz Baader and Tobias Nipkow. *Term Rewriting and All That*. Cambridge University Press, 1998.
- [Bra21] Edwin Brady. Idris 2: Quantitative type theory in practice. In *35th European Conference on Object-Oriented Programming (ECOOP 2021)*, volume 194 of *LIPICs*, pages 9:1–9:26. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2021.
- [BST07] Clark Barrett, Igor Shikanian, and Cesare Tinelli. An abstract decision procedure for a theory of inductive data types. *Journal on Satisfiability, Boolean Modeling and Computation*, 3(1–2):21–46, 2007.
- [CD18] Jesper Cockx and Dominique Devriese. Proof-relevant unification: Dependent pattern matching with only the axioms of your type theory. *Journal of Functional Programming*, 28:e12, 2018.
- [CDP14] Jesper Cockx, Dominique Devriese, and Frank Piessens. Pattern matching without K. In *Proceedings of the 19th ACM SIGPLAN International Conference on Functional Programming (ICFP 2014)*, pages 257–268. ACM, 2014.

- [CK16] Łukasz Czajka and Cezary Kaliszyk. Goal translation for a hammer for Coq (extended abstract). In *Proceedings First International Workshop on Hammers for Type Theories (HaTT 2016)*, volume 210 of *EPTCS*, pages 13–20, 2016.
- [CK18] Łukasz Czajka and Cezary Kaliszyk. Hammer for Coq: Automation for dependent type theory. *Journal of Automated Reasoning*, 61(1–4):423–453, 2018.
- [CK26] Łukasz Czajka and Cezary Kaliszyk. CoqHammer: Automated reasoning for Rocq (Coq). <https://coqhammer.github.io/>, 2026.
- [Coq92] Thierry Coquand. Pattern matching with dependent types. In *Proceedings of the Workshop on Types for Proofs and Programs*, pages 66–79, Båstad, Sweden, 1992.
- [CT96] Cristina Cornes and Delphine Terrasse. Automating inversion of inductive predicates in Coq. In *Types for Proofs and Programs (TYPES 1995)*, volume 1158 of *LNCS*, pages 85–104. Springer, 1996.
- [Cza18] Łukasz Czajka. A shallow embedding of pure type systems into first-order logic. In *22nd International Conference on Types for Proofs and Programs (TYPES 2016)*, volume 97 of *LIPICs*, pages 9:1–9:39. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2018.
- [Dyb94] Peter Dybjer. Inductive families. *Formal Aspects of Computing*, 6(4):440–465, 1994.
- [Ext26] Program extraction and specification extraction: A sound shallow translation of dependent type theory into first-order logic. Companion manuscript, 2026.
- [FP13] Jean-Christophe Filliâtre and Andrei Paskevich. Why3 — where programs meet provers. In *Programming Languages and Systems (ESOP 2013)*, volume 7792 of *LNCS*, pages 125–128. Springer, 2013.
- [GCST19] Gaëtan Gilbert, Jesper Cockx, Matthieu Sozeau, and Nicolas Tabareau. Definitional proof-irrelevance without K. *Proceedings of the ACM on Programming Languages*, 3(POPL):3:1–3:28, 2019.
- [Gim95] Eduardo Giménez. Codifying guarded definitions with recursive schemes. In *Types for Proofs and Programs (TYPES 1994)*, volume 996 of *LNCS*, pages 39–59. Springer, 1995.

- [GMM06] Healfdene Goguen, Conor McBride, and James McKinna. Eliminating dependent pattern matching. In *Algebra, Meaning, and Computation: Essays dedicated to Joseph A. Goguen*, volume 4060 of *LNCS*, pages 521–540. Springer, 2006.
- [Hur10] Chung-Kil Hur. Agda with excluded middle is inconsistent. Message to the Agda mailing list, <https://lists.chalmers.se/pipermail/agda/2010/001543.html>, January 2010.
- [Klo80] Jan Willem Klop. *Combinatory Reduction Systems*. PhD thesis, Utrecht University, 1980. Mathematical Centre Tracts 127.
- [KRV17] Laura Kovács, Simon Robillard, and Andrei Voronkov. Coming to terms with quantified reasoning. In *Proceedings of the 44th ACM SIGPLAN Symposium on Principles of Programming Languages (POPL 2017)*, pages 260–270. ACM, 2017.
- [Lei10] K. Rustan M. Leino. Dafny: An automatic program verifier for functional correctness. In *Logic for Programming, Artificial Intelligence, and Reasoning (LPAR-16)*, volume 6355 of *LNCS*, pages 348–370. Springer, 2010.
- [Let03] Pierre Letouzey. A new extraction for Coq. In *Types for Proofs and Programs (TYPES 2002)*, volume 2646 of *LNCS*, pages 200–219. Springer, 2003.
- [Let04] Pierre Letouzey. *Programmation fonctionnelle certifiée – L’extraction de programmes dans l’assistant Coq*. PhD thesis, Université Paris-Sud, 2004. HAL tel-00150912.
- [Let08] Pierre Letouzey. Extraction in Coq: An overview. In *Logic and Theory of Algorithms (CiE 2008)*, volume 5028 of *LNCS*, pages 359–369. Springer, 2008.
- [LW11] Gyesik Lee and Benjamin Werner. Proof-irrelevant model of CC with predicative induction and judgmental equality. *Logical Methods in Computer Science*, 7(4:5), 2011.
- [McB99] Conor McBride. *Dependently Typed Functional Programs and their Proofs*. PhD thesis, University of Edinburgh, 1999. Technical report ECS-LFCS-00-419.
- [McB14] Conor McBride. How to keep your neighbours in order. In *Proceedings of the 19th ACM SIGPLAN International Conference on Functional Programming (ICFP 2014)*, pages 297–309. ACM, 2014.

- [Mon10] Jean-François Monin. Proof trick: Small inversions. Second Coq Workshop, Edinburgh; HAL inria-00489412, 2010.
- [MP08] Jia Meng and Lawrence C. Paulson. Translating higher-order clauses to first-order clauses. *Journal of Automated Reasoning*, 40(1):35–60, 2008.
- [MW03] Alexandre Miquel and Benjamin Werner. The not so simple proof-irrelevant model of CC. In *Types for Proofs and Programs (TYPES 2002)*, volume 2646 of *LNCS*, pages 240–258. Springer, 2003.
- [QCBA25] Yicheng Qian, Joshua Clune, Clark Barrett, and Jeremy Avigad. Lean-auto: An interface between Lean 4 and automated theorem provers. In *Computer Aided Verification (CAV 2025)*. Springer, 2025. arXiv:2505.14929.
- [RB17] Andrew Reynolds and Jasmin Christian Blanchette. A decision procedure for (co)datatypes in SMT solvers. *Journal of Automated Reasoning*, 58(3):341–362, 2017.
- [SBF⁺20] Matthieu Sozeau, Simon Boulter, Yannick Forster, Nicolas Tabareau, and Théo Winterhalter. Coq Coq correct! Verification of type checking and erasure for Coq, in Coq. *Proceedings of the ACM on Programming Languages*, 4(POPL):8:1–8:28, 2020.
- [SHK⁺16] Nikhil Swamy, Cătălin Hrițcu, Chantal Keller, Aseem Rastogi, Antoine Delignat-Lavaud, Simon Forest, Karthikeyan Bhargavan, Cédric Fournet, Pierre-Yves Strub, Markulf Kohlweiss, Jean-Karim Zinzindohoué, and Santiago Zanella-Béguelin. Dependent types and multi-monadic effects in F*. In *Proceedings of the 43rd Annual ACM SIGPLAN-SIGACT Symposium on Principles of Programming Languages (POPL 2016)*, pages 256–270. ACM, 2016.
- [Sho67] Joseph R. Shoenfield. *Mathematical Logic*. Addison-Wesley, 1967.
- [SM19] Matthieu Sozeau and Cyprien Mangin. Equations reloaded: High-level dependently-typed functional programming and proving in Coq. *Proceedings of the ACM on Programming Languages*, 3(ICFP):86:1–86:29, 2019.
- [Tak95] Masako Takahashi. Parallel reductions in λ -calculus. *Information and Computation*, 118(1):120–127, 1995.
- [The24a] The F* Development Team. Understanding how F* uses Z3. The F* book, https://fstar-lang.org/tutorial/book/under_the_hood/uth_smt.html, 2024.

- [The24b] The Verus Development Team. Unsoundness from box/unbox axioms for empty structs and height axioms. verus-lang/verus issue #1366, <https://github.com/verus-lang/verus/issues/1366>, 2024.
- [The26] The Rocq Development Team. The Rocq prover reference manual. <https://rocq-prover.org/doc/>, 2026.
- [VSJ⁺14] Niki Vazou, Eric L. Seidel, Ranjit Jhala, Dimitrios Vytiniotis, and Simon Peyton-Jones. Refinement types for Haskell. In *Proceedings of the 19th ACM SIGPLAN International Conference on Functional Programming (ICFP 2014)*, pages 269–282. ACM, 2014.
- [Wer97] Benjamin Werner. Sets in types, types in sets. In *Theoretical Aspects of Computer Software (TACS 1997)*, volume 1281 of *LNCS*, pages 530–546. Springer, 1997.
- [Wer08] Benjamin Werner. On the strength of proof-irrelevant type theories. *Logical Methods in Computer Science*, 4(3:13), 2008.
- [Zha23] Tesla Zhang. Two tricks to trivialize higher-indexed families. arXiv:2309.14187, 2023.

A Confluence of $\lambda_{\square}$

We prove Theorem 4.3 by the Tait–Martin-Löf method in Takahashi’s formulation [Tak95], extended to the ι , μ and δ rules. The proof is that of the companion paper, reproduced for self-containedness; it depends on the constructor alphabet only through arities, so the indexed extension changes nothing. Recall that the δ -rule unfolds a constant to the erasure of its body, $c \rightarrow_{\square} \mathcal{E}(t_c)$, and that bodies mention only earlier constants; no property of this ordering is needed beyond the fact that each constant has a *fixed* unfolding.

Definition A.1 (Parallel reduction). The relation $\Rightarrow$ on $\lambda_{\square}$ terms is defined inductively:

1. $x \Rightarrow x$, $\square \Rightarrow \square$, $c \Rightarrow c$;
2. $c \Rightarrow \mathcal{E}(t_c)$ for a defined constant c ;
3. $\lambda x.e \Rightarrow \lambda x.e'$ if $e \Rightarrow e'$;
4. $e_1 e_2 \Rightarrow e'_1 e'_2$ if $e_i \Rightarrow e'_i$;
5. $c(\vec{e}) \Rightarrow c(\vec{e}')$ if $e_i \Rightarrow e'_i$ componentwise;
6. $\text{case}(e; \vec{b}r) \Rightarrow \text{case}(e'; \vec{b}r')$ if $e \Rightarrow e'$ and branch bodies reduce componentwise;

7. $\text{fix } f.e \Rightarrow \text{fix } f.e'$ if $e \Rightarrow e'$;
8. $(\lambda x.e) a \Rightarrow e'[a'/x]$ if $e \Rightarrow e'$ and $a \Rightarrow a'$;
9. $\text{case}(c_i(\vec{e}); \{c_j(\vec{x}_j) \Rightarrow b_j\}_j) \Rightarrow b'_i[\vec{e}'/\vec{x}_i]$ if $e_l \Rightarrow e'_l$ and $b_i \Rightarrow b'_i$;
10. $\text{fix } f.e \Rightarrow e'[\text{fix } f.e'/f]$ if $e \Rightarrow e'$.

Lemma A.2. (i) $e \Rightarrow e$ for all e . (ii) $\rightarrow_{\square} \subseteq \Rightarrow \subseteq \rightarrow_{\square}^*$. (iii) If $e \Rightarrow e'$ and $u \Rightarrow u'$ then $e[u/x] \Rightarrow e'[u'/x]$.

Proof. (i) Induction on e using clauses (1) and (3)–(7). (ii) For the first inclusion, a one-step reduction is a root rule applied inside a context: the corresponding root clause (2), (8), (9) or (10) with reflexive subderivations, propagated by the congruence clauses. For the second, induction on $e \Rightarrow e'$: each clause is realised by finitely many $\rightarrow_{\square}$ -steps. (iii) Induction on the derivation of $e \Rightarrow e'$. The variable clause $x \Rightarrow x$ becomes $u \Rightarrow u'$; clause (2) is unaffected because $\mathcal{E}(t_c)$ is closed; the congruence clauses follow from the induction hypothesis; the redex clauses use the standard substitution identities, e.g. $(e'[a'/y])[u'/x] = (e'[u'/x])[a'[u'/x]/y]$ for $y \notin \text{FV}(u') \cup \{x\}$. $\square$

Definition A.3 (Complete development). The complete development e^* is defined by recursion on e :

$$\begin{aligned}
x^* &= x, & \square^* &= \square, & c^* &= \mathcal{E}(t_c), \\
(\lambda x.e)^* &= \lambda x.e^*, & c(\vec{e})^* &= c(\vec{e}^*), & (\text{fix } f.e)^* &= e^*[\text{fix } f.e^*/f], \\
((\lambda x.e) a)^* &= e^*[a^*/x], & (e_1 e_2)^* &= e_1^* e_2^* \quad (e_1 \text{ not a } \lambda), \\
\text{case}(c_i(\vec{e}); \vec{br})^* &= b_i^*[\vec{e}^*/\vec{x}_i], \\
\text{case}(e; \vec{br})^* &= \text{case}(e^*; \vec{br}^*) \quad (e \text{ not a constructor form}).
\end{aligned}$$

(For a constant, the development unfolds once and does not recurse into the unfolding; for a fix, the root μ -redex is contracted once, its residuals in the contractum being new redexes that are not developed.)

Lemma A.4 (Triangle property). If $e \Rightarrow e'$ then $e' \Rightarrow e^*$.

Proof. Induction on the derivation of $e \Rightarrow e'$, with a case analysis on the last clause used.

- Clause (1): for variables and $\square$ immediate; for $c \Rightarrow c$ we need $c \Rightarrow c^* = \mathcal{E}(t_c)$, which is clause (2).
- Clause (2): $e' = \mathcal{E}(t_c) = c^*$, and $e' \Rightarrow e'$ by Lemma A.2(i).
- Clauses (3), (5), (7), and clause (6) with non-constructor scrutinee: by the induction hypothesis componentwise. Two subtleties: for $\text{fix } f.e \Rightarrow \text{fix } f.e'$ (clause (7)) the target is $(\text{fix } f.e)^* = e^*[\text{fix } f.e^*/f]$,

reached from $\text{fix } f.e'$ by clause (10) with $e' \Rightarrow e^*$ (induction hypothesis). For $\text{case}(e; \vec{b}r)$ with e not a constructor form, the scrutinee's reduct e'' may or may not be a constructor form, but the development $\text{case}(e^*; \vec{b}r^*)$ is reached by clause (6) either way.

- Clause (4) with $e_1 = \lambda x.b$: then $e_1 \Rightarrow e'_1$ is by clause (3), $e'_1 = \lambda x.b'$, and $e^* = b^*[a^*/x]$; from $e' = (\lambda x.b') a'$ apply clause (8) with $b' \Rightarrow b^*$, $a' \Rightarrow a^*$ (induction hypotheses).
- Clause (4) with e_1 not a λ : $e^* = e_1^* e_2^*$; apply clause (4) with the induction hypotheses.
- Clause (8): $e = (\lambda x.b) a$, $e' = b'[a'/x]$; by induction hypotheses $b' \Rightarrow b^*$ and $a' \Rightarrow a^*$, so Lemma A.2(iii) gives $e' \Rightarrow b^*[a^*/x] = e^*$.
- Clause (9): as clause (8), with the componentwise induction hypotheses for the constructor arguments and the branch, and Lemma A.2(iii) extended to simultaneous substitutions. The scrutinee $c_i(\vec{e})$ can only have been reduced componentwise (clause (5)), so the case analysis in Definition A.3 is stable.
- Clause (10): $e = \text{fix } f.b$, $e' = b'[\text{fix } f.b'/f]$; by the induction hypothesis $b' \Rightarrow b^*$, whence $\text{fix } f.b' \Rightarrow \text{fix } f.b^*$ by clause (7), and Lemma A.2(iii) gives $e' \Rightarrow b^*[\text{fix } f.b^*/f] = e^*$.

□

Proof of Theorem 4.3. By Lemma A.4, $\Rightarrow$ has the diamond property: if $e \Rightarrow e_1$ and $e \Rightarrow e_2$ then $e_1 \Rightarrow e^*$ and $e_2 \Rightarrow e^*$. A relation with the diamond property has confluent reflexive-transitive closure (tiling), and by Lemma A.2(ii) the closure of $\Rightarrow$ is $\rightarrow_\square^*$. The characterisation of $=_\square$ by common reducts follows by induction on the conversion zig-zag (Church–Rosser). □